



Prime Minister's Office

A GUIDE TO RISK MANAGEMENT

IN REGULATION AND PUBLIC POLICY

June 2018



Prime Minister's Office

A GUIDE TO RISK MANAGEMENT IN REGULATION AND PUBLIC POLICY

מדריך לניהול סיכונים ברגולציה ובמדיניות ציבורית

Editor: Guy Mor // Co-Writers: Gabi Lebel and Arnon Avital

June 2018 Version 1

Legal Clarification:

This Guide has been created to assist government decision makers in the management of risks, as part of their decision making process. Risk management is a flexible and multifaceted tool. Care should be taken to apply the tools presented in the Guide in a manner suited to the characteristics and circumstances of each decision. This Guide in no way qualifies or derogates from any statutory provisions including the provisions of the administrative law.

Governance and Social Affairs Department, Prime Minister's Office
<http://bit.ly/RegRiskManage>



PREFACE

Governments engage in the protection of the public from risks. The public faces many diverse risks throughout society encompassing health, social, economic, safety and environmental domains. It is therefore important for government to manage risks where possible in an efficient and effective manner, whilst taking into account ramifications impacting public policies.

This Guide aims to assist decision makers dealing with potential and realized risks to the public. The formulation of effective regulatory rules and public policy based on risk management enables better protection of the public, more efficient action and a reduction in the costs that maybe imposed on individuals and businesses.

The function of the regulator is to manage public risks by establishing and enforcing rules designed to address and/or prevent potential risks. As such, risk management is an important tool for regulators.

Risk management is intended to

assist decision makers in choosing and implementing accurate solutions, especially when dealing with ambiguous conditions and limited resources. The Guide's key objective is to provide optimal decision-making procedures for dealing with risks, as well as enabling decision makers to achieve greater consistency, uniformity and transparency in their processes. The method presented in the Guide is based on several fundamental principles: reliance on data (evidence-based approach), quantification of the effect of risks and government decisions, and the use of cost-benefit analysis. Some Israeli decision makers apply all, or some of the principles, of the method. The Guide enables comprehensive and consistent application of applicable risk management tools when formulating regulation and public policy.

The Guide is designed for practical use. It presents, in brief, what needs to be done and how to implement it through practical guidelines and limited theoretical context.

The Guide also provides tools for a diverse range of decision makers, be it policymakers considering the adoption of a new economic policy, a regulator concerned with establishing safety rules, an accountant examining invoices prior to payment, or a manager seeking to determine a method for supervision and enforcement on the ground,

The method used in the Guide is modeled on risk management methods of Western governments (notably the United States, the United Kingdom and Australia) as well as doctrine developed by international organizations such as the OECD and the EU.

The Guide's first chapter details the basic principles of public risk management. The second chapter deals with risk management in the development of regulatory rules and public policy. The third chapter focuses on risk management in regard to the supervision and enforcement of regulatory compliance.

EXECUTIVE SUMMARY

INTRODUCTION TO RISK MANAGEMENT REGULATION AND PUBLIC POLICY

Every country, business or private individual has assets, interests and values that are important to them (collectively referred to in the Guide as “assets”). An asset maybe a business project, physical property or a public interest (such as public health). The fact an asset is important to us means we want to prevent harm to it. Thus, anyone who has an asset of importance and/or value to them is exposed to risks. Almost any activity poses a risk that could harm a valued asset and/or someone else’s asset. Therefore, by default, any social activity, growth and development presents exposure to risks.

There is no such thing as “zero risk” – thus “zero risk” cannot be an objective. Risks, like assets, have always existed and will continue to exist. The role of decision makers is to manage risks based on the understanding that it is impossible to achieve “zero risk.” The severity and likelihood of some risks though can, and should, be mitigated when justified by cost-benefit considerations.

Risk management is a thinking process that must addresses six fundamental questions:

1. What are our objectives, and/or what is important for us to protect?
2. What may cause harm to our assets?
3. Which risks will we choose to deal with and which not?
4. How will we deal with each of the risks?
5. In retrospect, did the measures we implement mitigate the risk?
6. In retrospect, what has changed? Have the assets, the risks or the risk treatment measures changed?

Risk management is not a one-time task but a regular and constant mode of work and decision making.

There are three main strategies for dealing with risk:

1. Risk acceptance: Risk acceptance means identifying a risk and deciding not to take proactive measures to reduce the likelihood of its realization or the severity of its effect. This

strategy is suitable for minor risks when the cost of dealing with the risk is greater than the value of the damage expected.

2. Risk avoidance: This approach aims to remove the risk entirely by eliminating the activity involving the risk. Risk avoidance is a good strategy when the risk is likely to have a major negative impact and/or when the activity generates very little benefit for society. In the context of public policy, this is an extreme strategy, which for the most part is impracticable. It should be applied sparingly and only in extreme cases.

3. Risk mitigation: This is the primary risk management strategy involving action taken to reduce the likelihood of the realization of the risk and the severity of its effect. This strategy includes the acquisition of equipment and technology, instruction and training, improvement of work processes, along with supervision and control.

Risk management balances three parameters:

1. The activity and benefit from it: The benefit from the economic/ social activity that is being regulated (e.g., driving on a road, production of electricity from a power plant, importation of goods). The benefit from the activity also includes future positive outcomes and opportunities.

2. Potential risks: The cost to society if the risk occurs. Referencing the above examples this could be; a road accident, a work accident, smuggling of prohibited substances). The risks may cause death, injury or damage to property and society.

3. Cost of the risk treatment measures: There is a price to dealing with risk. For example, when mitigating road accident risks, this may involve the purchase of safety measures such as improved car brakes, lane departure warning system, etc. One can also choose to prohibit the activity entirely (e.g., banning cars or the prohibiting imports) – but choosing such an extreme risk treatment would result

in heavy social costs, since the entire benefit of the activity is lost (“throwing out the baby with the bathwater”). Risk treatment measures impose direct and indirect costs, reflected mainly in the interruption and disruption of the activity.

Risk management is a process that deals first and foremost with causality. Successful risk management is based on a solid understanding of the causal connection between the problem (the risk) and the factors that create it. It also requires an understanding of the causal connection between the problem and the solution, as well as understanding the causal connection between the chosen solution and its effects on society.

The principle of maximum aggregate welfare

In risk management of private projects (in which the decision maker manages their own risks) it is customary to refer to “risk appetite”. Risk appetite is the amount of risk an organization is willing to take against the expected gain resulting from the activity. This

methodology is less suited to the management of public risks. It's usually implemented by a private sector's project manager who internalizes the benefits and damages from the risk and the cost of the risk treatment measures. In the governmental domain, the government bears responsibility for mitigating risks, but the cost of risk treatment measures is imposed on the supervised entities. Also, the government does not benefit directly from the economic rewards of the activity. Traditionally, decision makers in the public sector do not focus on the cost of the risk treatment measure or on the benefit from the activity on the agenda. Regulators, in particular, are evaluated on the basis of risk reduction. Therefore, the definition of governmental/regulatory "risk appetite" is a loaded process, irreconcilable with the role of the government decision maker.

Accordingly, risk management in regulation and public policy should be performed based on the principle of maximum aggregate welfare. This means the government has a duty to maximize the aggregate social welfare. In this context, the decision maker must consider the following

three parameters described: the benefit from the activity, the damage from the risk, and the risk mitigation cost. The decision maker must choose a mode of action that will generate the maximum benefit for the public. From the governmental perspective, the loss of ILS 1 due to a reduction in beneficial activity is equivalent to the loss of ILS 1 due to a realized risk, as well as ILS 1 that was invested in preventing the risk. Therefore, the decision maker

must factor in all the costs to society and choose the alternative that minimizes the societal loss.

The decision maker must consider the overall damage, the cost to the government, the cost to the public and choose the most cost-effective alternative. The following table illustrates comparison performance according to the principle of maximum aggregate welfare:

	Expected annual damage due to the risk	Cost of risk treatment measures for the government	Cost of risk treatment measures for the public	Total cost
Current situation	-80	0	0	-80
Alternative A	-20	-30	-70	-120
Alternative B	-40	-10	-10	-60

The example in the table above illustrates the importance of the principle of maximum aggregate welfare. Seemingly, Alternative A is preferable because it results in a much greater reduction of the risk. However, Alternative A creates heavy costs for the public, resulting in a worse outcome than currently in place. Overall it is not cost-effective or viable. An examination of the overall picture (total costs to the public) demonstrates Alternative B would be the optimum choice.

Management of limited resources

Time, money, reputation and professional experience are all resources of limited availability. The government and the public are required to make decisions taking into account the limited nature of available resources. Since governmental and societal resources are limited, when dealing with risks one must choose where it is most effective to invest resources. In this sense, risk treatment is a "zero-sum game": any choice of an alternative comes at the expense of another alternative.

Any policy or regulation that is not risk based will not take into account the variance between different cases and their unique characteristics. Therefore, severe risks might be under-treated (allocating insufficient resources) while negligible risks will be over-treated (an over-investment in limited resources).

Risk management procedures enables decision makers to re-direct resources from low-risk items, for better application in dealing with high-risk items. The goal is that, to the extent possible, the policy should match the actual risks.

Uncertainty and incomplete information

In any decision-making procedure, information will be lacking concerning the past and present state, compounded with a level of uncertainty regarding the future. Risk management, which deals with decision making under conditions of ambiguity, helps the decision maker make better quantified decisions. The more data collected, and the more refined analysis methods are, the better one can prepare for future scenarios with an increased level of certainty. It is unfeasible to collect and analyze all relevant data available, instead an equilibrium must be reached between the resources that can be devoted to the process and the need to acquire sufficient information that can provide an adequate level of confidence. The guiding principle in this regard was determined by the OECD: Policy must be set "on the basis of the best available information" for the decision maker.

A decision to avoid an action cannot be taken merely because of future uncertainty or a lack of information. Uncertainty is a baseline situation that

must be dealt with and not used as a consideration in risk management decision making.

Decision making under conditions of uncertainty and incomplete information does not mean decision making without data. The analysis must be based on information and professional assessments.

The role of the government decision maker and the responsibility of the supervised entities

The government engages in public risk management, but the creators of public risks are generally players in the public sector. The governmental role is to establish regulation/policy that will mitigate risks and encourage compliance with the regulation/policy. Responsibility for avoiding excessive risks and complying with the regulation/policy is imposed on the public. The fact that the government manages the risks does not take away from the fact that risk creators are responsible for the risks they create and the damage they cause.

RISK MANAGEMENT IN THE DEVELOPMENT OF REGULATION AND PUBLIC POLICY

The process of risk management in the development of regulation and public policy consists of three components: Risk Assessment, Risk Treatment, and Execution and Monitoring.

Risk Assessment is understanding risks as a public problem that must be solved. Understanding the problem is a fundamental principle when formulating solutions. **Risk Treatment** deals with the development of solutions – for example; whether government intervention is suitable, the selection of any intervention method, to the formulation of detailed intervention procedures and policy. The choice between the alternatives is based on a cost-benefit analysis according to the principle of maximum aggregate welfare. **Execution and Monitoring** ensures on-going evaluation once effective measures have been implemented and facilitates the adjustment of regulation or public policy where applicable.

Risk Assessment	Context setting
	Risk identificaton
	Risk evaluation
	Risk analysis
Risk Treatment	Strategy selection
	Method selection
	Content selection
	Crisis management
Execution and Monitoring	Risk communication
	Monitoring and evaluation

Part One - Risk Assessment

Risk Assessment involves mapping all the relevant risks, defining the characteristics of each risk and performing a quantitative analysis of their effect. As the process progresses, the level of analysis and the use of data increases:

Context setting – Definition of the scope of work, the public interest under protection and the activity

involved in the creation of the risk. The framework of the discussion is delineated in this stage - an essential basis for the continuation of the process.

Risk identification – Screening the activity and processes in order to identify potential risks. Non-identified risks cannot be treated.

Risk evaluation – Evaluation of each risk’s characteristics (when it occurs, who causes it, can it be discovered in advance, etc.). This will define a clear scenario for the risk.

Risk analysis – Quantitative analysis of the likelihood of each risk and the severity of its effect, along with comparison and prioritization of the different risks.

Part Two – Risk Treatment

Risk treatment deals with the selection and application of policy tools based on work performed during the risk assessment:

Selecting a risk management strategy – Consideration of various strategies: risk acceptance, risk avoidance and

risk mitigation. In this stage, the risks that justify government intervention, as well as the risks that are/are not to be treated in the crisis management framework, are selected.

Selecting a risk management method – Comparison between regulatory methods and policy tools. The three principal categories are methods and tools based on permits and licenses, binding rules and advanced regulation (including non-regulatory tools). The comparison should include a cost-benefit analysis of the different alternatives.

Selecting the risk management content – Defining the exact content of the method that was selected in the preceding stage: who will be subject to the duties, how will requirements be defined, what sort of supervision and enforcement system will there be, etc. The comparison should include a cost-benefit analysis of different content alternatives.

Crisis management – Preparation and allocation of resources for dealing with risks following their actualization.

Part Three – Execution and Monitoring

Execution and monitoring should include tools that help the decision maker with the implementation of the risk treatment policy.

Risk communication – Communicating information to the public regarding the risks, their nature and the ways chosen to deal with them. Risk communication should also provide the public with apparatuses for coping with the risks ex ante and ex post.

Monitoring and evaluation – Evaluating risks and the effect of the policy or regulation. This includes reviewing the resultant effect of the policy after implementation and adjusting the policy as necessary.

RISK MANAGEMENT IN SUPERVISION AND ENFORCEMENT

Risk management in supervision and enforcement is the continuation of risk management in the development of regulation and public policy. To mitigate risks, the supervision and enforcement system must encourage and ensure regulation / policy compliance. Accordingly, supervision and enforcement actions must be reflective of the risk level, encourage compliance and ensure deterrence.

The risk management process in supervision and enforcement consists of three key components: defining the supervision objectives, establishing the supervision system, and execution and monitoring.

Defining the Supervision Objectives	Mapping requirements
	Mapping supervised entities
Establishing the Supervision System	Allocating supervision and enforcement resources
	Planning supervision and enforcement activities
Execution and Monitoring	Execution of supervision and enforcement activities
	Monitoring and evaluation

Part One - Defining Supervision Objectives

It is necessary to map out the following elements of the supervision and enforcement system:

Mapping the requirements – Identifying all the requirements and analyzing the effect of each requirement on reducing the likelihood and severity of the risk.

Mapping the supervised entities – Identifying the entities that are subject

to regulation and characterizing them according to the size of the supervised group, expected compliance level and major reason for noncompliance.

Part Two - Establishing the Supervision System

To select and apply the appropriate tools, the decision maker must rely on the information collected about each requirement:

Allocating the supervision and enforcement resources – This stage is the planning phase. The supervision system and the allocation of supervision resources among the various requirements and entities should be assigned in this stage. Firstly, all the supervision tasks should be divided into three equal-sized groups. Each group should contain the same number of supervision tasks, but the apportionment of resources is different between the groups – resource allocation to the groups should be based on the 50-30-10 model. As such, 50% of the supervision resources are allocated to the first group of supervision tasks (the most important), 30% to the middle third and 10% are allocated to the lower third (the least important group of tasks).

Planning the supervision and enforcement activities – In this phase it is determined which supervision tools to use, at what frequency, and how inspections will be carried out, as well as defining work methods. This should be done with the aim of increasing compliance with requirements. In order to determine the most suitable and effective supervision tool, the reason for noncompliance must first be identified. Noncompliance can be due to ignorance, misunderstanding the requirements, inability to comply, indifference or deliberate criminality.

Part Three - Execution and Monitoring

This phase calls for the application of appropriate tools and methods in which to operate the supervision and enforcement system:

Execution of supervision and enforcement activities – This involves: collecting information on the functioning of the supervision and enforcement system, examining the actual effect on the risk profile of the supervised entities, examining the resultant effect after implementation and adapting the system to changes.



CONTENTS

Executive Summary6

**Chapter A –
Introduction to Risk Management in
Regulation and Public Policy 17**

Risks and Risk Management18

Basic Concepts19

Private and Public Risk
Management..... 22

Basics of Public Risk Management. 24

The Principle of Maximum Aggregate
Welfare..... 26

Management of Limited Resources 29

Uncertainty and Incomplete
Information31

The Role of the Decision Maker and
the Responsibility of the Supervised
Entities 32

Principles of Risk Management in
Supervision and Enforcement..... 33

How to Use the Guide40

**Chapter B –
Risk Management in Development
of Regulation and Public Policy ..43**

General44

Context setting 46

Risk identification 49

Risk evaluation..... 53

 Guiding questions for
 characterizing the risks 55

 Sources of information for the
 Risk Assessment process 59

Risk analysis61

 Quantifying the severity of
 damage from risk..... 67

 Analysis and comparison
 between risks 74

Selecting a strategy and tools for Risk
treatment 78

 Selecting a strategy80

 Examination of regulatory methods
 and policy tools 82

 Examination of the content of
 the regulatory method and policy
 tools.....95

Crisis Management.....114

Crisis Management118

Monitoring and evaluation123

**Chapter C –
Risk Management in Supervision
and Enforcement..... 125**

General.....126

Mapping the Requirements129

Mapping of the Supervised
Entities132

Distribution of Supervision and
Enforcement Resources135

 Distribution of Supervision
 Resources by the 50-30-10
 Model.....135

Planning the Supervision System
to Suit Each Supervised Entity...137

**Bibliography and References for
Further Reading 159**

Annex A:
Asset Registry 168

Annex B:
Summary of the Current Situation 169

Annex
C: Mapping Relevant Stakeholders170

Annex D:
Instructions for Conducting a Public
Discourse: Receipt of Information,

Complaint-Based Supervision .. 140

Planning Supervision and
Enforcement Actions142

 Selection of Supervision Tools...142

 Selecting the Supervision
 Format 144

Consolidation of Supervision
Activities..... 146

Defining Measurable Targets and
Customizing the Supervision
System147

Participation, and Consultation.... 171

Annex E:
Risk Registry.....176

Annex F:
Comparison of Regulatory Methods
and Policy Tools according to the
Degree of Government
Intervention178

Annex G:
Guidelines for Choosing Regulatory
Methods and Policy Tools.....181

Annex H:

Execution of Supervision and
Enforcement Actions 148

 Uniformity and Consistency in
 Operating the Supervision
 System 148

 Certainty and Clarity for
 Supervised Entities 149

 Deterrent and Proportionate
 Enforcement151

Monitoring and Evaluation153

Rules for Identifying Possible Harm
to Competition.....192

Annex I:
Rules for Identifying Possible Harm
to Small and Medium-Size
Businesses..... 196

Annex J:
Supervision Task List 198

Annex K:
Links for Downloading Tools and
Accessories.....200



CHAPTER A – INTRODUCTION TO RISK MANAGEMENT IN REGULATION AND PUBLIC POLICY

RISKS AND RISK MANAGEMENT

Every country, business or private individual has assets, interests and values that are important to them (collectively referred to in the Guide as “assets”). An asset maybe a business project, physical property or a public interest (such as public health). The fact an asset is important to us means we want to prevent harm to it. Thus, anyone who has an asset of importance and/or value to them is exposed to risks. Almost any activity poses a risk that could harm a valued asset and/or someone else’s asset. Individuals, organizations and countries perform economic and social activities because the benefit from the activity is greater than any risk they pose. There is no social or economic benefit without risk. The greater the benefit accruing from the activity (more assets), the greater the risk it entails. Therefore, any advancement of society and growth or improvement in social welfare is accompanied by exposure to risk.

There is no such thing as “zero risk” – thus “zero risk” cannot be an objective. Risks, like assets, have always existed and will continue to exist. For example: if the aim is to totally prevent road accidents, then this would mean banning all methods of road

transportation; if the aim is to totally eradicate the risk of gas leaks, then there would have to be a prohibition on the use of gas; and aiming to totally prevent risks from the importation of products would call for an embargo on all imports. Thus, risk management does not mean the total elimination of risks, rather the mitigation of risks. The decision maker must manage risks based on the understanding that it is impossible to achieve a “zero risk” outcome. The severity of risks and likelihood of occurrence can and should be mitigated when justified by cost-benefit considerations. It is imperative to convey this principle to the public. It is important to convey validated expectations to the public and media. The public must be able to manage risks, supported by the knowledge and understanding behind the reasoning of decisions taken and implemented. This avoids friction in case of accidents or incidents arising from the decisions. Accidents and incidents will always occur; the question is whether the risk is being managed correctly. Decision makers and policymakers have a duty to manage public risks in the most effective way.

Risk management is a thinking process that must address six fundamental questions:

1. What are the organization’s objectives and/or what is important for the organization to protect? (assets)
2. What may cause harm to the assets? (risk identification and evaluation)
3. Which risks to deal with and which not? (risk analysis, strategy selection)
4. How to deal with each of the risks? (selection of a method and definition of the policy content)
5. In retrospect, did the measures taken achieve their purpose of mitigating the risk? (monitoring and evaluation)
6. In retrospect, what has changed? Have the assets, the risks or the risk treatment measures changed? (monitoring and evaluation)

Risk management is not a one-time task but an organization’s ‘modus operandi’.

BASIC CONCEPTS

Risk is an uncertain future event with a negative impact. A similar definition was adopted in ISO 31000:2009 – Risk Management: “effect of uncertainty on objectives.” A risk is by nature a defined factual event (scenario). Every risk is examined and characterized according to two basic criteria: likelihood and severity of effect. Regulation and government policy are intended to protect the public interest from diverse events. As such, the risk management process, in regulation and in government policy, enables the provision of a solution for all types of risks (e.g., operational, management, systemic, financial and reputational risks).

The **probability or likelihood** of a risk occurring reflects the degree of possibility that the risk will be realized. Since a risk is an uncertain event, every risk is rated for probability on a scale of 0 (the risk will never be realized) to 1 (the risk will always occur).

The **severity of effect or impact** of a risk describes the damage to society resulting from the risk, if and when it is realized.

The **expected damage** is calculated by multiplying the risk’s probability by the severity of its effect. This parameter reflects the social cost of

the risk, factoring in the likelihood that it will be realized.

Illustrative example:

A risk causes damage of ILS 50, the probability of the risk’s occurrence is 0.1 - therefore, the expected damage due to the risk is ILS 5 (ILS 50* 0.1).

There are three main strategies for dealing with risk:

Risk acceptance: Risk acceptance means identifying the risk and deciding not to take proactive measures that may reduce the likelihood of its realization or the severity of its effect. This strategy is suitable for minor risks and when the cost of dealing with the risk is greater than the extent of the expected damage. Risk acceptance can be supported with provisions for dealing with the risk if and when it is realized (crisis management).

Risk avoidance: This approach seeks to avoid risk entirely. This means eliminating the activity that may cause the risk. Risk avoidance is a good strategy when the risk is highly likely to have a major negative impact and when the activity generates very little benefit for society. In the context of public policy, risk avoidance means the prohibition of a certain activity.

This is an extreme strategy which for the most part is impracticable. It should be applied sparingly and only in extreme cases.

A risk may occur even if action has been taken to prevent it entirely. However, a sweeping attempt to avoid all risks is liable to lead to the opposite result as the absolute prohibition denies society information regarding the risk. If a certain activity is prohibited, the government, the public and businesses will not have information about the risks it entails. When an activity generates risk over time, appropriate measures or modes of action may be developed for its mitigation. An absolute prohibition on an activity prevents both the government and the public from developing technologies and work methods that could reduce the risk’s likelihood and severity. Total exclusion of a risk may result in the system being unable to cope with the after-effects should the risk occur.

Introduction to Risk Management in Regulation and Public Policy

Hypothetical Scenario: In an attempt to prevent all air accidents, the government decides to forbid the takeoff and landing of airplanes in Israel. Should an airplane crash (for

example, an airplane that flies over Israeli airspace), the system will not be ready to deal with such an accident.

The total prohibition of a certain activity may result in a market backlash. It may also result in the concealment of the prohibited activity making it difficult to supervise the activity. When a dangerous activity "goes underground," it is liable to lose the safeguards that were in place when it was permitted. As such, the risk it embodies increases.

Example: During the Prohibition Era in the United States (1920-1933), the production, import, distribution and sale of non-industrial alcohol was banned on grounds of morality and health. However, prohibition did not result in the cessation of alcohol production, trade and consumption. Instead, producers and bars went underground and thus were not subject to supervision. The need to conceal their activity often resulted in poor sanitary conditions and the use of inferior raw materials to prepare the beverages.

The sweeping ban aggravated the risk in other ways. In order to prevent the production of alcoholic beverages from industrial alcohol, the federal government issued an order requiring industrial alcohol to be made toxic.

As a result, thousands of alcohol consumers suffered from poisoning due to the consumption of alcoholic

beverages derived from toxic industrial alcohol.

Greater profits were earned on the trafficking of beverages with a high alcohol content. Accordingly, high-alcohol beverages accounted for the greatest proportion of alcohol smuggled into the country. As a result, the demand for high-alcohol beverages surged during prohibition. Ultimately, the consuming public switched to purchasing beverages containing a higher level of alcohol than in the period before prohibition when alcohol was legal.

Additionally, the outlawing of alcohol production and trade created a black market. The production and smuggling of illicit alcohol was the root cause for the boom in the organized crime underworld. This period saw the rise of some of the country's strongest twentieth-century criminal organizations.

Risk mitigation: This is the most common risk management strategy. This strategy calls for the implementation of action that will reduce the likelihood of a risk's actualization and the severity of its effect. This strategy includes the acquisition of equipment and technology, instruction and training, improvement of work processes, supervision and control.

The risk mitigation strategy also includes risk transfer, with the

decision maker transferring the effect and management of the risk to another body. The most widely used techniques for the implementation of this sub-strategy are insurance and outsourcing. Insurance: the insuring entity bears the risk in consideration for a premium. Outsourcing: the service provider is responsible for delivering the product or the service, while simultaneously bearing the risk.

Illustrative example: How will the three strategies be applied to the risk of road accidents involving electric bicycles?

Risk acceptance: Decision against government intervention, since there is no essential difference between electric bicycles and regular bicycles, and the cost of dealing with the risk exceeds the amount of the potential damage.

Risk avoidance: Imposition of a sweeping ban on riding electric bicycles (absolute prohibition or prohibition on riding in certain areas – for example in cities).

Risk mitigation: Taking steps to reduce the likelihood of accidents involving electric bicycles (obligation to install flashing lights, limiting the permitted traveling speed) or to mitigate the accident (wearing a protective helmet and body protectors).

Risk management balances three parameters:

The activity and the benefit from

it: The benefit from the economic/social activity that is being regulated (e.g., driving on a road, production of electricity from a power plant, the importation of goods). The benefit from the activity includes also future positive possibilities and opportunities.

Potential risks: The negative cost to society due to the realization of the risks involved in the performance of the activity (referring to the above examples: a road accident, a work accident, smuggling of prohibited substances). The risks may cause death, injury or damage to property.

Cost of risk treatment measures: There is a price to dealing with risk; for example – the purchase of safety measures against road accidents (improved car brakes, lane departure warning system) costs money. One can also choose to prohibit the activity entirely (e.g., banning cars or

prohibiting imports) – but choosing such an extreme risk treatment measure will result in heavy social costs, since the entire benefit of the activity is lost ("throwing out the baby with the bathwater"). Risk treatment measures impose direct costs (purchase of equipment, submission of documents for prior approval, etc.). They can also result in indirect costs reflected mainly in the interruption and disruption of the activity. Risk treatment measures aim to reduce the likelihood of the risk occurring and can mitigate the harmful effects if the risk is realized.

The United Nations Economic Commission for Europe (UNECE) guidebook "Risk Management in Regulatory Frameworks" explains that these three parameters create the "risk management triangle." The risk management triangle illustrates how a decision maker must balance the three parameters and position every decision taken within the triangle. Giving preference to one parameter in the triangle ("drawing closer" to one of the points of the triangle) will always come at the expense of another parameter (will "move us away" from the other points). Therefore, a decision to mitigate risks results in the additional cost of the risk treatment measures, compounded by a reduction in the benefit from the activity. On the other hand, a decision to maximize the benefit from the activity results in the likelihood of greater risks, etc.

Risk management tools enable rational decision making within the range of possibilities contained within the risk management triangle. Public policy or regulation is formulated by determining the balance to be struck between; the benefit from the activity, the degree of risk to be taken and the costs of risk treatment measures.

Risk management is a process that deals first and foremost with causality. Successful risk management is based on a solid understanding of the causal connection between the problem (the risk) to be addressed and the factors giving rise to it. It must also be based on an understanding of the causal connection between the problem and the solution, as well as a strong understanding of the causal connection between the chosen solution and its effects on society. Risk management also includes quantification of the risks and of the costs and benefits ensuing from the selected policy and regulation. But, at the core of any risk management process there is predominantly a cause and effect analysis.

The following part of the chapter explains the difference between private and public risk management. It deals with the three areas in which risk management assists the decision maker: cost-benefit considerations, allocation of limited resources and contending with uncertainty.



Figure 1: Risk Management Triangle

PRIVATE AND PUBLIC RISK MANAGEMENT

The risk management doctrine was originally developed for management of projects in the private sector but is also applied in the management of government-initiated projects. There is an essential difference between project risk management (which characterizes the private sector) and public risk management. This difference is demonstrated with the aid of the risk management triangle.

Risk management in projects is generally carried out by the project manager. The project manager creates the activity, enjoys the benefits generated and controls the risk profile arising from their actions. Additionally, the risk manager is exposed to risks because of the activity as well as bearing the costs of the risk treatment measures. As such, the manager of a project internalizes the benefits and costs entailed in every decision taken.

Unlike a project manager, the government decision maker neither initiates the activity (for example, they do not decide whether people will open a plant or export goods), nor creates a risk, or directly enjoys the benefits resulting from the activity.

The constituents of a society create risks vis-à-vis one another (e.g., a store that sells dangerous products does not endanger the government decision maker, and the profit generated by the activity is not transferred to the government decision maker). The government decision maker is responsible for risks, but only bears part of the costs of the risk treatment measures (they are partly affected by the costs to the government, and do not bear the costs imposed on the public). The government decision maker deals with public interests; the activities, risks and risk treatment costs with which they contend are not theirs alone, but also those of others.

Government risk management deals with determining the rules of the game for players in the private sector. Project risk management helps those same players decide how to act within the framework of the government-set rules. As such, the government decision maker establishes the framework for the decisions of the private project manager. The government decision maker does not create the risks; these arise from the business and social

activities performed by the public.

The key difference between project risk management and public risk management is that the public decision maker deals with the activities of others. They weigh the benefits and costs to the public, without internalizing all the costs entailed in the decisions taken by the project risk manager. The following table illustrates the differences between the project manager and the government decision maker:

Internalized Value	Project Manager	Government Decision Maker
Activity and its benefits	v	x
Risks entailed in the activity	v	v
Cost of risk treatment measures	v	xv

The following figures illustrate the characteristic point of equilibrium within the risk management triangle for the project manager and for the government decision maker:

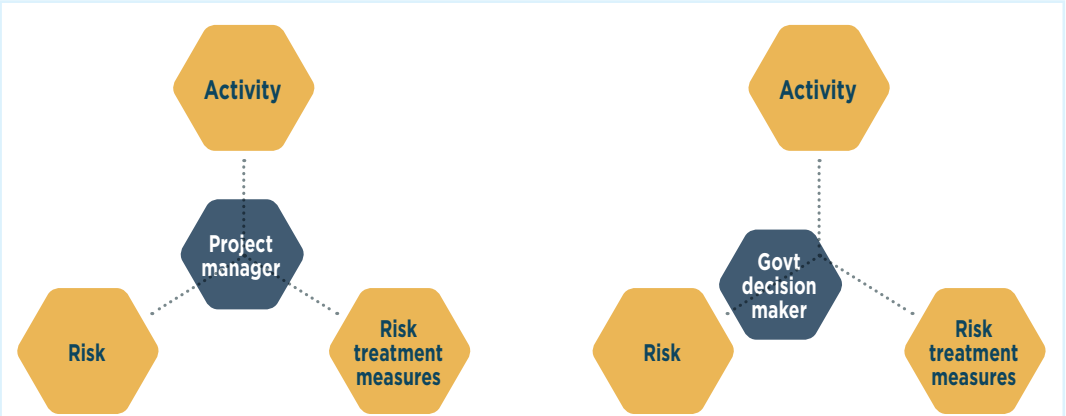


Figure 2: Risk Management Triangle Comparison – Private Risk Management versus Public Risk Management

A further difference lies in the risk treatment measures. A project manager is mainly able to take management and business steps. By contrast, the government decision maker tackles public risks that

threaten the market or society in general. Accordingly, the treatment tools are in the realm of public policy and macroeconomics. These include; the establishment of obligations to perform activities or not to perform

activities (prohibitions), as well as conditions or other limitations, and the setting of rules that confer an advantage or preference (e.g., tax benefits, recognition of property rights or other incentives).

The private sector has been performing project risk management for centuries, with the field blossoming during the post-World War Two era. Over the years, extensive knowledge and experience has been accumulated in project risk management. By contrast, risk management in regulation and public policy is still a nascent field, developing mainly over the last two decades. The key challenge of public risk management is the multiplicity of entities affected by the risks and by the risk treatment measures. Both the risks and risk treatment costs (whether direct – compliance with requirements, or indirect – the impact on economic and social activity) are not internalized by the government but dispersed over the entire economy.

BASICS OF PUBLIC RISK MANAGEMENT

Every risk management process in regulation and public policy includes the following four elements:

Understanding the problem –

Identifying and diagnosing the risks;

Considering risk treatment tools –

Considering different policy tools for reducing the likelihood of the risk occurring and/or mitigating the severity of damage ensuing from the risk;

Comparing alternatives –

Comparing the costs and benefits in each alternative, including the effectiveness of the risk treatment, costs to the government and costs to the public;

Application, monitoring and evaluation –

Monitoring the actual results following the application of the policy tools. It is necessary to determine whether the objective has been achieved, whether the costs justify the public benefit, and whether the risk map has changed (change in the risk profile, creation of new risks or the disappearance of

previously existing risks). Based on the monitoring and evaluation, the policy can be adjusted, and management of the risks continue.

To enable informed decision making and the development of effective action strategies, risk management should be performed based on data. Risk management requires information about risks, their causes and their impact. Therefore, the collection and analysis of data is an essential condition for the development of regulation and public policy.

It is important to distinguish between two main categories in the field of public risk management. The first category is risk management in the development of regulatory requirements or public policy (including any revision or repeal thereof). This category includes the identification and analysis of risks and the selection of an appropriate strategy and tools for dealing with them. The second category is risk management in the supervision and enforcement of existing regulation or policy. This category includes simpler and more delimited areas – the risks and tools have been defined, and the

question is now how to use them and manage the supervision system.

Risk management-based policy enables the government:

1. to deal effectively with public problems;
2. to effectively invest the limited resources in key problems;
3. to explain to the public the basis for the regulation or the public policy, thereby mobilizing the public, increasing compliance and strengthening public legitimacy;
4. to reduce the burden and costs imposed by the regulation or the public policy on individuals and businesses;
5. to create effective supervision and deterrent enforcement for dealing with violations and improving compliance with the regulation or the public policy.

Risk management facilitates threat monitoring and decision taking on whether to modify or adjust government policy. For example, risk management makes it possible

to determine that a certain risk has disappeared or been significantly reduced, thus enabling the elimination or curtailment of the regulation which addresses that risk. Risk management also supports decision making based on a consistent and documented method.

There is no such thing as zero risk, and every risk treatment measure has its attendant costs and prices. Consequently, a risk management-based policy is not intended to achieve zero risk but to balance between costs and benefits to achieve the best possible situation.

THE PRINCIPLE OF MAXIMUM AGGREGATE WELFARE

Risk management requires the definition of an organization's risk appetite (or risk tolerance). As explained earlier in this Guide, risk management in the private sector is simpler since the project manager weighs the benefit derived from the targeted activity against any potential damage and the risk treatment costs. Based on the balance between the benefit from the activity, the risks and the costs involved, private organizations define their own "risk appetite". They decide the amount of damage and loss they are prepared to bear (the level of risk it is willing to assume) in relation to the expected benefits and gains. Based on the defined risk appetite, the private organization manages its risks and decides which projects and risk treatment measures to execute.

By contrast, traditionally, decision makers in the public sector do not focus on the costs of the risk treatment measures or on the benefit from the activity in question. Regulators in particular are required to reduce risks and are evaluated on this basis. Unlike a private sector project manager,

who internalizes the benefits and damages from the risk and the cost of the risk treatment measures, the government bears responsibility for mitigating risks, but the cost of the risk treatment measures is imposed on the supervised entities. The government also does not directly benefit from the economic rewards of the activity.

The definition of governmental regulatory "risk appetite" is a loaded task that is irreconcilable with the role of the government decision maker. A business can decide that it is willing to accept a situation in which 10% of its customers may switch to competitors or 5% of its products are defective. However, the government cannot establish a similar "risk appetite" policy in which, for example, 10% of diners in restaurants will eat spoiled food. This poses a substantive difficulty in both professional and public/media terms. This is a policy decision that conflicts with the perception of the role of the state and the government regulator.

Risk management method in regulation and public policy does not require the decision maker to define in advance the extent of the

risk they are willing to assume. Risk management in regulation and public policy is performed based on the principle of maximum aggregate welfare. **According to this principle, government has a duty to maximize the aggregate social welfare.** The decision maker must consider the three parameters within the risk management triangle (the benefit from the activity, the damage due to the risk and the risk mitigation cost) and choose a mode of action that will generate the maximum benefit for the public. As a public agent, the decision maker must examine the effects of the risk and possible risk treatment measures. In principle, from a governmental perspective, the loss of ILS 1 on a beneficial mitigating activity is equivalent to the loss of ILS 1 on a realized risk, as well as ILS 1 invested in preventing the risk. Therefore, the decision maker must factor in all the costs to society and choose the alternative that minimizes the societal loss.

To do this, the decision maker must examine different hypothetical situations (risk without government

intervention, action mode A, action mode B, etc.) and total the costs to society. The cost of each hypothetical situation consists of three components: the damage created by the risk, the cost of the risk treatment measures to the public treasury, and the cost of the risk treatment measures to the public.

1. Expected annual damage due to the risk – This component reflects the damage to the public due to the realization of the risk. It is calculated based on the actual damage for one year or by multiplying the severity of the risk's effect by the probability of its occurrence during the year. If there are several risks, they should all be factored into the sum.
2. Cost of risk treatment measures to the government – This component reflects the government spending required for dealing with the risk. It includes every government expenditure, notably the allocation of manpower (positions) and financial outlay for procurement and services. Governmental resources are funded by the citizens and residents of the state. Every

government expenditure imposes a cost on the public.

3. Cost of risk treatment measures to the public – This component reflects the regulation or policy costs imposed on the public. Direct costs include; the costs in implementing the public policy or complying with the regulatory requirements: compliance with standards, reporting, submission of documents, etc. Indirect costs include; the broad impact of the policy or regulation: avoidance of activity, curtailment of the beneficial activity (curtailment of

the benefits arising from it), creation of entry barriers, impairment of productivity, effects on related sectors, harm to competition, etc.

Dealing with public risk is intended to protect and benefit the public. It is important to consider the aggregate amount of the above three costs. The public treasury is funded by the taxpaying public – which is the public the government seeks to protect. It is the public who will bear the costs arising from the regulation or the public policy. Therefore, in the final analysis, it is immaterial whether the public incurs damage of ILS 1 million,

	Expected annual damage due to the risk	Cost of risk treatment measures for the government	Cost of risk treatment measures for the public	Total costs
Current situation	-80	0	0	-80
Alternative A	-20	-30	-70	-120
Alternative B	-40	-10	-10	-60

or government spending is increased by ILS 1 million. The regulation imposes on the public, compliance costs amounting to ILS 1 million. The government decision maker must act according to the principle of maximum aggregate welfare and examine all the costs to the public.⁽¹⁾

This involves an equation with three negative values. The decision maker must select an alternative in which the sum of the costs is the lowest. The table on the previous page illustrates the performance of the comparison according to the principle of maximum aggregate welfare:

Seemingly, Alternative A will result in an improvement in the situation because it reduces the expected annual damage resulting from the risk from 80 to 20. However, it creates an additional cost of 100 (for the public treasury and for the public). In total, choosing Alternative A will worsen the situation of the public (increasing the total cost to the public from 80 to 120). An examination of the overall picture (total cost to the public) shows that Alternative B should be selected.

Use of the principle of maximum aggregate welfare allows decision

makers to consider the full implications of the regulation or the public policy and its effect on the public. This principle is also consistent with a cost-benefit analysis and with the legal rule of proportionality (which requires congruence between the objective of any step and its positive and negative implications).

1) For the sake of accuracy, it should be noted that the different costs are imposed on different societal groups. The public affected by the realization of the risk is not the public that bears the cost of the risk treatment measures (e.g. drivers that endanger bicycle riders) or the public that funds the state budget (e.g. prevention of risk to children, who do not pay taxes). But in the final analysis, society bears the cost of risks, both in public spending and private spending. As a public trustee, the government decision maker has a duty to reduce the overall societal loss and promote the aggregate social welfare. There is extensive empirical writing about the model as well as various developments on it. Additional variables influence the decision whether to violate norms (some of which are detailed in this chapter).

MANAGEMENT OF LIMITED RESOURCES

Time, money, reputation and professional experience are all limited resources. The government and the public are required to make decisions taking into account the limited nature of available resources.

Since governmental and societal resources are always limited, it is important to wisely choose where best to channel them. In this sense, risk treatment is a "zero-sum game": any choice of an alternative comes at the expense of another alternative. Choosing to invest resources in supervising a certain market means not investing those resources in the supervision of another market. A decision to use supervision and enforcement resources for inspecting a certain item (building, business, shipment, product) means they will not be allocated for the inspection of another item. Risk management makes it possible to compare and prioritize risks based on an understanding of the need to choose which actions to focus on and which to forgo.

Choosing to inspect all items would result in heavy costs to the public. It would also impair the effectiveness of the inspection system, since it is not possible to inspect all items thoroughly

(the dispersion of resources over a large quantity of items harms the quality of the supervision and enforcement).

Different items have different risk levels (the risk level consists of the severity of the damage and the probability of the risk's occurrence). Policy tools and supervision methods must be adapted to each risk. Policy or regulation that is not based on risk management will not serve its purpose. The charts on the following page provide a general illustration of this point. Regulation or

a supervision method may be uniform or differentiated. If the regulation is not adapted to the different risk levels, a uniform supervision rule or method will be selected for all cases (e.g., uniform requirements for small and large businesses, uniform supervision and enforcement for all roads in the country).

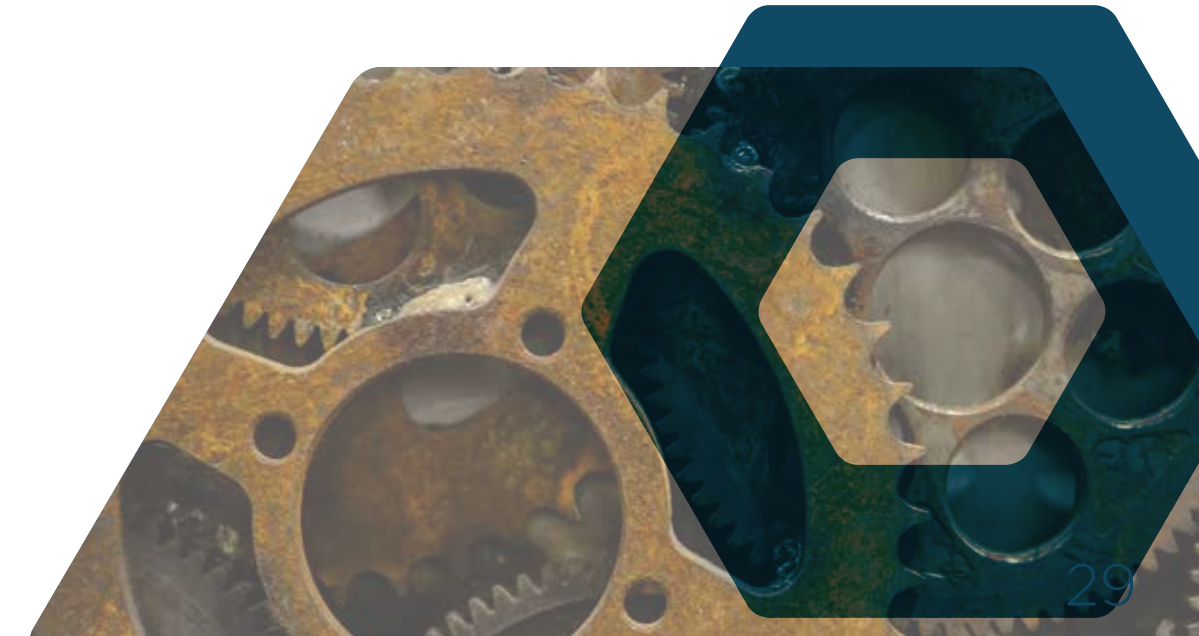


Figure 3: Different items have different risk levels. The below chart shows the different risk levels (blue line). In the example shown, regulation is not based on risk management and is therefore uniform in all cases (and may appear to be arbitrary). It is presented as a horizontal line that does not change according to the risk level of each item (black line). The black line describes the degree of stringency of the requirements (regulation content), the scope of the supervision or the enforcement policy.

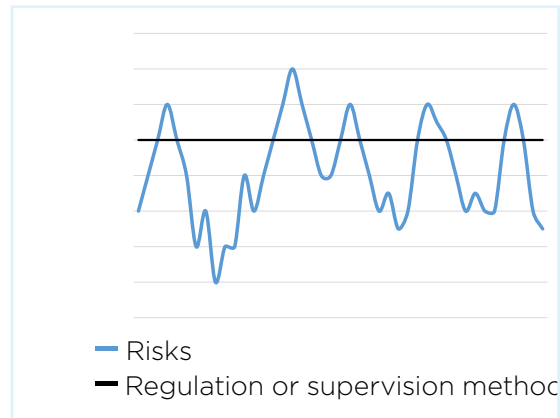


Figure 4: An unsuitable supervision method may be too lenient for serious cases (high risk level), resulting in the public being inadequately protected against the risk (the areas of the graph marked in yellow). In the absence of risk management, resources may be allocated in an erroneous manner and fail to provide effective protection for the public interest.

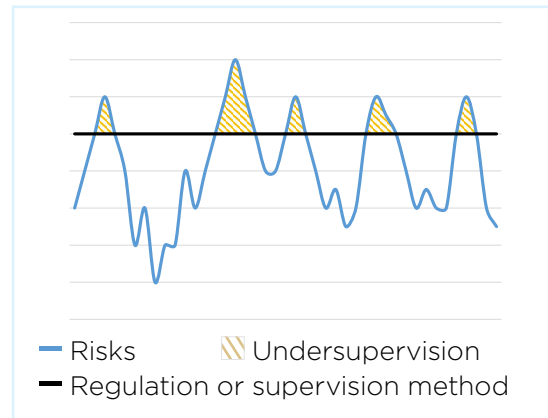
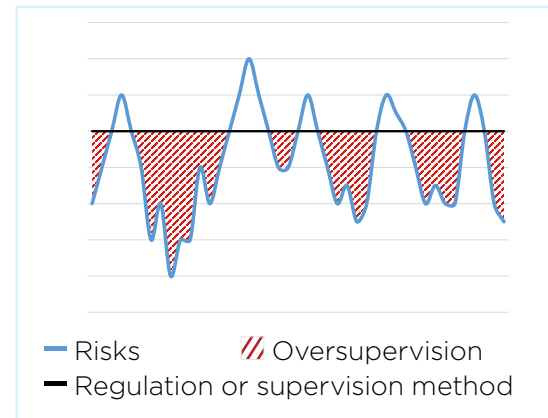


Figure 5: Unsuitable policy or regulation may be too stringent for minor cases (low risk level). In the absence of risk management, the policy or regulation may impose unnecessary costs on the public (the areas of the graph marked in red).



These figures show that risk management is necessary for managing resources that are in short supply. Figure 4 shows underinvestment in items with a high-risk level, while Figure 5 shows overinvestment of resources in items with a low risk level. The use of risk management would have enabled the redirection of resources invested in low-risk items (marked in red) for the supervision of high-risk items (marked in yellow). The aim is for the policy to be as pertinent and suitable as possible in relation to the actual risk – that is, the black line (content of the policy or the supervision method) should follow the blue line (the different risk levels) as closely as possible. Risk management ensures the protection of public interests and enables more effective use of the resources available to society.

UNCERTAINTY AND INCOMPLETE INFORMATION

In any decision-making process, information will be lacking regarding the present and the past. There is always uncertainty regarding the future. Risk management, which deals with decision making under conditions of uncertainty, helps the decision maker cope with uncertainty. The more data collected, the more refined the analysis method. This facilitates a better preparation for future scenarios with a higher level of certainty. Obviously, it is not possible to collect and analyze all relevant information. Information is among the costliest resources to produce (it has been observed that "information is the oil of the twenty-first century"). As such, it is necessary to reach an equilibrium between the resources that can be devoted to the process and the need to acquire sufficient information at an adequate confidence level.

The guiding principle in this regard was determined by the OECD: Policy must be set "on the basis of the best available information." This principle is applied by decision makers in the United States, Europe, the European Union, New Zealand and other first-world countries.

Illustrative example: Let us assume that we want to predict the height of the next person who walks into the room. People vary in height and this variance creates uncertainty. We have a sense or notion of the reasonable range (not less than half a meter and not more than two meters), but we will be able to make a more accurate prediction only after we have examined statistical data or performed empirical measurements. If we measure the height of the entire population, we will have a complete understanding of the average height and distribution (standard deviation). This is not a practical solution. We must therefore decide how many measurements to perform and on what pre-existing information to rely on, in order to arrive at an informed decision.

The risk management process always involves uncertainty, regarding the past, present and future. A decision to avoid a certain action cannot be taken due to uncertainty regarding the future or a lack of information. Uncertainty is a baseline situation that must be dealt with and not used as a consideration in the decision-making process.

The rationale for selecting a risk management strategy and tools must be based on analysis of available information. Risk management provides an organized methodological framework for factoring all the available data, including probability and uncertainty, in order to make an informed decision. The share of available resources to be devoted to collecting information must be commensurate with the issue and decision at hand. It is important to bear in mind that it is impossible to obtain 100% of all available data. One can, and should, arrive at decisions even when faced with conflicting data or missing information. It is necessary to determine how much information is needed in order to make a well-informed decision. Another method for coping with incomplete information is estimating the range of possibilities that may be revealed when factoring in missing information and determining how and if these possibilities would impact any decision. Sometimes it becomes apparent that the missing information would not materially change the outcome of any decision taken based on available information.

THE ROLE OF THE DECISION MAKER AND THE RESPONSIBILITY OF THE SUPERVISED ENTITIES

An important component of the method is ex post monitoring and evaluation. This can determine if the forecasts were accurate and whether the previously selected tools achieved their projected result. The combination of ex-ante analysis and ex-post examination provides the decision maker with feedback that enhances decision making over time.

Decision making under conditions of uncertainty and incomplete information does not mean decision making without data. The analysis must be based on information and professional assessments. If the decision maker lacks information, decisions can be applied and enacted for a limited period and/or to a limited scope of activities (trial periods or pilot test cases). In case of an objective lack of material, preliminary regulation can be imposed with the aim of collecting data (such as the duty of reporting).

The data collected and analyzed can then be used to examine the key decision at hand.

The role of the government decision maker is to manage risks based on the principle of maximum aggregate welfare. When the decision maker seeks to mitigate risks, they establish policy or regulation affecting the manner in which the public acts. After the policy or regulation has been set, the decision maker can encourage compliance by means of directives, instruction, enforcement and punishment.

The government engages in public risk management, but the creators of public risks are individuals, companies and organizations in the public sector. The role of the government is to establish regulation or policy that will mitigate the risks and encourage compliance with the

regulation or policy. Responsibility for avoiding excessive risks and for complying with regulation or policy is a public requirement. Even though the government manages the risks, the risk creators are still responsible for the risks they create and the damage they cause.

It is important that the supervised entities bear responsibility for compliance as well as responsibility for the risks they create. The imposition of responsibility is important from the legal, media and public standpoints.

It is also essential that the government encourages compliance with regulation and public policy. The government is responsible not only for the establishment and enforcement of rules, but for ensuring supervised entities comply with the rules, as well as providing mechanisms that encourage and increase compliance.

PRINCIPLES OF RISK MANAGEMENT IN SUPERVISION AND ENFORCEMENT

Meta-goal: effective and efficient compliance. The overriding purpose of the supervision and enforcement system is to increase compliance in a manner that mitigates risks, at minimum cost to the public and the government. The term "supervision" refers to all activities aimed at increasing compliance with requirements. The term "enforcement" refers to all activities involved in the imposition of sanctions or penalties for noncompliance with requirements.

It is in the interests of both the government and the supervised entities to have an effective, consistent and deterrent supervision and enforcement system. Such a system is needed by the government to ensure compliance with the regulatory requirements. Supervised entities also have an interest in an effective and consistent supervision system. If there is no control over compliance violations – law-abiding supervised entities will be in an inferior position to supervised entities that violate the law (resulting in potential gain). The most notable example is that of two business competitors: one who is law abiding (and devotes resources and finances to this), and the other who is not.

Adjustment to the risk level

An effective supervision and enforcement system creates an incentive for supervised entities to act in accordance with the regulation or policy, saving governmental and public resources on compliance and supervision.

There is no logic to exercising the same supervision over all entities as not all entities are comparable. For example – the risk level may warrant a greater or smaller number of supervision actions; the nature of the activity may call for different supervision actions (requests for information, visit to the supervised entity, laboratory tests); and the nature of the supervised entity and its history may justify different sanctions at varying levels of severity.

Example: The environmental control authorities in England and Wales implemented a 75% reduction in the number of supervision actions at low risk waste disposal sites. This enabled the allocation of greater resources for the supervision of higher-risk sites. The number of inspections at high-risk waste disposal sites increased from 70 to 1,500 per year.

Not all existing activities can be

supervised. The government cannot be everywhere, all the time, and examine every action, person or object that is involved in risk-creating activities. The supervision system is based on the premise that not all supervised entities can be monitored. Regulators and enforcers must often choose which entities to supervise and which not.

Risk-management-based supervision is sometimes described as random supervision. This is a mistake. In risk management-based supervision, the decision whom to monitor depends on the level of risk and compliance. There is no logic in random supervision. Random supervision is usually an outcome of an attempt to supervise everyone, a priority doomed to fail. Supervision actions should be undertaken only if there is a clear reason and need. The reason must be based on an examination of the risks arising from various violations and various supervised entities. It must also take into account a cost-benefit analysis versus other possible supervision actions (for the optimal utilization of resources). It is therefore necessary to be familiar with the supervised entities. It is necessary to identify the activities, people and



objects being supervised and to have knowledge of their characteristics. The supervision system must be managed flexibly and adapted to the relevant characteristics of the supervised entities in terms of the risk and compliance level. All the components of the supervision system should be adapted to the risk and compliance level of the different supervised entities, in terms of number of actions, type of supervision, depth of inspection, punitive measures, etc.

The purpose of the supervision and enforcement system is not to catch offenders but to bring about compliance that reduces risks. The emphasis is on changing the behavior of supervised entities. As such, it is important that supervision focus on more serious violations (associated with significant risks in terms of severity and probability) rather than on simple supervision actions or violations that are easily enforced but do not significantly impact the risk level. To do this, it is necessary to identify dangerous activities based on the probability and severity of violations and determine the expected compliance level of each supervised entity (noncompliance is the problem the supervision is seeking to resolve).

Additionally, it is important to be cognizant of the effectiveness of governmental intervention.

A supervised entity's compliance record influences the supervision format and the frequency of the supervision actions. A supervised entity that has fully complied with all requirements during a defined period maybe less closely monitored in the future. A supervised entity with a record of violations should be more closely monitored. In the case of a violation, even if the violating entity has complied with requirements in the past, forbearance should not be given. The sanction must match in strength the severity of the violation – the risk created, the violating entity's gain, as well as any previous history of violations.

Supervision actions necessitate extensive governmental resources and place a burden on the supervised entities. Studies have shown that the execution of frequent supervision actions does not necessarily improve compliance and that the imposition of stronger sanctions does not necessarily protect the public from risks. Supervision and enforcement are not only a matter of quantity (number of inspections, amount of

the fine), but also depend on quality and performance of actions. As such, an improved supervision and enforcement system is necessary when resources are limited.

Should there be supervision?

Since it is not possible to supervise all activities at all places and times, it is necessary to determine where not to exercise supervision. Consequently, it must be considered whether there are non-governmental mechanisms that can help supervised entities comply with requirements and avoid creating excessive risks.

Government norms are constantly enforced by means of other mechanisms, such as market forces and civil lawsuits. The market forces mechanisms give supervised entities an incentive to comply with regulatory requirements (e.g., consumers will leave a communications company that does not provide nationwide coverage; consumers will opt to buy clothes from a manufacturer that is known for its high quality). The market forces' mechanisms depend on the existence of a functioning competitive market. Civil lawsuits provide the public with a powerful tool. If such a suit is accepted, the creator of the risk bears

the cost of the damage caused and the injured party receives compensation. Such a mechanism deters creators of damage and encourages them to take preemptive measures against risk, while also compensating the injured party.

When supervised entities have a sufficiently good incentive to comply and the ability to do so – supervision activities can be reduced and even discontinued. Sometimes, the required government intervention consists of enabling the mechanism of market forces or civil lawsuits, after which there is no need for supervision activities. The need for governmental supervision decreases when in-built supervision mechanisms come into play (naturally or with the help of the state). Contract law and corporate law contain numerous mechanisms that enable the various players to supervise each other. This reduces the need for governmental supervision and enforcement activities.

Encouragement of compliance

The intention of supervision activities and sanctions is to change the behavior and actions of supervised entities which will lead to a reduction in the risk they pose to the public. In order to abolish noncompliance by supervised entities, appropriate supervision and enforcement tools must be utilized. Noncompliance maybe a result of ignorance or misunderstanding of the requirements, distrust of regulation or policy, inability to comply with the requirements, refusal to comply due to high compliance costs, or deliberate criminality. A fundamental condition for compliance is familiarity and comprehension of the requirements. An excessive number of requirements or vagueness in their formulation makes compliance more difficult.

In the case of a conscious choice to violate a requirement – there must be a deterrent that deprives supervised entities of any incentive to flout the norm (this is expanded upon in the following sub-chapter).

Traditional supervision and punishment are not the optimal solutions regarding violations arising

from ignorance or misunderstanding of requirements. Any supervised entity may commit such violations, but they are more prevalent among supervised entities who are inexperienced (e.g., new drivers or new business enterprises) or supervised entities with limited resources (small businesses). Traditional supervision that does not take this into account is liable to harm such entities, without improving the level of compliance. It is difficult for small businesses and new business enterprises to understand and cope with multiple, complex requirements. As such, they are liable to violate the requirements and incur punishment due to ignorance or misunderstanding. In such cases, supervision should be based on instruction and explanations to the entities. They need to fully understand what the requirements are, what their purpose is, how to comply with them, and whether they are in compliance. These tools increase the compliance level and can be used without imposing sanctions on the supervised entities.

Example: Several empirical studies conducted in the Netherlands found that instruction and assistance to supervised entities are no less effective tools than the strengthening of enforcement. One of the studies dealt with regulation in the agricultural sector (planning and construction and plant health). In one district, the supervision and enforcement regulators were instructed that first-time offenders should receive instruction on the purpose of the regulation and explanations on how to remedy the situation and comply with the rules. In another district, the supervision and enforcement regulators imposed immediate fines for a first violation. The level of compliance with the regulatory requirements was much higher in the first district. These findings held true both in the short and long term. Beyond assistance in complying with the requirements, the instruction provided by the regulators mobilized the supervised entities and provided a psychological incentive to cooperate with the regulators.

The classic supervisor-supervised relationship is one of confrontation and conflict of interests. If in addition to the inspection activities, the supervisor can also help the supervised entity comply with the requirements, a

dynamic cooperation can be forged encouraging the supervised entity to achieve regulatory compliance.

Another factor that influences the level of compliance with the requirements is the supervised entities' identification with the regulation or policy. People prefer to comply with requirements they perceive as being right and justified. Conversely, when a requirement is seen as meaningless, arbitrary or formalistic - the willingness of the supervised entities to comply is low. Added to this is another behavioral aspect: studies have found that people do not perceive themselves nor wish to perceive themselves as criminals. If the individual feels they can justify the violation (even partly), their level of compliance will decrease. This effect is known as bounded ethicality. To ensure a high compliance level, it is important to verify that the purpose of the requirements is known and clear and the public can identify with it. A key method of achieving this is through the communication of risks (as elaborated upon previously).

A major compliance encouragement tool is the publication of information and creation of compliance incentives, without the use of punishment. An example of this is the release of inspection results or publication of

the ratings of supervised entities (all of them or only those who strayed significantly from the requirements). Such a mechanism provides information and is based on the reputation of the supervised entities. It can create a strong incentive to comply with the requirements, and enables other entities (customers, suppliers, consumers, public opinion) to exert pressure on the supervised entities.

Example: The Israeli Ministry of Environmental Protection publishes the Environmental Impact Index for Public Companies. The index includes indicators for risk (e.g., emission of pollutants, use of hazardous substances, site of an activity), indicators for regulatory compliance and indicators for positive and desirable environmental behavior. Published once a year, the index relates to industrial, energy and biotech companies. It is also used by the Ministry to set inspection intervals.

Example: For several years, the New York City Health Department has been operating a sanitary grading system for the city's restaurants (Letter Grading for Restaurants). The grading is based on the number of violations

found in a restaurant in their most recent inspection. It also takes into account the severity of the violation. Each restaurant is assigned one of three letter grades: A (the highest), B and C. The Health Department had always kept track of the number of violations by a restaurant, but the new grading method requires each restaurant to post its current grade on the facade of the business, where it can easily be seen by passer-byes.

An additional tool is reliance on the supervised entity's track record in setting the scope and format of the supervision. This tool encourages long-term compliance. Supervised entities must devote resources in regard to supervision activities. By proving compliance with the requirements, they can benefit from more limited supervision at longer intervals, thus saving resources. As a complementary component, supervised entities may be given an opportunity and means for improvement and for demonstrating an increase in the compliance level.

Supervision, enforcement and deterrence

When a decision is made to carry out supervision, the supervision and enforcement actions must create deterrence against violations.

Deterrence serves as a means for regulating the behavior of supervised entities that have no interest in conforming to norms, thus increasing the level of compliance.

The economist Gary Becker laid the foundations for economic analysis of enforcement and deterrence.⁽²⁾ According to Becker's model, when a supervised entity is considering to violate a norm, it asks itself whether it is more "worthwhile" for it to comply with the norm or to violate it. The supervised entity weighs the gain that would accrue from violating the norm against the extent of the expected sanction (multiplication of the strength of the sanction by the probability of its actual imposition on the supervised entity). In fact, supervised entities perform their own risk management vis-à-vis the government. They examine the risk of the imposition of a sanction, taking into account its severity and the probability of its realization. On this basis they decide whether or not to violate the norm.

Example: Supervision and enforcement against parking offenses in city centers is based on low fines. Nevertheless, effective deterrence is achieved when the sanction is perceived as being certain (high probability). The system used by the Tel Aviv-Jaffa Municipality is considered to have special deterrence value. This is not an outcome of higher fines but of a higher probability of the imposition of sanctions.

2) The model is presented here in a simplified and non-exhaustive manner in order to describe the influence of the likelihood of the imposition of a sanction and its severity on the level of deterrence. There is extensive empirical writing about the model as well as various developments on it. Additional variables influence the decision whether to violate norms (some of which are detailed in this chapter).

The likelihood of a sanction being imposed depends first and foremost on the supervision system, whereas the severity of the sanction depends on the punitive system. The multiplication of these two variables dictates the deterrence. Therefore, if we wish to maintain a high level of deterrence, lenient punishment entails strict supervision, and vice versa.

In order to create an effective deterrent system, it is important to understand the considerations of the supervised entity in regards to violations: what benefits it believes it will gain from the violation (increased profit, reduced costs), which violations are easier to commit and which are more worthwhile. Additionally, it is important to ascertain which sanction will be imposed on the violating entity and what is the likelihood of its imposition (identification of the violation, collection of sufficient evidence, final decision to impose a sanction).

Rules for strengthening deterrence

To ensure effective deterrence, supervision activities must be unexpected and irregular. Flexible supervision creates uncertainty among supervised entities. Research shows that such uncertainty increases the deterrence of these activities.

On the other hand, punishment must be certain and consistent. The sanction – the “price” that is collected for the violation – must be clear. It must be known in advance, consistent and proportionate (to the extent possible). Accordingly, the supervising entity must ensure transparency regarding the type and severity of the sanction that will be imposed for any violation. Ambiguity regarding the sanction lessens its deterrent effect. The combination of irregular supervision and guaranteed enforcement strengthens deterrence and contributes to increased compliance with the requirements.

It is important the supervised entity sees a close connection between the violation and the sanction. The best way to create such a connection is by means of a swift supervision, enforcement and punitive process. When the process is protracted, the deterrent message is weakened and the connection between the violation and the sanction is lessened.

The fine and financial sanctions must be commensurate with the damage the violation may cause or the profit the violating entity will gain from the violation. Both these alternatives reflect principles of risk management. A fine in the amount

of the damage reflects enforcement that is suited to the severity of the damage resulting from the violation. A fine in the amount of the profit from the violation is designed to deter the supervised entity and to influence the “risk management” performed by the supervised entity when deciding whether to violate or to comply with the norm. If the fine is lower than the expected profit from the violation – the violation becomes worthwhile and rewarding for the supervised entity. In such a case there is no deterrence.

The supervision and enforcement system must convey a market-wide message that noncompliance is not a legitimate option. To this end, consideration should be given to a policy of publication of violations and of the sanction imposed in each case. Such a policy could expand deterrence from the individual violating entity to encompass all the supervised entities.

Just as the supervision system must be based on risk management and adjusted to the type of violation and type of violating entity, so too the punitive instruments must be varied to enable their adaptation to differing circumstances. A broad hierarchy of punitive instruments should be created. This enables flexibility according to the severity of the violation and its

circumstances (repeat violation, small business, private individual, deliberate violation versus technical violation).

Professionalism and use of information

For supervision and enforcement activities to be effective and efficient, the supervision and enforcement system must be extremely professional.

The professionalism of inspectors and enforcement personnel includes not only their professional discipline (engineering, medicine, financial markets, consumer rights), but more importantly the qualifications and skills required for the performance of supervision and enforcement. These skills include: sampling and testing, data collection and analysis, risk management-based decision making, compliance promotion methods, risk communication, treatment of violations, etc. Additionally, it is important for the supervision and enforcement personnel to be familiar with related government systems, so that their actions do not conflict with the policy of another government entity. Inspectors should be provided with an understanding of the business logic in the supervised areas. They must also be aware of the constraints and incentives of the supervised

entities and how public opinion affects them (broad public discourse, NGOs, customers, suppliers, etc.). Of equal importance are training and certification courses for supervision and enforcement personnel – imparting both professional knowledge and supporting skills.

The use of inspectors raises management-related questions regarding headquarters versus field operations. The inspector in the field will always have discretion, thus the training they undergo must give them the ability to act independently in implementing the policy, with the possibility to consult with management or supporting professionals.

No incentives should be given for the identification of violations. Additionally, it should be ensured that the supervision unit does not profit from the imposition of fines so as not to distort the structure and operation of the supervision system. Management of the supervision and enforcement system should focus on improving compliance, with special attention to the key issues.

One of the main challenges of supervision and enforcement systems is ensuring uniformity in the application and interpretation of the requirements. A lack of uniformity among inspectors

hinders the achievement of the objectives, as well as lessening the deterrence of the supervision and enforcement system. Moreover, a lack of uniformity can result in a feeling of arbitrary supervision. This may reduce the supervised entities’ motivation for compliance. This situation creates a double challenge: Firstly, it is necessary to ensure that rules and requirements are clear and may be implemented in a uniform manner. Clear requirements – for both supervising and supervised entities – are an imperative condition for the success of the regulation or the policy. Cases of ambiguity should be investigated, and the rules revised accordingly (there is room to exercise discretion, but it must also be delimited and defined). Secondly, it is necessary to establish a training and instruction system that imparts to all inspectors equal knowledge and skills, as well as equal familiarity with the rules and a uniform basis for their interpretation.

Just as the setting of a policy and establishment of a supervision system must be based on data, so too the management of a supervision and enforcement system. The results of the supervision and enforcement actions should be measured from two aspects: the effect on the compliance level (or correction of defects) and the effect on the risk level (events

that occurred or nearly occurred). The decision maker must monitor the results and determine if they conform to the system's objectives.

In addition to an examination of the results of the supervision and enforcement actions, information on the dedicated resources must be collected and analyzed. For example; at what frequency are supervision actions carried out, what is the number of supervised entities, how much government and private time is devoted to the supervision actions and what sanctions have been imposed.

HOW TO USE THE GUIDE

This Guide is not a textbook on risk management or public policy. The following chapters provide practical guidelines for implementing the public risk management method. The Guide describes the stages of the process in detail, including examples and suggested work tools.

The following chapters present the stages of the risk management process. It is recommended to devote to each stage a separate meeting (one at least). Preparatory work ahead of the meetings increases the efficiency of the process and shortens the time required for its completion. It is advisable for the process manager to set the agenda in advance of each meeting and ensure staff members prepare ahead of time. A significant part of the work can be carried out independently.

Each stage in the process concludes with the production of an intermediate product – for example, a list of identified risks or a risk comparison matrix. Each product serves as a basis for the following stages of the process. It is important to document the analysis – not only for future use by the process manager but also for

retrospective evaluation of the process and adjustments to be implemented. Documentation is a tool that assists the organization. There is no need to generate formal and verbose documents. Orderly and documented analysis enables peers, experts and public stakeholders to share in the process. Such sharing enriches the decision maker with information and confers legitimacy on the process.

For the risk management process to be effective, it is important to adhere strictly to the following principles:

Professional – methodical, objective and rational – based on data and analysis.

Flexible – the analysis and decision-making process should be adapted to the relevant context, to the level of uncertainty, to the professional complexity and to the tradeoffs between risks, costs and other benefits.

Consistency – in the analysis method, especially in the application of the policy and the regulation. Consistency prevents errors and ensures decisions will differ only when there is a justified reason and not because of differences

in the analysis method.

Transparent – the basis for decisions should be known and open to the public. Transparency also facilitates efficient communication and an understanding of the policy or the regulation.

Dynamic – it is important to adapt the process and the policy tools to diverse situations and to the timetables available to the decision maker, while studying and improving work methods.

During the period when the process is being implemented and integrated, it is recommended to have a moderator follow the process. The moderator should not be a decision maker or a content expert in the relevant policy area. Preferably, the moderator should be a "consultant" who is knowledgeable in public risk management methods or an expert in regulation policy.

The scope and depth of the process depends on the decision at hand, the assets and the risks. The method presented here is flexible and can be adapted to each case. The preliminary stage of "context setting" is intended, among other things, to help determine

the scope of the work to be carried out and the amount of resources devoted to it.

In order to stay away from theoretical discussion, additional sources are listed for further study at the end of the Guide. Additionally, the annexes to the Guide contain templates for the work tools referred to in the Guide and can help the decision maker in the performance of risk management.

Throughout the Guide, two kinds of examples are detailed: examples – that present cases from law, regulation or existing practice; and illustrative examples – that present purely hypothetical cases and do not include true data.



CHAPTER B – RISK MANAGEMENT IN DEVELOPMENT OF REGULATION AND PUBLIC POLICY

GENERAL

As a rule, the development of regulation and public policy should be carried out in accordance with the RIA (Regulatory Impact Assessment) method. This method includes defining the targeted problem, determining goals, examining alternatives and comparing benefits and costs. The risk management process is a central part of the RIA method, but not all of it.

The process is based on the principle that the problem must be fully understood in order to develop an effective and efficient solution. The risk management process in the development of regulation and public policy consists of three main parts: Risk Assessment, Risk Treatment and Execution and Monitoring. The process structure is described in the following figure.



Figure 6: Risk management process in the development of regulation and public policy

The first part – Risk Assessment –

This enables the decision maker to map all relevant risks, define the characteristics of each risk and quantitatively analyze their impact. As the process progress, the level of analysis and use of data increases. This part – Risk Assessment - consists of the following stages:

- Context setting – defining the scope of work, the public interest to be protected and the activity involved in creating the risk. This stage defines the framework of the discussion and constitutes an essential basis for the continuation of the process.
- Risk identification – reviewing the activity and processes to identify possible risks.
- Risk evaluation – examining the characteristics of each risk in order to define a clear scenario for it.
- Risk analysis – a quantitative analysis of the probability and severity of the impact of each risk, comparison of different risks and their prioritization.

The second part – Risk Treatment –

The decision maker utilizes the information collected for each risk in order to select and apply the appropriate policy tools.

- Selection of a risk management strategy – examining different strategies: acceptance of risk, risk avoidance and risk mitigation.

At this stage, the risks requiring government intervention are selected, as well as the risks that will not be addressed.

- Selection of a risk management method – comparison of regulatory methods and policy tools. The main groups are tools based on permits and licenses, binding norms and advanced regulation (including non-regulatory tools). The comparison includes a cost-benefit analysis of the various alternatives.
- Selection of the risk management content – defining the precise content of the method selected: who will be subject to the obligations, how will the requirements be defined, what the system of supervision and enforcement will be, etc. The comparison includes a cost-benefit analysis of the various content alternatives.
- Crisis management – preparation and allocation of resources for coping with risks after their realization.

The third part – Execution and Monitoring –

This includes tools that will assist the decision maker during the implementation of the risk management policy.

- Risk communication – providing information to the public about the risks, their nature and the selected methods of managing the risk. This stage provides the public with tools for coping with the risks in advance and in case of actualization.
- Monitoring and evaluation – carrying out an evaluation of the risks and the impact of policy or regulation. This includes lessons learned and policy updates.

The remainder of the chapter details with the implementation of each stage. At times, organizations choose to split the first two parts of the process, with a "risk assessor" leading the Risk Assessment, and the decision maker or "risk manager" leading the Risk Treatment. This division is suitable for certain organizations and processes.

The scope and depth of the process depends on the nature of the decision – the risks and assets. If part of the risk management process has been carried out, there is no need to repeat it. It is possible to start directly from the next stage (given that it is complete, up-to-date and thorough enough to be relied on). For example, if Risk Assessment has already been carried out and the regulatory system has been selected, the work process can commence from the content selection stage.

It is important to document the process at all stages: to consolidate the data, write a profile for each risk, document the work assumptions, etc. Writing helps thinking, enables communication within the organization and ensures documentation of the process, that can be used by the organization in the future. Information is the most basic work tool of a risk manager.

One of the most important sources of information is discussion and consultation – both within the government and the public. Annex D presents guidelines for conducting a public discourse.

CONTEXT SETTING

Every government decision maker operates within a demarcated environment that defines the risks they face. In addition, every government decision maker handles various processes, works with several units within the government and with different target audiences (directly supervised entities and indirectly the affected public). Therefore, the first stage in the process is context setting. This establishes a framework for the discussion of the risks.

Defining the type of assets and the type of risks to be treated

- The assets must be defined. It is also recommended to define the type of risks. By defining the assets and risks, the decision maker clarifies the purpose of the policy or regulation and the framework of the discussion.
- The assets can be tangible, such as a plant or money, and can be abstract, like the health or safety of a defined population. In the world of public risks, assets are public interests (public health, safety at work, etc.). It is important that the definition for each asset be as precise as possible.

Illustrative example: the definition "health of children under one year of age of non-breastfeeding mothers" clearly delineates the framework of the discussion and the purpose of the policy to be formulated. The definition "safety of workers on the assembly line in wood processing plants" is also a good definition.

This definition clarifies that the discussion will deal with the safety of employees and not with other interests (such as welfare, health or equality). It clarifies that the policy will only address workers on the assembly line, and not administrative employees or employees in the transportation system, etc.

- The assets and risks may already be defined by legislation. Generally, the government unit is authorized to deal with a specific type of asset or risks. For example, the Ministry of Health deals with the protection of public health (people) rather than animal health – this is a specific type of asset. The Antitrust Authority is a lateral regulator covering all branches of the economy but deals only with the prevention of risk of harm to competition.

- The assets must be concentrated in the Register of Assets (see Annex A). The Register of Assets must specify all the assets the decision maker will protect, as well as prioritizing them. A description of the asset can also be added, such as its level of vulnerability or other characteristics, (for example, implications for the country's foreign relations). The Register of Assets defines the framework. In a later stage of the process, we will examine the effect of the risks on the assets specified therein. At this stage, prioritizing different assets and determining which is more important in terms of government and private resource allocation is recommended.

Example: Veterinary services are responsible for supervising animal husbandry in order to protect three main interests: public health, animal health and animal welfare. These are close assets (interests), but a situation may arise where the decision maker will be required to protect one of the interests, at the expense of the others.

For example, imposing a quarantine order on an afflicted farm in order to prevent public contagion. This may come at the expense of harming animal welfare.

After the decision maker has identified the protected assets, they must be prioritized. In this example the "public health" asset is more important than the "animal welfare" asset.

Defining processes, players and activities

- The information must be summarized in a concise document with general data, flowcharts and a verbal / written description of the existing activity (see Annex B). The evaluation of the factual setting is done through the following actions:
- Evaluation of the existing social or economic activity: What actions are being carried out – by whom, to what extent, why, when, and their results (benefits to society, interactions with responsibility industries, etc.).
- Mapping of work processes: For example, when supervising the industrial sector, the stages of the work should be mapped and evaluated (receipt of input,

sorting, processing, packaging and distribution). Understanding the order of the work flow is essential in order to understand where risks originate and how to intervene. It is possible that in order to understand the process, visits to the field and meetings with the relevant parties will be required. It is easy to summarize the evaluation of work processes using flowcharts.

- Mapping players and interfaces: identifying the players involved in the existing activity (businesses that trade in the products, the bank and clearing company, workers on the assembly line, etc.) and their reciprocal relations (seller-buyer; recurring or one-time relationships; whether there is cooperation or rivalry among the various players, etc.).

Defining the assets' vulnerability

- The vulnerability of the assets to the risks must be examined. Vulnerability awareness helps focus later discussions. Vulnerability describes circumstances where assets are more vulnerable to risks and what variables increase vulnerability (for example – when and where the level of vulnerability

increases significantly). Vulnerability analysis can also help identify inherent weaknesses of existing assets or systems.

Illustrative example: circumstances where there is limited vulnerability. For example, driving at night increases the vulnerability to the risk of traffic accidents (increases the risk of a traffic accident); outdoor work (for example on scaffolding or on the roof of the building) is more vulnerable when the weather is stormy (increases the risk of a work accident).

Illustrative example: inherent weaknesses. For example, transport risks – drivers suffer from a "dead zone" where visibility is low (when driving, the right-rear corner of the vehicle). Risks in the field of safety at work – since employees' attention is related to the level of fatigue, there are more work accidents at the end of the shift. Examples of vulnerability from other fields: fire components in an electronic system; Water-sensitive paper documents; Wood structures sensitive to fire.

- The vulnerability level of different assets can be due to various reasons. Often, the elderly is defined as a vulnerable population because

their health is more fragile, and they find it difficult to move around (for example, evacuation from a natural disaster area). Infants and young children are also more vulnerable in terms of health, but their vulnerability stems from the fact that they are largely unaware of the consequences of their actions (for example, playing with dangerous objects, swallowing small objects). Therefore, the vulnerability of infants and young children is related to the fact that they are not careful in the face of self-evident risks and may endanger themselves.

Defining the existing legal and regulatory framework

- The existing legal-normative framework should be evaluated, and the information should be concisely detailed in Annex B. Most markets and sectors are subject to some legal or governmental regulation. Regulation may be designated for that industry or apply broadly to the economy (for example, an extension order for all industries granting social benefits to employees). The legal framework may also include international obligations.
- The assets or risks may already be covered by some regulation or

government policy. It is important to be aware of this and not create contradictions, redundancies or overlaps. In addition, understanding the legal situation helps understand the market action and the room for maneuvering of the various players. Sometimes the risks to be addressed are the result of government regulations. The normative framework may also include non-legal norms, such as social expectations. For example, public expectation that government encourage small businesses by easing bureaucratic requirements, or the expectation that public companies should be stricter regarding compliance with tax laws, compared to non-public companies.

Mapping stakeholders relevant to the process

- Relevant stakeholders should be mapped and collated in the form in Annex C. The stakeholders are a source of qualitative and quantitative information. Their integration into the process is extremely beneficial to the decision maker. Stakeholders can be divided into three circles: the internal organizational circle (the ministry or the authority), the entire

government circle, and the circle of the public at large.

- Stakeholders from the organization and the government include all related professional approaches to the risks or assets in question. In addition, consideration should be given to integrating parallel policy makers. For example, a regulator engaging in the supervision of food imports should consult with a regulator in the field of cosmetic imports. Other entities with whom a discussion should be considered are units with in-depth specialization: the Tax Authority, the Commissioner of Standardization, the Antitrust Authority, etc. It is also recommended to turn to those who have previously served in a relevant position and have completed their term of office – their accumulated experience and knowledge may be beneficial.
- Stakeholders from the public may have a direct connection to the issue: risk creators, those affected by the risk, those with an affinity to the assets, and those who are expected to be directly and indirectly affected by the decision. In addition to the stakeholders, it is also important to contact representative

organizations – employers' organizations (The Federation of Israeli Chambers of Commerce, the Manufacturers' Association), workers' organizations, third sector organizations (NGOs promoting environmental policy, etc.) and consumer organizations.

- The decision maker will especially benefit from discourse that includes experts from academia, research institutes, the government, foreign countries and international organizations.
- The identification of the stakeholders is an ongoing process that should continue even after the completion of this stage, as further information becomes available.
- At this stage it is possible to decide on the extent of participation and consultation that will be conducted with each of the parties, and at what stage of the process to integrate them. Guidelines for conducting a public discourse are detailed in Annex D.

Defining the scope of the work and the complexity of the analysis

- Finally, the scope of the decision and its economic importance should be briefly defined (the scope of the impact both with and

without intervention). By using this data, the decision maker will be able to determine how thorough the process will be, how many resources will be devoted and what the schedules will be.

- A definition of the scope of work must also include a time range for the risk management process. Is the Risk Assessment intended to deal with the risks in the next five years, the next decade, or with risks that could occur in 30 years? This definition is essential in order to determine the depth of the analysis and the consideration of long-term trends (climate change, demographic changes, construction, etc.). In this context it is important to map current and predicted trends that may affect the analysis. This will enable the Risk Assessment to be complete even subject to gradual changes in the environment of the assets and risks. Trends and changes are usually identified using the following cross-sections: political, economic, technological, environmental, legal and social.

RISK IDENTIFICATION

- The next step in the process is to identify the risks. At this stage, the decision maker must identify possible risks, so they can be evaluated later on. Unrecognized risks cannot be addressed. Therefore, a dedicated stage is devoted to recognizing all relevant risks. The main challenge in identifying risks is methodological: there is concern that not all risks will be identified. When an orderly risk identification process is not conducted, there is a high probability that some risks will not be identified or addressed.
- It is important to use the mapping that was carried out during the context setting stage – to examine the stages of the various processes (examine what risks are created or developed at each stage), the players involved in the activity, the various stakeholders and the effect of the legal-normative framework (how state intervention impacts the creation of risks).
- The guiding rule at the risk identification stage is that there is no 'bad idea' and no suggested risks can be ignored. It is important the

risk screening stage is carried out without any censoriousness. Risk identification is a group process, carried out together, to encourage brainstorming and identification of as many conceivable risks as possible. At the end of this stage, the risks will be clarified.

- For complex fields, identifying the risks requires investing a great deal of resources. It is recommended to carry out a preliminary examination. A preliminary examination will enable the decision maker to determine which areas to focus on for the purpose of data collection, risk identification and evaluation.

Example: Flood hazard identification required the running of complex hydrological models and calculations. Therefore, it was decided to carry out a preliminary examination of areas with settlements and roads (assets). This examination was done simply and without dedicating a large number of resources. The hydrological models were then calculated only for areas with settlements and roads.

Following are possible methods for identifying risks, the main methods are presented first. Several methods can be used to cover the entire range of possibilities.

Method 1: Semi-structured brainstorming

(including breakdown and intersecting):

The key question is which events may endanger the public interest? The method intersects assets and potential sources of risk.

First, the activity must be broken down according to various cross-sections (relating to the needs of the decision maker and the nature of the activity). The breakdown is based on the existing factual situation as defined at the context setting stage. Possible breakdowns include: chronological progression, identifying players involved in the process, and types of activities (construction, ongoing operation), etc.

Afterwards, a brainstorming session is conducted with respect to each cross-section and intersected with the various assets. Every step of the process, action and/or player must be examined – and intersected with the protected assets. At every intersection between a risk source and a protected asset, the question of what might occur should be asked.

Each round deals with a cross section (for example, chronological, geographic, etc.). At the end of

each round, an examination must be conducted as to whether the identified risk exists or could occur in another context – (at another stage of the process, with another player, etc.

Illustrative example: In a brainstorming session on hospital risks, a risk identification round is first conducted for identifying risks in the operating room. After the round, it is necessary to examine whether the risks identified are also relevant to other areas. For example, are the same risks relevant to the maternity ward? Are the risks relevant in the post-operative recovery ward? Etc.

An intersection between the cross-section examined (geographical, chronological, etc.) is then carried out with other cross-sections, along with additional principles guiding the decision maker (environmental protection, transparency, equal opportunity, etc.). This is done in order to identify lateral risks. Intersecting makes it possible to determine if a particular risk is detrimental to both the business owner and the customer, to identify a risk that threatens additional assets or to identify the risk as related to the protection of another public asset.

A semi-structured brainstorming

process defines a systematic thinking process, preventing the omission of important issues. The breakdown also helps identify small, elusive risks. It reflects the factual situation and the intersection with the guiding principles expresses the desired policy. The combination of the breakdown and intersecting helps identify the gap between what is desired and the reality – the risks.

Method 2: "Guided brainstorming"

(or SWIFT – Structured "What-If" Technique).

This is a semi-structured associative method.

One of the team members acts as a "moderator", presenting a keyword from a pre-prepared list. The rest of the team should then bring up risk scenarios connected to the association presented. For example, if the moderator presents the keyword "merchandise," a possible question might be: What happens if there is water in the crates, or what happens if there is a problem with commodity handling procedures in the products' country of origin. The keyword can be general (water, fire, delay) or targeted according to the relevant content world (electrical short circuit, breach of fiduciary duty of a director).

Illustrative example: In identifying risks associated with transporting gas tanks, the moderator presents a general keyword: "human error". A discussion then follows regarding the risk involved in the careless loading of tanks on trucks and the driving of the truck driver. A more targeted keyword can also be used such as: "loading" or "speed."

Illustrative example: In the context of identifying risks related to restaurants, the moderator may present a keyword such as: "the restaurant door". This will lead to a discussion of the risks associated with the use of a transparent glass door that cannot be seen and the risks of the door not being closed completely – which may, for example, enable stray animals to enter the kitchen area.

Method 3: Root Cause Analysis (RCA)

This method starts with the undesirable result and traces its causes through a technique of reverse engineering. This helps to breakdown and analyze the scenario behind a defined result and is useful in identifying the root causes of the known results. The disadvantage of this method is that it is a narrow, focused method that limits analysis. It is useful when wishing to examine in-depth the causes of a particular type

of outcome.

In the first stage, the undesirable results should be defined (factory catching fire, plane crash at takeoff, etc.).

Afterwards, each outcome is analyzed, and an attempt is made to identify as many as possible conceivable scenarios and causes. The emphasis in this method is in identifying key root causes in which failure can generate risks.

Illustrative example: the decision maker fears that a particular virus may enter a food product. Using the method, a discussion may be held on the sources and scenarios that can cause the virus to enter the food.

Method 4: Examining historical data

This is a cost-effective and time-saving method that depends on the quality of documentation of past events. The disadvantage is this method restricts the decision maker's thinking only to events that have already occurred. It therefore prevents them from preparing for events that have yet to materialize. Accordingly, the weakness of the method emerges in dynamic areas that are rapidly changing and developing, and in areas where there is a wide variety of risk possibilities. .

In the framework of this method, past events are investigated in order to identify risks that have occurred or were identified and avoided.

Historical data can also be examined in other areas or in other countries. This data can help identify similar risks by analogy.

It is advisable not to rely entirely on historical documentation and to use past data as a starting point for brainstorming regarding potential additional risks. The principle of this method can be used once the risk identification stage has been completed. It then filters out identified risks and validates risk identification. When examining identified risks, one may ask – whether the risk occurred (locally or worldwide) and whether there was a real concern that the risk would realize or if it was a distant and theoretical risk.

Method 5: Dialogue with the public and stakeholders

Collecting information with the help of the public and relevant stakeholders enlarges the source of information and knowledge available. A consideration maybe to publish a short questionnaire on the main risks involved in the process: "What risks may harm [an interest] in the

process [area/process]?" In contrast to conducting a discussion with the public, discussions with stakeholders or with content experts may be focused and qualitative, but also subject to biases and distortions. This is due to the interests and perspectives of the various parties. For example, an engineering expert will focus on engineering failures. Dialogue with the public may focus on risks that are known to the public ("known risks") and not necessarily on the actual main risks.

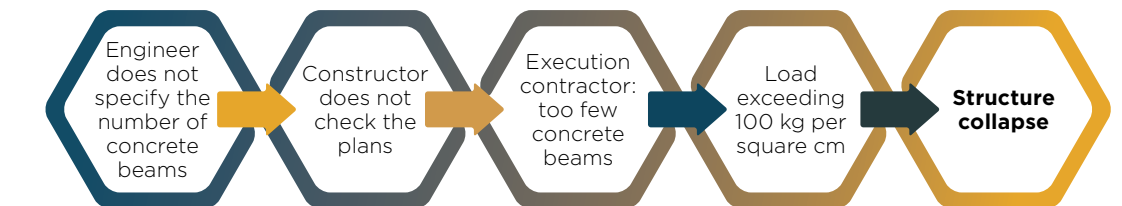
Summary and sorting: Once the sorting and Risk Identification are complete, they should be examined. This is done to filter out irrelevant risks and theoretical risks (risks with a high likelihood of not occurring) and to examine whether the examination of all the risks identified raises additional risks. At this stage it is also possible to split a single identified risk into a number of risks, or to consolidate a number of similar risks. Similar action will be taken at the end of the risk evaluation stage.

RISK EVALUATION

Once the risks have been identified, each risk can be assessed in order to characterize it as a clear scenario. Risk is a defined scenario, a "story" with a beginning (root causes), middle and end (the realization of the risk). The goal is to define risk scenarios/ situations for the process of the risk manifestation and realization. In other words, to "tell the story of the risk". It is recommended to use the mapping performed in the context setting stage to identify the characteristics of every risk in a cross-section of processes, players involved, and the impact of the existing legal/normative framework.

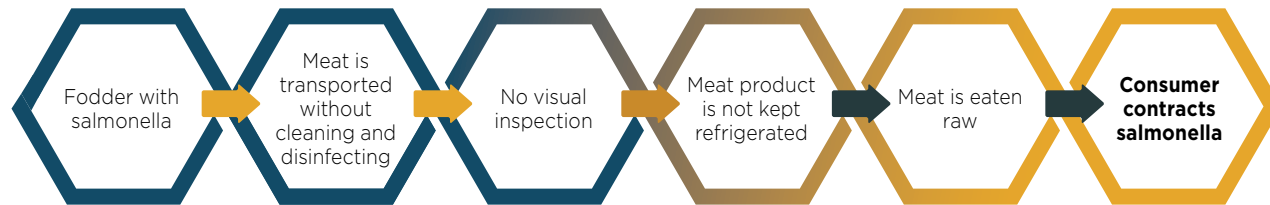
Good characterization contains the chronology of the event: from the root causes that trigger the event, through the occurrence, ending with the realization of the damage and the harm to the protected asset. Breaking down the event into stages helps identify the actions that lead it to taking place, as well as the players involved, and the existing controls.

Illustrative example: The process of realization of a building collapse risk, due to a construction defect. The process analysis shows the risk scenario consists of the following stages: (1) The engineer does not specify in the plans the number of horizontal concrete beams required; (2) The constructor does not check the plans; (3) The execution contractor orders the construction of fewer concrete beams than is needed; (4) A load exceeding 100 kg per square cm is exerted on the structure; (5) The beams do not withstand the load and the structure collapses.



Illustrative example: the process of contracting illness due to meat consumption.

The risk scenario consists of the following stages: (1) The cattle feed contains salmonella pathogens; (2) The meat is transported to a factory without cleaning and disinfecting; (3) The meat does not undergo visual inspection at the factory; (4) The food product is not kept refrigerated; (5) The meat is eaten raw.



Risks evaluation is a qualitative examination of the risks. At the risk evaluation stage, a risk log must be created. The Risk Register is a table concentrating all the information on the various risks (see Annex E). The control and monitoring of risks is carried out through the Risk Register. The risk evaluation stage should turn an identified abstract risk into a specific scenario that can be addressed and discussed. During the risk evaluation, it may be found that a risk identified during the risk identification stage is actually a number of separate and distinct risks.

The evaluation of risks is performed by answering the guiding questions detailed in this chapter. In order to answer the questions (and in order to continue the analysis), the decision maker will need to gather information about the risk and the risk environment. The remainder of the chapter includes references to sources of information and instructions for their use.

Illustrative example: An abstract risk could be the entry of foot and mouth disease into Israel. The risk evaluation process is designed to accurately evaluate the risk in order to generate a specific scenario. For example: the entry of foot and mouth disease to Israel through beef shipments from Australia. It is possible that the evaluation of the risks will reveal that the general risk identified should be expanded to include additional specific risks. For example, the introduction of mouth and foot disease to Israel due to the transportation of viruses from the Gaza Strip, and through the passage of infected wild boars through the Syrian border at the Golan Heights.

Guiding questions for characterizing the risks

What are the characteristics of the risk? How, when and where the risk may occur. The characteristics describe the typical circumstances in which the risk may occur. They express a correlation ratio: the characteristics provide information that is true in a significant number of cases. It is rare that a characteristic is true in all cases of risk realization. The risk characteristics can refer to time (over a year, week, day), location, circumstances, stage of the process, type of population, type of business, extent of damage, and any other characteristic of an event where the risk is realized.

The most convenient way to define the characteristics of the risk is to answer the 'WH' questions about it:

What? What exactly is the event?

Who? Who creates the risk? Who is affected by the occurrence of the risk?

Who is involved in creating the risk? In addition to the risk creator and the risk victim, there may be other factors involved in the process. Business and logistics processes involve various players who may contribute to the risk's creation, mitigate it, or influence the course of events.

Another question is whether there are professionals involved in the risk realization process. For example, the use of prescription drugs requires a meeting with a doctor or pharmacist. The purchase of securities usually involves interfacing with a stockbroker, investment house, or a banking corporation, etc. Characterization of the professionals involved in the process will enable the identification of intervention points for the treatment of risk.

When? At what time of year, week or day does the risk occur? Does the risk occur before or after another event?

Where? Where does the risk occur? Is the risk typical of any region or environment? Does the proximity of an object increase or decrease the severity or probability of occurrence? (For example, proximity to a border, proximity to water reservoirs, near a mountain slope, etc.). It is important to place the risk within the processes and affiliations defined at the context setting stage.

Illustrative example: gas transmission. A natural gas transmission system consists of several components and sections: marine receipt stations (from off-shore platforms, ships and foreign pipelines); transmission lines from suppliers; a marine transmission segment and major land transport segments and regional gas pipelines. In addition to the geographic variance (marine/ground; north/south), each section contains different components and is characterized by different factors and actions. These questions will not only help characterize where the risk occurs, but also the causes for the risk, the players involved in creating the risk, the effects of the risk, and more.

How? How will the risk be realized? Each risk is a defined scenario – an event with a beginning, middle and end. What is the risk's "story"? The goal in this question is to find out the causal sequence leading to the occurrence of the risk. Understanding the causes and processes leading to the realization of the risk is necessary for developing a risk management mechanism.

What is the harm? A verbal/written description of the risk's effect: Who will be affected and how?

First, the extent of public exposure should be defined. Does the creator of the risk expose only themselves to danger, do they expose one person to the risk or do they endanger many? When a person risks only themselves – the need for government intervention is small. When a person risks his environment – the more assets he exposes (people, property, etc.) to risk – the greater the need and justification for government intervention in managing the risk.

Illustrative example: Vaccines against diseases. When the Ministry of Health is considering whether or not to subsidize a vaccine, it examines the extent to which the disease harms the health of the patient, but no less importantly – how infectious the disease is. If the disease is almost non-contagious – it means the person who chooses not to be vaccinated does not significantly endanger their environment. In such a situation, the need for government intervention will be small. On the other hand, if it is a very contagious disease, potentially endangering anyone within a 10-meter radius of the infected individual – the scope of exposure to the public is large and accordingly there is an overwhelming need to manage the risk.

The type of harm and type of damage

should then be characterized::

Damage to property: description of the direct damage (destruction of equipment, accelerated wear and tear) and indirect damage (if any) such as shutdown of activity (closure of a factory, road blockage, delay in international trade, etc.). The extent of the property that may be damaged when the risk is realized (all the businesses in the area, etc.).

Personal injury: description of the direct damage (morbidity and injuries), and indirect damage (if any), such as loss of work days. The number of casualties must be considered.

Loss of human life: What is the number of casualties?

Other damage: damage of the type that has an indirect effect: violation of international obligations, damage to public trust (in a particular sector or in the country), harm to animal welfare, damage to the reputation of a commercial sector, insulting religious/cultural sensibilities, etc.

Illustrative example: Contracting a disease in a hospital.

Characteristics: contracting a disease usually occurs during the winter/morning shift, in patients hospitalized for an extended period of time, in the internal ward of the hospital, etc.

Illustrative example: a traffic accident involving a head-on collision between two vehicles.

Characteristics: these accidents usually occur at night/ on weekends/ on winding roads/ on dark roads, etc.

Is it possible to discover the risk in advance? To what extent can the risk be discovered before injury occurs? This characteristic is designed to help us distinguish between hidden risks and risks that can be detected and prevented. In the event of hidden risk, the point of intervention should be earlier. When risks can be easily discovered, it is useful to provide the public with information in order to equip consumers, those involved, and relevant stakeholders and with tools to take part in pre-empting the risk.

Illustrative example: The Ministry of Health aims to prevent food poisoning from spoiled food. It is known that there are no preliminary signs before chicken meat is spoiled, but on the other hand, beef changes its color and odor before becoming rotten. It is therefore possible to identify in

advance the risk in beef and thus prematurely prevent consumption of the spoiled meat. This characteristic can help the enforcement agents (inspectors in the markets) and help the consumer avoid consumption of spoiled meat.

What are the conditions necessary for the risk to be realized? Among the characteristics identified, the necessary conditions for its realization must be refined.

The necessary conditions express a causal relationship without which the risk will not be realized. Understanding the necessary conditions is important. Effective prevention will integrally focus on the conditions necessary to create the risk. Because decision makers often deal with many different risks, identifying the necessary conditions allows focusing on scenarios with the same necessary conditions. This can then provide a focused response to the source of a number of risks.

Illustrative example: Contracting the HIV virus in a hospital.

A necessary condition for the risk is the direct passage of infected blood. Possible conclusions: Use of syringes and blood transfusion are dangerous; on the other hand, saliva and urine samples do not create the risk.

Illustrative example: a traffic accident involving a head-on collision between two vehicles.

A necessary condition for the risk is the swerving of at least one of the vehicles into the opposite lane. Possible conclusions: A problem with the vehicle's steering system or human error may cause the risk (while the lack of fuel or a defect in the vehicle's mirrors does not create the risk).

An important distinction between the necessary conditions relates to the question of whether this is a risk stemming from human error or a technical failure. This distinction makes it possible to focus on solutions. For example, is technical regulation required, or should human action and the use of mechanical equipment be regulated.

What is the normative cause of the risk? In order to understand the nature of the mechanism causing the risk, the risk should be attributed to one of the following normative causes:

Force majeure – an event that takes

place without human interference such as natural disaster ("pure accident").

Market failure – the relevant players operate legitimately but due to problems of externalization, problems of information flow or problems of coordination, an inefficient outcome is obtained – the risk. ⁽³⁾ The subgroup of market failures of the type of externalization can be classified as cases of negligence. These are cases in which the damage is not desirable in the eyes of the one causing the risk but caused by lack of professional or managerial rigor.

Deliberate action – at times the risk is knowingly caused by deliberate action. The most prominent example is criminal behavior such as theft, smuggling or intentional damage to property.

Government policy – at times, the risk results from government policy. Government policy may

3) Externalization occurs when a player makes a decision that affects others, and the player does not internalize the full implications of the decision (a typical example is air pollution). Problems with information flow include a situation of asymmetrical information or decision making based on partial information. Problems occur when coordination is required between a number of players, for example, adopting a uniform technological standard. These problems may cause a lack of uniformity or an inferior standard being selected.

be the obligation to take action, a prohibition on taking action, the setting of conditions, or the creation of incentives (positive or negative). In some cases, the activity creating the risk stems from "compliance" with the norm or policy set by the state. It is important to identify risks arising from government policy, as the government itself is the source of the risk. These risks can be resolved by changing existing policies without the need to create and enforce new norms. Since the government is involved in the country's economic activity, many risks are affected, directly or indirectly, by existing governmental policies or regulations.

Illustrative example: A risk in the use of carcinogenic material when disinfecting factories has been identified. Factories use this material because the government requires business owners to use only the carcinogenic material. Therefore, the normative cause is government policy.

Illustrative example: Using the tax system can contribute to risk creation. A safety hazard that results from the use of obsolete equipment rather than new and safe equipment has been identified. Tax benefits though are granted when purchasing the obsolete equipment, therefore government policy encourages the risk.

During the risk evaluation process, it sometimes becomes apparent that risks should be split or grouped together. If the risk characteristics vary widely, it may be preferably to split the risk into a number of risks – so that each one is treated according to its characteristics. On the other hand, there may be several risks with similar characteristics. If the risks are similar, it may be preferable to group them together. Splitting risks improves the accuracy of the analysis, and merging risks simplifies the analysis. This enables the decision maker to carry out the process more efficiently. These are methodological aids that may be helpful in improving the process.

Illustrative example: During a risk evaluation process, it is discovered that a risk of a gas leak may occur in the transmission lines from the gas facilities to the customers' premises. These are alternative characteristics to the location of the risk's occurrence. Due to the large differences between them, a new evaluation must be carried out for two separate risks: (1) gas leakage on the transmission lines; (2) gas leakage from the customer's internal facilities.

Once a risk evaluation has been completed, it is recommended to hold a dialogue with the public. It is important to understand how the public perceives and understands the risk. Such a discourse is relevant to stakeholders in both the risk creators' group, as well as the public affected by the risk. This dialogue sheds light on how the public perceives each of the risks. It is possible that the public, for example, perceives serious risks as marginal, or that negligible risks as particularly threatening. If the public perceives the risk inaccurately, the first step in mitigating the risks will be to explain the risk to the public enabling them to respond and prepare accordingly (see – the "Risk Communication" chapter).

Sources of information for the Risk Assessment process

Risk management is based on information. Collecting and processing information takes time, and it is impossible to collect 100% of all the information. The aim should be to gather as much reliable information as possible. Following are the main sources of information for the risk management process – especially in consideration to the risks evaluation stage and the risk analysis stage.

In addition, it is recommended to conduct a comparative international review of data and information from other countries. Such a review can provide comparative information about the problem, the solutions, the effects of solutions attempted, etc.

For every source of information, it is important to examine how and where it is collected from and determine if the information is empirical or based on assumptions and estimates. Assumptions can be used, and the information can be processed, but it is important to note this in the analysis. In addition to the three sources of information listed below, the public and stakeholders are an important and useful source and should be integrated as much as possible.

First priority: data from authorized sources.

- In order to make an informed decision, an evidence-based decision is preferable.
- Collect and review official/governmental sources of information: The Central Bureau of Statistics, epidemiological reports of the Ministry of Health (the study of diseases in the public), reports of the State Comptroller, annual reports of government authorities, data in the possession of the Accountant General, the Population Authority, the Registrar of Companies and the Tax Authority.
- Consult with a professional unit or the regulator that handles or regulates a specific market or activity that is relevant to the risks to be managed. In addition to information, these entities have expertise and familiarity with the same market or activity. Inter-governmental cooperation is a fertile ground for quality data. People who have previously held positions in a government unit are a potential source of valuable information. For example, the predecessor of the decision maker may know historical information, as

well as its professional context.

- Additional information can be found in academia and research institutes.
- Information is also available within the public or business sector. Such data is published in reports of public companies traded on the stock exchange, collated by banks, etc.
- Insurance companies can also provide information. Insurers deal with risk management and specialize in carrying them effectively. They specialize in weighting information about expected and unexpected risks and pricing them. Insurance companies can greatly contribute to understanding existing risks – for example, insurance premiums analysis is strong evidence of how the insurance company perceives the risk

Second priority: Data from international sources

- When there is no data relating to the State of Israel or when wishing to enrich the supply of information, data from abroad can be used. It is important to select, analyze, and apply this data in a balanced and careful manner, taking into account the factual and legal differences

compared to Israel. Therefore, the Israeli context (defined at the beginning of the process) should be compared to the foreign context.

Factual differences can be significant. For example, an attempt to rely on an analysis of environmental risks in Europe. There is a difference between the heavy industries operating in Europe and those in Israel (the absence of large shipyards and the automobile industry in Israel versus a developed military industry). Agriculture in Israel is different from that in Europe (for example – raising cattle and sheep versus pig farming). The main characteristics of the State of Israel are its geography, geo-political environment, and demographic.

Government and legal differences must also be taken into account – government involvement has a greatly influences the risk environment and impacts societal activity. Political culture in Israel (Israel is not a federal state, and one must take into account the role played by the central government), the legal culture (the central role of the court in interpreting and developing the law), and of course specific legal and regulatory arrangements are of great importance. It is also important to take into account indirect influences such

as the tax system and other economic incentives (National Insurance and fees for government services).

A preferred source of information for evaluating and analyzing risks are foreign risk assessment bodies. For example, within the European Union there is the European Food Safety Authority (EFSA), which deals with food safety risk assessment; the European Medicines Agency (EMA), which deals with risk assessment of medicines and medical products; and the International Energy Agency (IEA), which deals with energy security and sustainability. These organizations have extensive professional infrastructure and the information and analysis performed by them is a valuable resource.

Data from international sources can be collected from foreign governments, international organizations (such as the EU, WTO, OECD, WHO), academic institutions and foreign research institutes, as well as information held by private entities.

Third priority: expert opinions in the absence of hard data, expert assessments can be used.

It is important to rely on at least two opinions and make sure the opinions are independent (independent

experts and opinions based on a separate analysis).

Upon completion of this part of the process, the Risk Register should include the name of the risk, the description of the risk scenario, the type of risk, the type of asset, the necessary conditions, the normative cause of the risk and the description of the harm. At this stage, the risks identified can be sorted. Negligible risks omitted, duplicate risks consolidated, and general risks split into more specific risks. Risk sorting depends on the discretion and circumstances the decision maker is coping with – sometimes the difference between two risks is important (a decision may be taken to split the risks), and sometimes the difference is not significant, and it is possible to consolidate the risks or "lump" them together.

The risks evaluation and the Risk Register should be presented to the decision maker or to a senior manager, in order to discuss and confirm they are complete.

RISK ANALYSIS

Risk analysis includes a quantitative analysis of the probability of risk actualization and its potential impact on the protected assets.

To ensure consistent and uniform analysis, severity and probability should relate to events occurring within a one-year period. It is important to define the probability and severity in relation to a specific object or to the individual scenario characterized at the risk evaluation stage (quantitative analysis cannot be carried out for an abstract risk that has not been defined).

Illustrative example: Road safety

When analyzing the risk of a road accident involving pedestrians, there is no significance to the overall probability of road accidents in general.

Illustrative example: Water leaks

When examining the probability of leakage from pipes that carry water from desalination plants, one should ask what is the probability that "these pipes will burst". The type, quality, location and use of the pipes must be defined in this scenario. It is not enough to examine the probability that "general" pipes will burst.

The use of quantitative data is important, but there is usually a variability between events (for example, a scenario that does not repeat itself in the same manner). The goal is not to empirically measure the severity of the risk or to calculate the probability with high accuracy. Risk analysis is intended to allow comparison of the risks and a comparison of the expected damage resulting from the risk to the cost of coping with it. The more accurate the data, the better, but risk analysis does not require accurate empirical calculations. This is a process that includes assumptions, evaluations and the use of statistical information. It is important to document the assumptions and sources of information used by the decision maker for the purpose of risk analysis.

For the purpose of risk analysis, sources of information presented in the chapter on risk evaluation should be used: data from authorized sources, data from international sources and expert opinions.

At times, information is only available about a situation that does not exist in practice, such as data collected prior to the existing regulation in the relevant area. As such, information

is based on data that does not take into account the business dynamics in reality. In these cases, the data should be adjusted to the actual state of affairs. It is important to distinguish between the inherent risk and the residual risk. The inherent risk is the theoretical risk of a "clean" world situation, while residual risk is the actual risk considering the effect of safety measures, economic incentives, and other factors that affect the severity and probability of the risk. It is important the analysis be performed in respect of the residual risk. If the data describes the inherent risk, it must be adjusted so that it describes the residual risk to the extent possible.

Illustrative example: residual probability

The production standard of valves in water pipes determines that the valve can withstand the pressure applied to it and will collapse with the probability of $\frac{1}{100,000}$ uses. But when building contractors install the valves they surround them with pads reducing the pressure and prolonging the life of the valves. As such, they collapse in the probability of $\frac{1}{200,000}$ uses. Therefore, residual risk probability is $\frac{1}{200,000}$ and not $\frac{1}{100,000}$.

Illustrative example: residual severity

In the case of a head-on collision, the main damage expected to the driver is a severe head injury resulting from a crushed skull. This is a life-threatening and even fatal injury, which is defined as high severity. Since cars have airbags installed, the actual damage from the accident is a light to moderate neck injury, with no damage to the skull. Therefore, residual risk severity is light to moderate and not high.

This chapter presents the principles for quantifying the probability and severity of the various risks. The guide does not cover all quantification methods for all types of risks. The tools and examples presented serve as a guiding principle. Some risks will require the development of individual models to quantify their probability and severity.

The outcomes of the risk analysis stage are quantification of the probability of the risk occurring (the number of risks realized in a year, and the relative prevalence of the risks that occur), as well as quantifying the severity of the risk. Since risk management is a process that deals with uncertain events and assessment based on partial information, it is recommended

that alongside such data, the degree of confidence in the outcome of the analysis is noted. Reflecting the level of confidence enables the decision maker to explicitly express the degree of uncertainty involved in the analysis.

After quantifying the probability and severity, the data must be entered in the Risk Register. In addition, the expected damage from the risk must be calculated – multiplying the probability by the severity of the risk (remember that the severity of the risk is the product of the number of damage from a single event). The expected damage will be used later in order to compare the risk to the coping method. In essence, a coping method is chosen if its cost is lower than the damage it seeks to mitigate.

It is important to carry out a dynamic analysis. When estimating the severity and probability of a future risk, historical and expected trends should be examined. One should not only rely on a single figure (for example, a figure from the past year). How the risk will be reflected in the future must be taken into account, according to the trends and changes that have occurred and are predicted. For example, if in the past year there were 20 events, but trends predict an

increase in the number of events – it should be taken into account that the future number of events will not be 20, but a larger number – based on the growth trend. The following figures demonstrate what dynamic analysis is, versus static analysis (which ignores trends and changes, and assumes that the data does not change over time). It is important to focus on dynamic analysis, as the policy and regulation that will be determined are forward looking and should suit, to the extent possible, the future state of affairs.

The example in figures 7 and 8 show a change in the data (severity/

probability): the level of risk decreases with time. Therefore, one must take into account the historical trend and assume that even if we do not intervene, the historical trend will continue.

Quantifying the probability of realization of the risk

The probability of the risk occurring expresses the number of events in the year that are expected to happen, compared to the total relevant activity. It is easy to describe the probability using a simple fraction: a numerator and a denominator. The numerator is the number of risks that occur each year (accidents, incidents) and the denominator is the extent of the activity that creates the risk.

Illustrative example: One in 100,000 births has a case of cleft lip. In this case, the probability that a baby will be born with a cleft lip is $\frac{1}{100,000}$, which is 0.001%.

- The number of risks realized (numerator): This is the number of negative events occurring during a one-year period (car accidents,

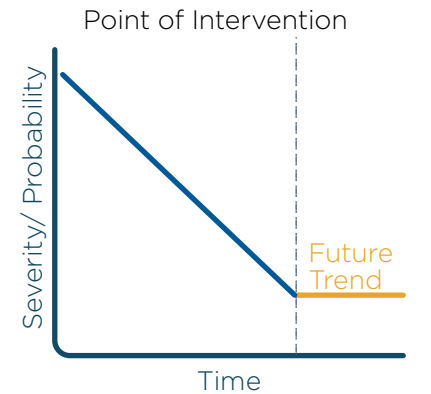


Figure 7: Static analysis - an erroneous assessment of future development of risks

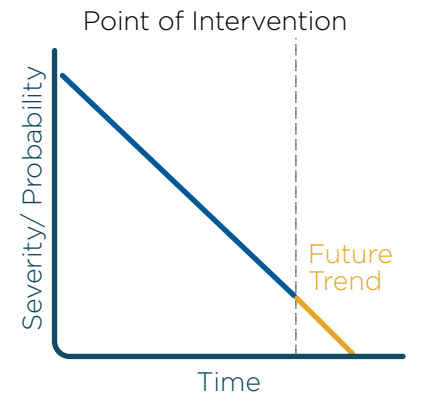


Figure 8: Dynamic analysis - a correct assessment of the future development of risks

workers falling from scaffolding, competitors that created cartels and coordinated prices). The more accurately the risks are characterized, the easier it is to define them in the risk evaluation stage and determine their number.

- The scope of activity (denominator): First, the activity that creates the risk must be defined. "Activity" is a count of actions, not a count of objects (factories, crossing points, businesses). Actions can be: driving, manufacturing food products, construction of residential buildings, importing containers with toys. Afterwards, a unit of measure must be selected to measure the activity on a nationwide scale over a one-year period. For example, driving is measured by mileage. The volume of imports of goods can be measured by the weight of goods or the number of containers imported per year. Water consumption can be measured by the amount of water (cubic meter). The use of electrical appliances can be measured by the electric current (kWh).

Illustrative example: In one year there were 16 cases of professional negligence in audits performed by auditors in annual reports of public companies. Israel has 14,000 accountants and 530 public companies that submit annual reports. The number of accountants is not relevant to the probability. Of the 530 annual reports audited, 16 incidents of professional malpractice occurred. Therefore, the ratio between the number of incidents and the volume of activity is $\frac{16}{560} = 0.0301$. Hence, the probability that professional malpractice will occur in preparing an annual report is 3.01%

There are two techniques for estimating the probability of risks occurring in the future: (1) Statistics of the realization of risk in the past; (2) Probabilistic analysis of risk actualization.

Statistics of the historical realization of risk

When dealing with an existing risk or one that has occurred in the past, it is possible to examine statistics on the realization of risk in the past. There should be sufficient reliable data on risks that have occurred over a significant period of time. A future trend should be extracted from the data (growth, decrease or other change in the pattern of realization

of the risks) and an assessment established. In addition, it is important to take into account the limitations of existing statistics (for example, the pattern fluctuating due to the adoption of new technology or change in the volume of activity affecting the risk).

Illustrative example: There is accurate information on the annual incidence of influenza. Existing data refers to morbidity over the past 20 years. If the number of patients is fixed over several years and is "known in advance" at a reasonable level of certainty, statistics can reliably assess the extent of morbidity in the future.

It is important to consider the limitations of past data: if the data indicates a change in the scope of morbidity, such as a 2% compounded annual decrease, morbidity can be estimated according to the trend identified. In addition, it is important to consider external variables: Has a vaccine recently been developed to reduce morbidity, or due to an increase in international trade, more virulent strains of influenza are expected to enter the country.

Probabilistic analysis of the feasibility of risk realization

If historical data on the frequency of realization are insufficient – the

probability must be quantified according to the stages of the risk scenario. The stages of the risk scenario are cumulative conditions for realization, therefore the probability of each step can be multiplied with each other.

The likelihood of occurrence is based on multiplying the probabilities of all stages of the scenario. Therefore, if one of the stages has a probability of 0 – this means that the scenario will not be realized.

Illustrative example: A child's injury from a damaged plastic toy is a risk arising from three stages: (a) During the production process of the toy, too much air was injected into the plastic; (b) The plastic with the air was not properly fastened to the toy base; (c) When using the toy, pressure exceeding 2 kg was exerted on the toy, resulting in the fragmentation of sharp plastic pieces.

According to risk evaluation, the scenario consists of three events (stages). In order to know the probability of the occurrence of the risk – the multiplication of the probabilities must be calculated:

[Probability of occurrence of event A] * [probability of occurrence of event B] * [probability of occurrence of event C] = probability of realization of risk.

For example: in 20% of the production line, an excessive amount of air is injected into the plastic. Faults in the fastening system cause 10% of toys to be fastened incorrectly. Children between the ages of 3-6 play with toys and only 50% of them are able to apply pressure of 2 kg. Assuming that children always apply their full force on plastic toys (a strict assumption), the probability multiplication is: 20% * 10% * 50% = 1%. The overall probability of the risk being realized is 1%.

The probability correlation between different factors or stages should also be considered. If the realization of risk is contingent on two cumulative factors, it is necessary to examine the probability of the two factors occurring simultaneously (each of them separately is not sufficient to cause the risk).

Illustrative example: The Electricity Authority is preparing to cope with power outages. The risk scenario was defined: At peak consumption of electricity, a backup generator of the Israel Electric Corporation is disabled

due to flooding by rainwater. The two cumulative risk factors are the flooding of the generator due to rain and peak electricity consumption. In Israel, although there is rain in the winter, peak electricity consumption occurs mainly in the summer. In this case, the probability of rain is high, and the probability of peak electricity consumption is high, but there is no correlation between the two factors.

Therefore, the probability of the risk scenario occurring will be low. In the previous example of the toy factory, perfect correlation was presented.

The generator example shows an inverse correlation.

Whether using the statistical method (past data) or the probabilistic method, the frequency of the risk relative to the scope of the activity should be identified. This must be done in order to understand whether the risk is a common or rare phenomenon. Even if there are complete statistics on the realization of the risk, the prevalence of the risk out of the total relevant activity is important

Illustrative example: There are 300 traffic accidents each year. The figure is relatively stable, with slight deviations. It is also possible to calculate the severity of the damage – which does not change significantly from year to year. This data is not enough in itself to determine policy. In addition, it is important to understand the extent of travel, in the course of the accidents occurring.

There will be a material difference if the figure refers to the number of accidents in Israel or in the United States. The reason for this is the need for the decision maker to understand the social context, the magnitude and relative frequency (according to the volume of travel or the size of the population).

When examining the data on the realization of the risk or on its various stages, it is important to take into account the information will be partial. This is due to the phenomena of underreporting malfunctions or accidents. Adding a "factor" to the reported data (usually between 1% and 10%) should be a consideration. The use of a factor and its size depends on the type of data, the reporting population, and the accuracy of the data. If a factor is added – it is important to note throughout the work documents that an assumption

of under-reporting has been factored in.

Illustrative example: In the earlier toy example, let us assume that toy factories under-report faults and malfunctions in their production line, but the data relating to children playing with the toys is reliable and complete.

As such, an additional factor of 5% was added to the data reported by the toy factories. Accordingly, the probability of introducing an excess amount of air into the plastic increased from 20% to 21% (an increment of 5% to the original 20%) and the probability of malfunctions in the fastening system increased from 10% to 10.5%. The probability of applying pressure while using the toy did not change.

Therefore, the probability multiplier is: $21\% * 10.5\% * 50\% = 1.1025\%$. The overall probability of realization of the risk is now 1.1025%.

Quantifying the severity of damage from risk

The severity of damage from a single incident should be multiplied by the number of events occurring each year (the number of accidents, the number of people contracting the disease, etc.). Mapping the severity of damage from the risk refers to the impact of all incidents occurring within within one year.

Illustrative example: In a grocery store robbery, on average an amount of ILS 5,000 is stolen. Over the past decade, each year has seen an average of thirty robberies in grocery stores. Therefore, the total annual economic damage is ILS 150,000.

In quantifying the severity of the damage, it is important to identify the real value that the risk poses to society as a whole. It does not matter whether the one harmed by the risk was insured against the risk (and therefore received indemnity) or if an employer participates in the payment of expenses of an employee who fell ill. The decision maker must analyze the real damage of the risk to the economy, regardless of who bears it.

The economic expression of the risk's impact is necessary to activate the principle of maximum aggregate welfare. Quantifying the damage enables comparison to damage from other risks, as well as the costs arising from the means of coping with the risks.

One risk can cause a number of injuries – for example, air pollution can cause allergies and cancer. In such a case, the severity of all damages must be quantified. In addition, one risk may result in different damages with different severity levels. This is a common condition. In this instance, it

is necessary to examine the extent to which the damages differ from one another. If there is extreme variance, it may be a separate risk. As such, the risk should be divided into a number of different risks. If variance is small – the average severity representative of the severity level for the risk must be selected. This can be the midpoint, the mean, the median, or the characteristic damage from the risk. In examining the characteristic level of severity, it is important not to take into account extreme cases and extreme values that do not reflect the rule.

Illustrative example: Marine accidents. As part of the risk management for the "Titanic" luxury vessel, the risk of the ship being hit by a glacier was identified. Damage caused by a glacier hit may range from a superficial scratch on the ship's paint to the creation of a large hole in the ship hull. Since these are materially different damages, it was decided to split the risk into two separate risks. The split makes it possible to examine them separately and to examine the differences resulting in the different severity of damage – the size of the glacier, the angle of impact and the velocity of travel at the time of impact.

Illustrative example: Motorcycle theft. In the framework of risk management for law enforcement in property offenses, a risk of theft of motorcycles parked on streets was identified. Motorcycle prices range from ILS 3,000 to ILS 15,000 (excluding luxury motorcycles, which are generally rare, not parked on streets, and therefore few in number). The above range is not extreme, and the value of most motorcycles ranges from ILS 6,000 to ILS 8,000. As such, it was decided to carry out the quantification on the basis of ILS 6,500 value per motorcycle. This is reflective of the typical price range of stolen motorcycles

In the last example, two steps were taken. Luxury motorcycles were "eliminated", which is the exception that does not testify to the rule (and may even be an asset that is not threatened by the risk in question). Then the range of motorcycle prices was examined and a typical price reflecting the common value was selected.

The effect of the risks varies from field to field. However, most types of damage can be quantified using three basic components: (1) The cost of restoring the situation to its original state before the damage occurred; (2) The cost as a result of the interruption of activity; (3) Loss of human life. We

will show how these components are used to quantify the severity of the most common damages: property damage, personal injury, and damage to human life.

In the event of a risk of damage to property, the cost of returning the situation to its former status usually reflects a repair of property damaged or the purchase of replacement property. The cost of disabling activity is usually expressed in the prevention of social or business activity (for example, closing down a plant).

Illustrative example: rioting on the street could result in the risk of throwing a stone through a shop window. The financial damage is divided into two parts: (1) restoration of the situation: the cost of repairing the window (including cleaning the glass fragments, purchasing a new window and paying for its installation); (2) Cessation of the business activity: loss of income of the store for due to the fact the store must be closed until the window is repaired.

In case of risk of personal injury, the cost of restoring the condition to its previous state is usually expressed in medical expenses (including examinations, treatments and medical follow-up); The cost of cessation of activity is usually expressed in loss

of income due to illness, injury and disability. Personal injury can be classified as acute harm and chronic harm. Acute harm lasts for a limited period of time. Chronic harm involves an incubation period, breakout and repeated occurrences.

Illustrative example: In a work accident at a construction site, a worker falls from a height and breaks his hand. The cost of restoring the situation to its previous state: the cost of medical treatment (including x-ray, surgery, a cast and hospitalization), as well as periodic examinations and medical follow-up. Cost for cessation of activity: During the treatment period, hospitalization and recovery, the worker is denied his wages. It is important to note that even if the worker is insured and does not bear this cost directly, it is nonetheless a cost to the economy that must be taken into account. In addition, as a result of the injury, the employee may suffer from disability that will reduce his future earning levels. This damage too constitutes cost of cessation of activity.

In the case of risk to human life, the quantification of damage is calculated according to the Value of Statistical Life index (VSL) as calculated by the OECD. According to this index, the statistical value in Israel for preventing one case of mortality is ILS 12.830 million (USD 3.301 million at 2015 values).

Value of Statistical Life index (VSL)

In recent decades, governments have been increasingly using cost benefit analysis (CBA) to formulate public policy. They also take into account the risk of mortality when making decisions on energy, infrastructure, health care, transportation, employment, environment, food, etc.

Even when the risk of mortality is not explicitly estimated, many public decisions involve an implicit assessment of the economic value of mortality. For example: if the government decides to adopt a plan that will reduce one case of death for every ILS 1 million invested in the program's budget, it means the government is estimating the economic value of human life at more than ILS 1 million.

According to the OECD, decisions made on the basis of an implicit assessment of the value of human life tend to be inconsistent and vary from case to case. The value depends on the level of decision makers' information and by the influence of emotional baggage that characterizes each case. Therefore, the use of an explicit, consistent and uniform economic value ensures better quality decisions.

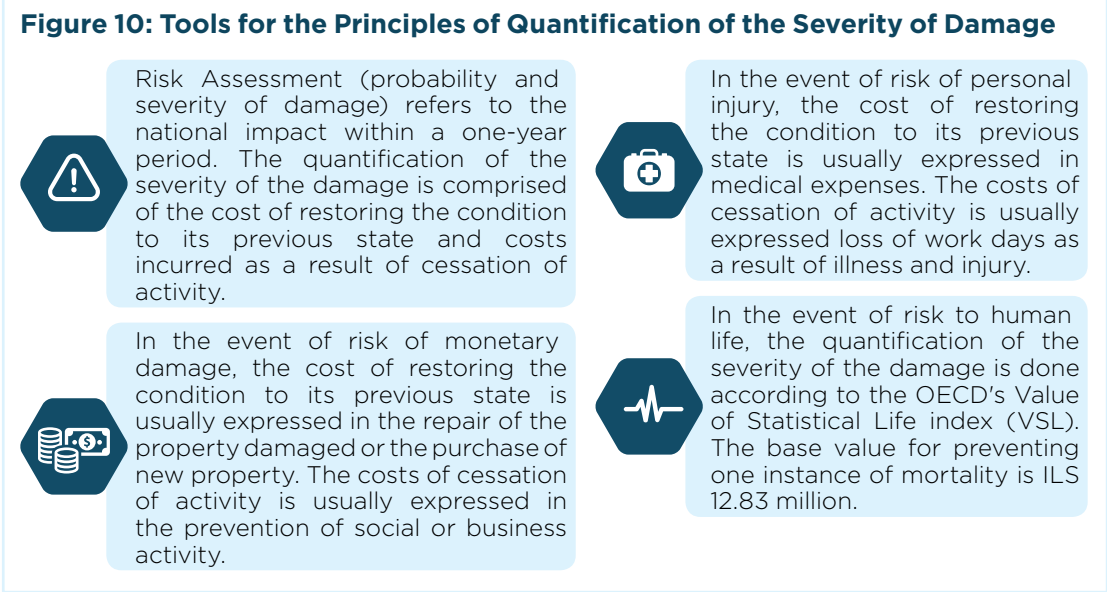
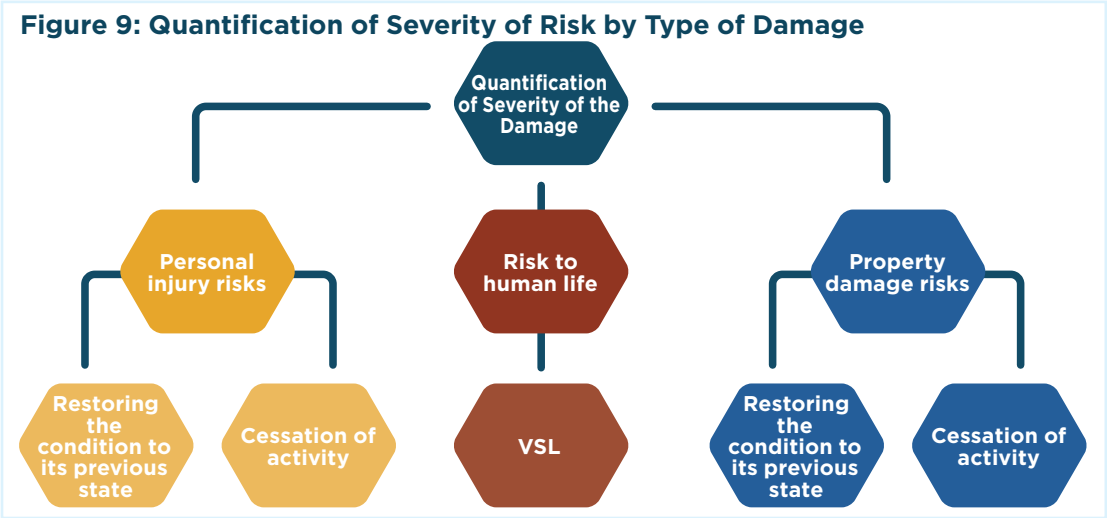
The Value of Statistical Life index (VSL) is used by many countries to formulate public policy. In the United States, the Office of Management and Budget (OMB) uses it to guide the various ministries and authorities on what economic value to attribute to human life when examining policy. The Value of Statistical Life index of the OECD was based on extensive studies in OECD countries of the damage caused by a single case of mortality. It is premised on a person's willingness to pay (WTP) to reduce the risk of death.

The use of the Value of Statistical Life index also ensures uniformity and equality: the value is constant and uniform for the entire population – regardless of age, origin, place of residence or level of earnings. Using an index that varies by age or place of residence will lead to a systematic bias of the decisions – for example, a willingness to endanger adults or expose residents in the periphery of the country to greater risks than residents in the center. This is a problematic result; therefore, the Value of Statistical Life index expresses a desire for consistent and egalitarian decision making. The index does not seek to define the value of human life, but rather the benefit to society as a result of reducing the risk of mortality. It is a statistical calculation designed to help decision makers make an informed cost-benefit analysis in order to produce the optimal policy.

Life index expresses a desire for consistent and egalitarian decision making. The index does not seek to define the value of human life, but rather the benefit to society as a result of reducing the risk of mortality. This is a statistical calculation designed to help decision makers make an informed cost-benefit analysis in order to produce the optimal policy.

How to use the index: The number of deaths should be estimated and then multiplied by the Value of Statistical Life index – ILS 12.83 million.

The following figures summarize the manner of quantifying the severity of the damage:



At times, a risk may cause many small damages. A single event has little or no effect, but cumulatively it may create a significant effect. It is important to take into account such damages – to multiply the effect of each occurrence by the total number of events.

Example: The phenomenon of sending e-mail spam involves harassing the recipient, along with unwanted e-mail that takes up space on web servers. A single spam messages creates negligible harassment and only slightly increases occupancy on web servers. However, the number of messages is vast, estimated to be in the billions. According to the EU Commission, the phenomenon has had a devastating impact on a massive scale: this is apparent when multiplying the damage from one message (a waste of seconds and the occupancy of Web servers) by the enormous number of spam messages sent.

At times, it appears that the risk threatens abstract interests, but even harm to abstract values often expresses quantifiable economic damage. Quantifying these risks may require indirect damage analysis.

Illustrative example: Fiduciary duty

There is a risk that an investment manager will breach his fiduciary duty towards the customer (for example, due to a conflict of interest). Ostensibly, this is a matter of ethics and abstract damage. In practice, if investment managers violate their fiduciary duty, they will not manage their portfolios optimally. As a result, the return on customers' investments will be harmed. Damage to the yield on investments is an economic damage that can be quantified.

Illustrative example: Environmental pollution

Environmental damage such as air pollution and water pollution create a risk to public health. Therefore, it is not only an abstract ecological issue but also tangible damage to human health. Even ecological damage that does not harm the public can lead to concrete damage. For example, harm, to the quality of seawater will damage tourism in the same area (negatively affect employment and income of hotels and restaurants). One of the methods for quantifying environmental effects is the Impact Pathway Approach. This method can also be applicable for other types of damages.

Illustrative example: Damage to biodiversity

Damage to biological diversity is not purely environmental damage. For example, the penetration of an invasive species can result in the spread of diseases (the Asian Tiger Mosquito that entered Israel is the carrier of many diseases), or severely damage flora and fauna (the palm weevil causes severe damage to the palm growing industry). Damage or extinction of a biological species can cause deformation of the biological equilibrium. For example, the extinction of a species has an indirect effect on the environmental system.

Quantification of the severity of the damage can be exhausted in direct damages and can be extended to indirect damage and effects. The scope and depth of the analysis of indirect damages depends on the discretion of the person executing the analysis, in accordance with the issue at hand and available resources.

The method presented here is intended to be a uniform and simple basis for implementation, suitable for quantifying various types of risks. There are other models for quantifying damage severity, including models adapted to different areas of practice

and risks (for example Israel's Ministry of Transport's Transportation Projects Procedure, 2012). This guide provides a methodological basis for possible models for quantifying the severity of damage. Of course, there is room to develop and implement dedicated models for individual fields. The decision maker must choose the most appropriate cost quantification model according to the professional field in which they are involved. They must take into account the need to create uniformity and consistency within the government, with the aim of making the risk management process effective. Following are models for quantifying damages with unique characteristics.

Risks with general impact

The hedonic pricing method helps compare a "damaged" asset to a protected asset. This method is applicable in cases where it is difficult to examine the effect of the risk. The hedonic model is useful for examining "soft" impact such as noise, odor and heat nuisances, mild environmental damage, blocking of sunlight, and the elimination of beneficial resources. The model examines the harm to enjoyment or value that can be derived

from the protected asset.

- According to the hedonic model, the value that can be derived from a damaged asset should be compared to the benefit that can be derived from a protected asset. The hedonic model does not focus on the harmful action, but on the outcome comparison between a protected and a damaged asset.

Example: Many countries have rules in regard to the timbering of trees. These rules are based on the perception that trees provide environmental and social value. Trees provide shade, trap carbon, produce oxygen and have a scenic contribution. However, it is difficult to place a value on the environmental and social value of a tree or the economic damage caused by cutting down a tree.

Using the hedonic model, it is possible to compare the price of a house surrounded by trees to a similar house with no trees in its vicinity. A study by the University of Washington found that buyers are willing to pay a premium of 7% - 19% for properties with mature trees around them. The

model is not based on the fact that tree cutting causes direct damage of 7% - 19% in the value of the property, but a comparison using the hedonic model enables quantification of the benefit from the trees.

- The main limitation of the hedonic model is that there is not always information on the value of similar assets (damaged and protected) to compare with.

Risk with long-term effects

- There are risks whose main effect is "acute" (exposed immediately). Other risks are characterized by "chronic" injury (revealed over time and sometimes after long periods). The classic example is morbidity due to prolonged exposure to a harmful environment. The damage is realized only after a long series of events.

Example: Daily exposure to a carcinogen over 20 years causes cancer. Single exposure does not cause morbidity, neither does daily exposure over one year.

- Quantifying risks with long-term effects requires the use of two tools to calculate the damage in one year

(like the rest of the risk analysis): attribution of damage to one year and discounted cash flow. There are more complex and accurate models for quantifying chronic damage, but the model presented here is simple and can be applied to a variety of damages.

- Attribution of the damage in one year - In the previous example, the damage forms only after exposure over 20 years, therefore the contribution of a specific year to the damage is only 1/20 of the total damage caused, (a year of exposure is 5% of the total damage). This is a linear model, but of course a more complex model can be applied that relates to nonlinear years. Such models exist mainly in the field of health and the environment.
- The damage must then be discounted. The economic cost of the damage can be estimated according to the general principle presented at the beginning of the chapter. Discounting means the translation of the value of money from one period to another. If we examine damage that took place in 2010, but the risk causing it

occurred in 1990 - there is a gap of 20 years between the two dates. In order to translate the value of money paid for medical expenses in 2010 into 1990 values, it must be discounted. The discounting is based on the use of interest rates. The discounting formula is:

$$PV = \sum \frac{C_T}{(1 + r)^T}$$

C (Cost) - the future cost as a result of the risk (the damage)

T (Time) - the period for which the discount is made

r - Interest rate

PV (Present Value) - Present value of future money

When performing such a calculation, it should be taken into account that not everyone exposed to the risk will reach the level at which the risk occurs. For example, if 20 years of exposure to hazardous material is required, part of the population will be exposed to the material only 13 times and will not develop chronic damage (or develop damage at a low level of severity).

Moral risks

Sometimes we aim to reduce a risk that does not involve harm to human life, health or property. These risks express a desire to preserve protected values or interests, even though they cannot be translated directly into economic damage to assets. For example: the protection of religious sensitivities and the right to demonstrate.

When the government is required to protect interests and values that do not necessarily have an economic price, it is possible these are moral and ethical questions.

- It is important to try to quantify such risks. Coping with such risks involve costs to the government and society - and ultimately a cost-benefit analysis is required, between the risks and the means of coping with them.
- One of the main methods of quantifying the economic value of those interests is to conduct a "willingness to pay" study. These empirical studies examine how much a person is prepared to pay in order to prevent or mitigate harm to a particular interest (usually on the level of probability of injury). Such

research will reveal how much the public is willing to pay to address the problem. The "price" the public is willing to pay reflects the severity of the risk the public perceives as harming of the interest.

Illustrative example: animal welfare

It is difficult to estimate and quantify the severity of the damage caused by holding cows in overcrowded cells in dairy farms, as opposed to holding them in more spacious cells. The difficulty is that animal welfare is an abstract value and does not involve economic damage.

A study of willingness to pay will examine how much the public is willing to pay to convert the crowded cells into more spacious cells. Typically, the various alternatives are known and have a clear economic price.

Improving animal welfare entails farming and production costs, which will lead to an increase in food prices. A study of willingness to pay will examine how much more the dairy consumer is willing to pay for a liter of milk, in order to ensure larger holding cells for the cows. This price increase reflects the economic value that the

public attributes to maintaining that value. This is the price that society is willing to pay.

- The approach behind the studies of willingness to pay reflects democratic logic: policy makers must protect values that are important to the public, according to the degree of importance the public attributes to them. Since dealing with every risk involves a social price (government budget, private expenditure), it must be determined how much money the public is prepared to spend on an issue, in order to develop a policy that suits the public's will. Willingness to pay studies are a complex field and not an exact science. When deciding to conduct such research it is important to enlist and consult experts in the relevant fields.

- Coping with moral risks is not simple. It is precisely for this reason that important policy in these areas be developed on the basis of an in-depth public examination and discussion, reflecting the benefits and prices that society derives from the policy.

Analysis and comparison between risks

The culmination of the risk analysis stage is the analysis and comparison. The goal is to examine the information gathered about the risks and to prioritize. The need to prioritize risks is mainly due to the limited resources (both the resources for dealing with the risks and the resources to formulate a solution for each risk). In addition, prioritization indicates in which order to deal with the risks. Risks are examined in two main axes: quantitative and qualitative.

The quantitative axis ranks the risks and prioritizes them, based on the probability and severity of each risk. This axis is designed to compare the risks according to severity, probability and expectation. It is recommended to rate the risks according to their expectation and first deal with risks expected to result in greater damage. A deeper analysis can then be performed using a two-dimensional risk:

- The severity and probability data obtained in the earlier risk analysis stage should be placed in the matrix. The x-axis describes the

probability of the risk being realized (the number of risks occurring in a year, divided by the volume of activity that creates the risks). The y-axis describes the severity of the annual damage to the economy as a result of the risk.

- In principle, the x-axis ranges from 0% to 100% but the matrix only displays a range between 0% and 50%. It is important to note that risks, by definition, are uncertain future events. Therefore, risks will usually appear on the left side of the graph (below 50% probability). When the probability approaches or exceeds the 50% threshold, it is an event whose probability of occurring is higher than the probability of not occurring – and the risk begins to become a "fact". Therefore, the matrix is focused in the range of 0% to 50%. Risks with a probability of more than 50% will appear at the end of the matrix, similar to risks with 50% probability.
- The analysis is performed using two matrices. Matrix (1) is a uniform matrix – the maximum value of the y-axis is ILS 1 billion. This matrix makes it possible to compare risk

management processes between different decision makers. This is a uniform government matrix that enables decision makers

in government ministries and authorities to conduct analysis on a national uniform and comparable basis.

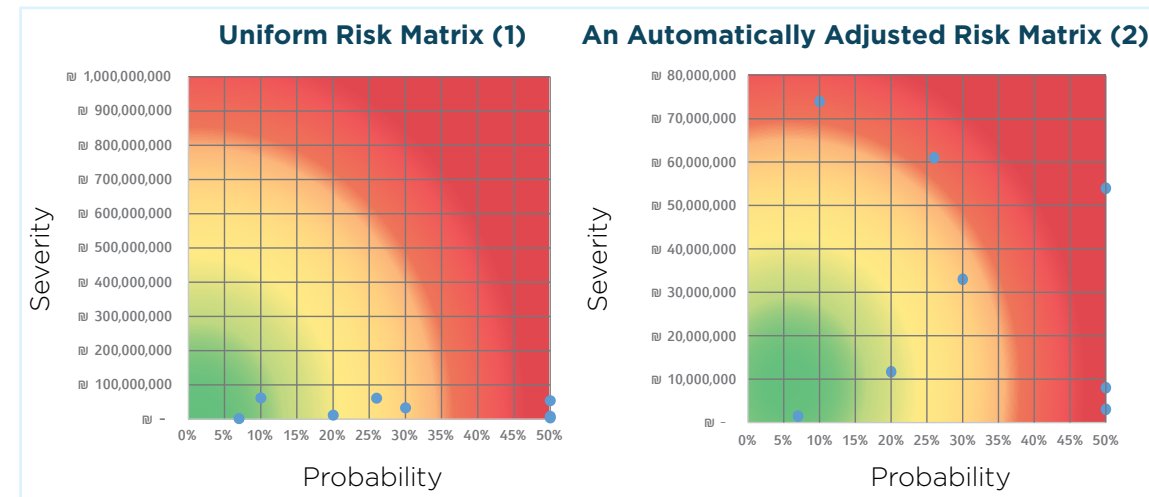


Figure 11: A uniform risk matrix

Figure 12: An automatically adjusted risk matrix

Matrix (2) is the same as matrix (1), but the y-axis is not constant and allows automatic adjustment of the chart layout to the risk scale on a case-by-case basis. Matrix (2) is adapted to the values that emerged in the analysis of individual risks and allows a focused look at the scale relevant to these values.

The analysis using the two-dimensional matrix enables us to examine the data and ask the following questions:

What risks should be ranked as central (upper right corner) and what risks should be considered as negligible (bottom left corner)?

Are there groups of similar risks (at a similar level)?

Is there room to distinguish between serious and rare risks (top left corner), which justify insurance or crisis management strategies, and minor and common risks (lower right corner) that justify slight routine preparation?

Where are the main risks? What characterizes most of the risks?

As part of the quantitative analysis it is important to pay attention to and avoid two major cognitive biases:

- Attributing great weight to severity at the expense of probability. This means that risks with high severity and low probability are given greater weight than risks with low severity and high probability.
- A tendency to attribute great weight to unusual events, and to

new or prominent events (referred to as the salience bias). Unusual or new cases capture the attention and disproportionately shape the decision-making environment to their importance. This means that "anecdotes" or unusual incidents dictate general policy that applies to many cases where almost no exceptional events occur. Such a policy may not be appropriate to solve the problems that characterize common cases as it was designed for exceptional and rare cases.

The combination of these two biases leads to decisions that may be made on the basis of extreme and unique cases, characterized by high severity but low probability. As such, a decision may be made that ignores the probability of the risks and is based solely on their severity. In such a situation, the policy may be determined in relation to the exception and not to the general rule. In this case, common risks are not treated and the public bears the burden of focusing on exceptional events as if they were routine events.

Example: The public fears flying and feels safer driving cars. The "gut feeling" of most of us is that it is safer to travel in or drive a private vehicle. Data unequivocally demonstrates that airplanes are much safer than cars.

According to the American National Transportation Safety Board (NTSB), the risk of being killed in a car accident is 1 in 98. The risk of being killed in a plane crash is 1 to 7,718.

Rationally speaking this is a distortion, a mistaken decision of the public. In terms of safety, it is better to use airplanes than cars.

Despite the importance and clarity that quantitative analysis provides, in order to obtain a complete picture, the qualitative axis must also be examined. This axis examines the risks through additional considerations and systemic thinking. Naturally, there are a variety of qualitative considerations, and this is not a closed list.

The qualitative axis is intended to examine the risks according to the information collected at the stage of risk evaluation. This axis consists of four parts:

- Identifying common trends and characteristics - is there a necessary condition common to a number of risks? (If so, dealing with one essential condition may address a number of risks). Can the risks be divided into groups according to other criteria? (Normative cause, process, factors involved, place, time, etc.). Are there distinct groups of risks with similar characteristics? What is the ratio between the qualitative groups and the quantitative analysis of the risks? (For example, is there a set of risks with similar qualitative characteristics, whose damage is high or low).
- Risky actions or risk-sensitive steps - are there steps or actions in the process that are more exposed to risk than others? Identifying potential failure points is important so as to focus policy tools on them, and also supervision and enforcement.
- Key Risks - are there risks that require extra attention, even if this is not indicated in the quantitative data? Are there serious risks that are

difficult to express in an economic value (for example, violation of basic rights, competition or international relations)?

- The feasibility test - prioritizing the treatment of a risk that can be dealt with by simple, applicable and effective means (rather than a risk which entails a cumbersome regulatory framework, questionable effectiveness and/or significant risks or costs). This is a preliminary examination of possible treatment methods. At this stage the feasibility test can be used to prioritize the risks.

It is important to implement the conclusions from the risk analysis stage in the Risk Register. Completion of the risk analysis stage completes the Risk Assessment. The Risk Register should contain data and analysis of the various risks after prioritization. It is necessary to examine which risks justify treatment through government intervention and which risks do not. The latter group should be taken out of the process.

It is important to present the conclusions and hold a discussion

with the decision maker or another senior official. This discussion is necessary to validate the information and conclusions and critically examine them. It is recommended to hold a dialogue with the public to ensure that the picture received is complete and accurate. Dialogue with the public is also a useful tool in validating and refining the analysis.

It is recommended to summarize the Risk Assessment (including their rating and prioritization) in a risk assessment report and the Risk Register. It is recommended to present these documents for the approval of the ministry's or the authority's management. Management approval at this stage is important. It is vital that the risks are properly defined, and that management agrees to proceed with treatment of the prioritized risks.

SELECTING A STRATEGY AND TOOLS FOR RISK TREATMENT

The process of risk treatment consists of three main stages: selecting a strategy for dealing with the risk or a group of risks; selecting a regulatory method and policy tools; selecting regulatory or policy content.

The strategy selection stage is a simple and fundamental stage. The primary purpose of this stage is to identify whether a risk or a group of risks will be addressed by government policy, or whether there is no reason to intervene.

The next two stages are the culmination of the risk management process: the stage of selecting the regulatory method and the policy tools that will deal with the examination of various risk treatment tools. Choosing a method for treating risk is akin to matching a solution to the problem. This stage is based on understanding the risks as assessed and analyzed. First, 2-4 possible methods should be selected according to guidelines for matching policy tools to the type

of risk. Then, alternatives must be compared and the alternative that best serves the principle of maximum aggregate welfare should be chosen.

The content selection stage is similar to the method selection stage. First, the main questions and issues relating to the chosen method must be identified. For each issue, a number of alternatives should be compared and the most appropriate alternative for the principle of maximum aggregate welfare chosen. The following figure shows the Risk Treatment process.

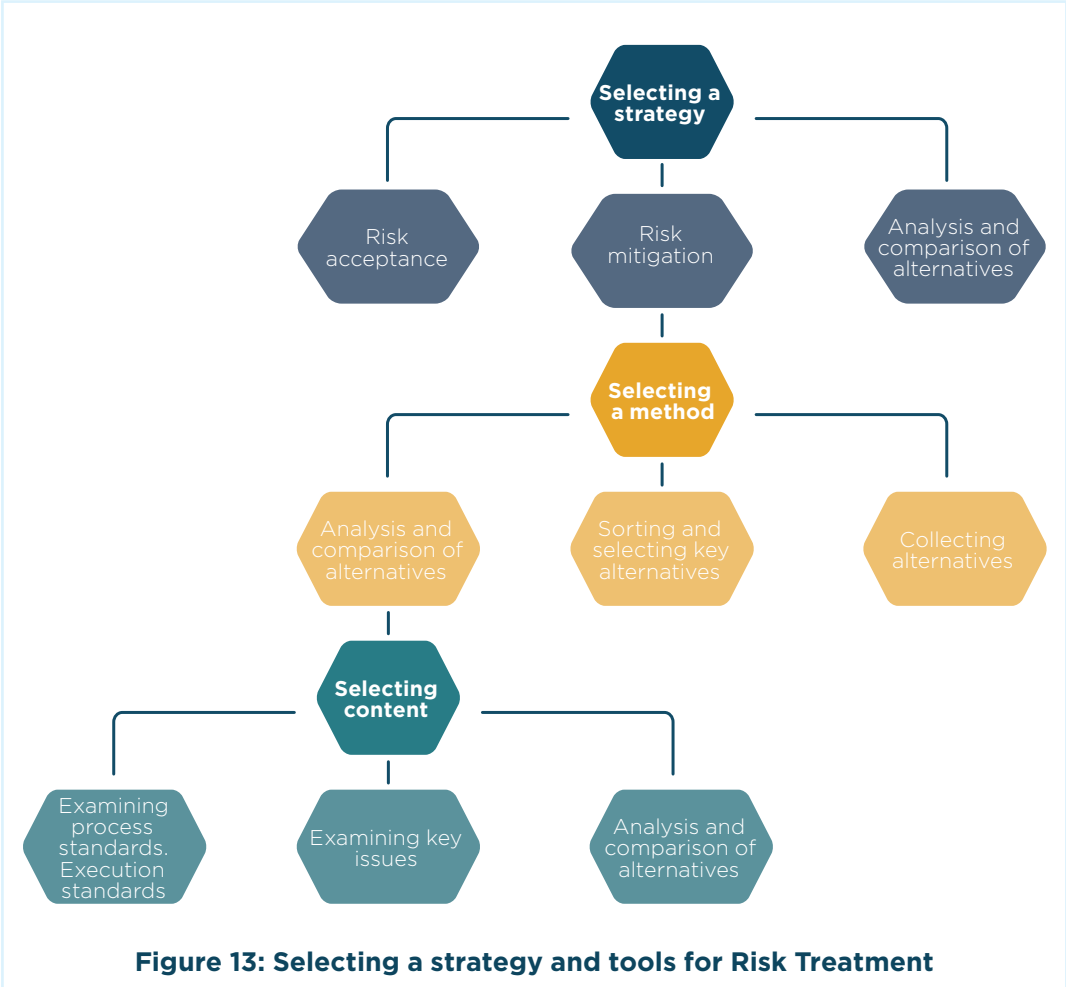


Figure 13: Selecting a strategy and tools for Risk Treatment

Selecting a strategy

As detailed in the guide's introduction, there are three main strategies for risk management. At this stage it is necessary to decide which strategy will be implemented in relation to each risk.

Risk acceptance

- This strategy is often implemented as public resources are limited, and government intervention has a price. All risks cannot be addressed and there is no public or economic rationale for government intervention in relation to many of the potential risks.
- Acceptance of risks is a very common possibility – most risks that may occur are not treated by governments. The government has no interest in dealing with trivial matters or risks that society can solve on its own. In many cases, government intervention is not an efficient or effective means of coping with the risk. After completing the Risk Assessment, it sometimes becomes apparent that some risks are not as dangerous as they first appeared.
- Risk acceptance means the

government does not take steps to mitigate in advance the probability of occurrence or severity of the risk. The decision to accept a risk means that from a government perspective, the risk can be tolerated, or the public can cope with it on its own – and it is better to devote public resources to treating other risks. In addition, it is possible to allocate monies to a compensation fund for the treatment and recompense of casualties if/when the risk is realized.

- This strategy is suitable for hazards with a low level of severity on a national scale (an annual damage of ILS 100,000 may sound like a high sum to a private individual, but from the state's perspective this is not a significant amount). It is also a strategy for consideration when dealing with severe but rare risks. When a serious risk rarely occurs – it means that in the overwhelming majority of cases, tools for coping with the risk impose a social price without any benefit. This is the case of rare but severe events such as violent earthquakes. There is value in preparing for earthquakes that occur frequently (mainly

through building standards), but there is no justification for building all the structures in the State of Israel to withstand an earthquake measuring 9 on the Richter Scale (which occurs worldwide once every 10-50 years). The reason for this is twofold: since this is a serious incident, many resources will be required to deal with it; and since this is a rare event, there is little risk of benefit from the same resources.

- Since there is no justification for trying to deal with those risks in advance, it is preferable to implement a risk acceptance strategy. In the area of private risk management, these events are not treated, or insurance tools are used. For example, the premium to the insurance company is significantly lower than the cost of coping in advance with such severe events, that have a low chance of occurrence. Sometimes the state can also use apparatuses used by the insurance industry, as part of its risk acceptance strategy. Even when the government chooses not to proactively take action in order to mitigate the severity or probability of a risk, it may prepare tools to deal with the risk if it occurs. This

field is called crisis management, and it includes handling the risks in retrospect and providing a response to restore the situation to normalcy. Crisis management tools can be integrated with each of the strategies. This issue is explained in the next chapter.

Risk avoidance

This is an extreme strategy aimed at completely avoiding the risk. Since we are dealing with public risks arising from social activities – this strategy involves imposing a sweeping prohibition on a certain action or activity, in order to completely avoid the risks involved. This strategy includes the imposition of a total prohibition on a given activity, a significant reduction of that activity, and a determination that only the state or its institutions will be permitted to engage in the activity.

Illustrative example: Risk of losing money as a result of investing in the stock exchange. If a private individual wishes to avoid risk, they can decide not to invest in stocks. But if the government wishes to protect the public from the same risk with a risk avoidance strategy, it must implement a sweeping ban on investing in stocks

– closing the capital market. The government deals with the interests and actions of society as a whole, and accordingly, with the systemic risks that society creates and is exposed to. Therefore, in order to prevent all of society from being exposed to the risk of loss in stock investment – the state will have to prohibit investment in shares. The decision to prevent the entire society from carrying out any activity has far-reaching effects, much more than a private decision. In addition, imposing a ban on an action may create a black market and attract criminal activity into the field. Unlike private risk management, in public risk management, a risk avoidance strategy is very rare.

As detailed in the introduction chapter, this strategy is extreme and impractical. In the real world, this is almost a theoretical strategy, as absolute avoidance of a certain type of risk requires the imposition of sweeping prohibitions on the public. It also demands supervision and enforcement to ensure the activities are not carried out (since any deviation from this will actually turn the strategy from avoidance to risk mitigation). It is important to understand that

choosing this strategy means that even mitigating the risk will be defined as failure.

This strategy has four main drawbacks. The first is that this strategy strives for a goal of "zero risk", which is impractical and therefore the goal of this strategy is never fully achieved. The second drawback is that imposing a ban on a certain activity makes it illegal and accordingly, hidden from the eyes of the state. The activity then "goes underground" and a black market is created. Once the ban is imposed, the state will not be able to effectively monitor activity that will now be conducted illegally. As such, procedures cannot be implemented to make the activity safer. Therefore, the decision maker's risk management ability will gradually grow weaker, and as time passes, the less data there will be on the risk and related activities. Making an activity illegal also creates a black market and invites criminal elements to engage in the illicit activity and profit from it. The third drawback of the avoidance strategy is the significant costs to the public. It is a blatant intervention in the economy and a ban on activity prevents the public from carrying out the activity even when it is beneficial and

necessary. Such a ban also impedes social, economic and technological development, leaving the society and the economy behind. The fourth drawback of the avoidance strategy is that many government resources are required for the supervision and enforcement of a total prohibition. In addition, due to the aforementioned information limitations – the government will be required to allocate resources for intelligence and investigations.

Because of these shortcomings, the use of the avoidance strategy should be extremely limited and used only in risks characterized by great severity (at a national level) and high probability of occurrence. A selection of this strategy can only be made after a thorough examination and consideration of the costs to the public and the actual risk that this strategy will encourage criminal activity.

Risk mitigation

- This is the main strategy for dealing with risks. This will be discussed later in this chapter. A risk mitigation strategy is found between risk acceptance and risk avoidance. This strategy is designed to mitigate the impact of risk on assets in the most effective manner but will not

prevent the risk in its entirety.

- There are various tools for dealing with risks in order to mitigate them. A risk mitigation strategy can be directed at both components of each risk. One component is to reduce the probability that the risk will occur. The second is to reduce the severity of the damage when it occurs. When examining policy tools and regulatory methods it is worthwhile to consider; which component will be affected, if there is a way to influence the other component, and which will be the most effective.

Strategy selection

Strategy selection is based on the risk analysis.

Each of the risks (or risk groups consolidated at the Risk Assessment stage) should be examined and the appropriate strategy for that risk determined. Risks that were ranked low (both due to severity and probability) will be examined first as candidates for a risk acceptance strategy. In rare cases, a decision will be made to adopt a strategy of risk avoidance.

Generally, most risks with significant

impact will be classified under the risk mitigation strategy. The treatment of these risks should be carried out in the following stages: selecting the method and choosing the content. At these stages, a cost-benefit analysis should be executed, based on the principle of maximum aggregate welfare. This analysis challenges the various alternatives to the treatment of the risk. In some cases, the analysis of the various alternatives may indicate that there is no appropriate regulatory alternative and that it is preferable to adopt a risk acceptance strategy.



Examination of regulatory methods and policy tools

Key groups of regulatory methods and policy tools

- The regulatory methods and the policy tools can be divided into three main groups. Each group has its own typical advantages and disadvantages. For further details on the various regulatory methods and policy tools, see Annex A to the Government's Guide to Regulatory Impact Assessment (RIA Guide): <http://regulation.gov.il/RIAGUIDE>.
- Permits and licenses: This group includes rules requiring approval from the state prior to carrying out the supervised action. Permits and licenses allow the state to ensure compliance with the standard that has been determined prior to commencing activity. This ensures anyone who enters the market or is about to carry out an action possesses the skills, knowledge and equipment required to do so. It is important to emphasize that permits and licenses are a preliminary bureaucratic mechanism that applies to the entire population. It does not constitute supervision designed to encourage retroactive compliance.

Examples include: a license to open a business, a building permit and permit to carry out an import operation.

- Binding rules: This group includes binding and enforceable rules and penalties. This is a diverse set of rules and policy tools. The rules can prohibit the execution of an action, the duty to perform an action or conditions for carrying out an action. Everyone is entitled to operate and enter the market, provided they comply with the rules. There is no preliminary inspection and no permit and license is required from the state. Supervision and enforcement are carried out routinely and in retrospect. Examples: a binding engineering standard for a product, a ban on dumping waste on the street, a ban on smoking in restaurants, a public company's obligation to publish information to the investing public.
- Advanced regulation and nonregulatory tools: This group includes tools designed to change behavior without imposing a duty and without imposing compliance. The tools in this method are designed to influence the incentive

system of the various players, to improve the flow of information or to grant civil rights enabling parties to exercise their rights effectively, without the active involvement of the state. The main advantage of these rules is the lack of coercive directives that the public is required to comply with. Therefore, these rules do not require a supervision and enforcement system. The method includes providing information about the risks and ways in which the public can cope with them. This is done by using voluntary (non-binding) standards and solutions based on a market economy. Examples include: government subsidization (direct or via tax benefits) for the purchase of cars emitting little greenhouse gas, granting a right to file a class action, using a behavioral economy to shape a choice and disseminating information to the public about the risks and coping mechanisms.

Consideration of the cost of methods and policy tools – ensuring effective policy

- Since risk management is based on a cost-benefit analysis, methods and policy tools that maximize the benefit and reduce the cost to the

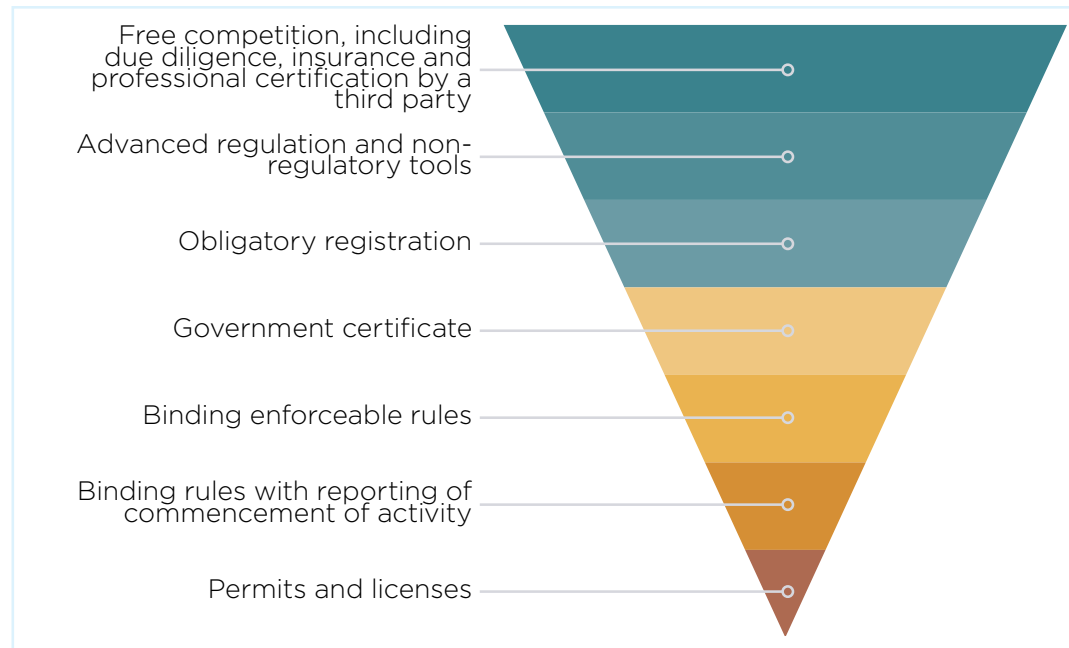


Figure 14: The Inverse Pyramid – Regulatory Methods and Policy Tools by Degree of Government Involvement

government and public should be chosen. In-depth government involvement requires dedicating government resources, as well as imposing costs and restrictions on the public. Therefore, in order for the risk treatment method to be cost-effective, it is generally preferable to examine methods that minimize government involvement.

- Figure 14 lists groups of methods and policy tools according to the degree of government involvement. The figure is hierarchically structured. Methods and policy tools with the lowest government involvement stand at the top of the figure and methods and policy tools involving the greatest degree of government involvement at the bottom.
- In many cases, the risk can be successfully addressed by means of little government involvement at a low cost. The inverse pyramid figure enables the government decision maker to first examine the methods with the lowest public cost and gradually move to examine more burdensome methods. Annex F lists the various methods presented in the inverse pyramid figure.

Matching the methods and policy tools with identified risks:

- Matching the methods and policy tools to the risks is based on an understanding of the causal mechanism of each risk. The data is mapped at the risk evaluation stage. This includes; the conditions necessary for the realization of the risk, the identity of the risk creator, the identity of those affected, the time and place in which the risk is realized, the normative cause of the risk and other characteristics detailed in the risk evaluation chapter. For example, a risk that occurs due to the introduction of unskilled players into the market requires a solution at an early point in time, while a risk that occurs after a while – requires a different solution. Guidelines for this purpose are set forth in Annex G.
- Sometimes, a basic analysis of the risk matrix allows the identification of key methods suitable for the treatment of risks. The various corners of the matrix describe different types of risks. Accordingly, there are varying strategies and methods for dealing with the different types.

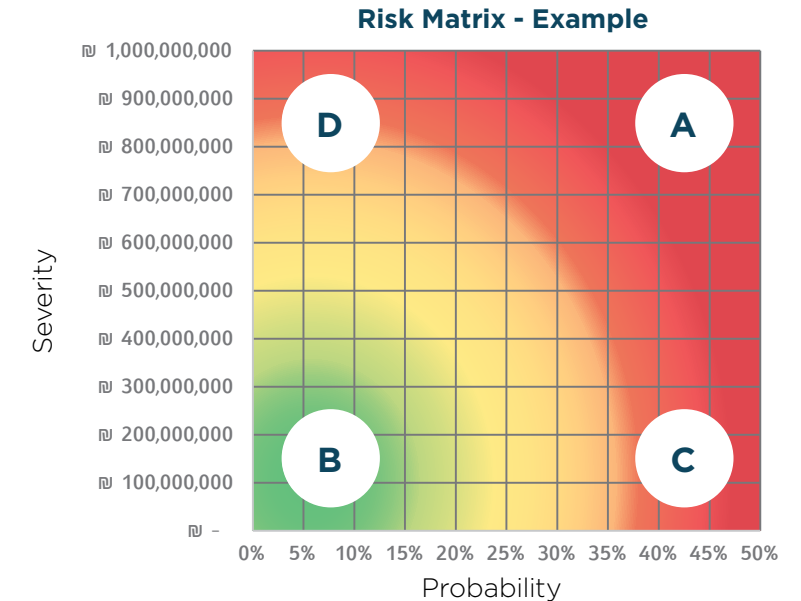


Figure 15: Mapping of distinct areas in the risk matrix

High severity/high probability (A – upper right corner): These risks are the risk manager's top priorities. It is usually necessary to deal with them through binding rules or permits and licenses. A system of supervision and enforcement is mandatory.

Low severity/low probability (B – lower left corner): These are minor risks, which should be at the bottom of the risk manager's priorities. A risk acceptance or a risk mitigation strategy may be selected – with little intervention. Usually there are more

significant risks that require treatment and therefore the risk manager may elect not to deal with minor risks. With regard to these risks, one must ask whether government intervention is required at all.

Low severity/high probability (C – lower right corner): These are "routine" risks that occur frequently but cause very little damage. Since these are common events they justify intervention but because their severity is minimal, they justify minor intervention (and the dedication of little resources). It is preferable to first examine the use of advanced regulation and non-regulatory tools, as well as specific use of binding rules that involve minimal costs.

High severity/low probability (D – upper left corner): These risks are the mirror image of type C risks: These are rare events with high impact, such as rare, natural disasters. The probability of these risks occurring is very low, and most likely not within the foreseeable future. As such, there is no justification for taking measures that would require the allocation of extensive resources (governmental or public), as they may be invested in preventing a risk that has a significant likelihood of not occurring. Due to the low probability

of occurrence, there is no justification for dedicating regular and routine resources in order to pro-actively deal with the risk. However, their severity requires a response. The preferred approach to treating these risks is through insurance methods. Insurance has costs, but they are significantly lower than the costs involved in mitigating in advance the probability and severity of the damage from the risk. In the framework of public risk management, insurance methods include three main applications: crisis management (retroactive treatment), imposing an obligation to purchase an insurance policy against the damage (through binding rules) and subsidizing an insurance policy against damage (non-regulatory tool). The three insurance applications allow mitigation of the harm caused by the occurrence of the damage, without dedicating resources for dealing with the risk in advance.

Gathering a wide range of alternatives to regulatory methods and policy tools

- In order to develop alternatives, all the options must be reviewed. Such a review makes it possible to identify solutions that go beyond the obvious - "outside the box"

type considerations. When the decision maker has a variety of possible policy tools, they may decide to combine some of them. In the discussions on the proposed policy (vis-à-vis entities inside and outside the government), the decision maker will be asked to explain why other alternatives were not selected. They must be able to justify their choice of alternative – providing a quantified and robust answer regarding their decision.

- Searching and reviewing alternatives to regulatory methods and policy tools is similar to risk identification: it is carried out without critical thinking, so as not to rule out any possibility.
- The natural tendency of professionals is to focus on problems and solutions from the fields / industries they are familiar with. The discussion of risk and policy tools may be focused on the content world and the solutions the decision maker knows. It is important to expand the review and discuss additional options as well.
- The guiding questions are: what can affect the causes of the risk

and what can affect the behavior associated with the risk. Annex G provides guidelines for matching regulatory methods according to risk characteristics. These rules help identify methods that are relevant to the risks involved. It is recommended these rules be used when examining regulatory methods and policy tools.

- Choosing the means for the treatment of a risk is akin to finding a solution to a problem. Before choosing one tool or another, one must fully understand the risk and its components which we wish to influence. After the Risk Assessment, the decision maker will have extensive information about the risk. Information such as: who causes the risk, who is affected by the risk, when is the risk realized, what stages lead to the realization of the risk, what are the conditions necessary for the realization of the risk and what is its normative cause. Understanding the failure that causes risk is key to developing a policy that addresses the risk in a focused, effective and efficient manner.

Illustrative example: Risk of electric shock from an electrical outlet in residential apartments.

The treatment of the risk will vary depending on the source of the problem. An important distinction is between a risk resulting from a technical fault and risk arising from human behavior.

If the cause of the risk is poor quality outlets (technical faults), or outlets that are unsuitable for the local electrical current, standards for the manufacture and import of electrical outlets should be established.

The treatment of each risk will be different. The first risk concerns the manner of use of the outlet. To prevent electric shock as a result of children playing, access to the outlets should be restricted. It can be done in a variety of ways – coercive and non-coercive. The second risk relates to the skill level of the professional who installed the outlet.

Regulation and policy tools will vary from case to case, be it technical faults, playing with the outlet, and an unreliable professional. Selecting a tool that does not suit the risk will not mitigate the risk.



- The review must include a wide range of possible tools. These may include; "hard" regulatory methods, "soft" solutions based on market mechanisms (advanced regulation, economic incentives, etc.), methods used in the field, methods that have not been used in the past, and innovative tools such as behavioral economics. For further details on the various regulatory methods and policy tools, see Annex A to the Government's Guide to Regulatory Impact Assessment (RIA Guide): <http://regulation.gov.il/RIAGUIDE>.

When examining the methods and policy tools, it is also important to examine tools available to other units in the government in order to expand the range of possibilities to their areas of specialization as well.

Illustrative example: The Ministry of Health is coping with the risk of a low level of service in the HMOs. It was found that the essential condition for the risk is the lack of competition between the HMOs. The Ministry of Health can use the usual tools at its disposal as a regulator and publish instructions regarding the manner of managing the HMOs. Another alternative is to take measures to encourage competition in order to solve the problem in a targeted manner. In order to develop and implement policy tools to encourage competition, the Ministry of Health will cooperate with the Antitrust Authority.

- It is also recommended to examine a change in the existing policy: for example, to reduce the level of intervention or streamline the existing mechanism (whether it is possible to treat risks in an easier, more efficient manner, that is cheaper for the public, or to improve the existing supervisory and enforcement mechanisms). Sometimes the most appropriate tool for dealing with risk requires a change in the existing mechanism.

Illustrative example: The Supervisor of Banks recognizes that one bank holds a 95% market share in the banking market and is concerned that this may endanger the stability of the financial system and harm the banks' customers. The regulator decides to examine the possibility of imposing restrictions to ensure the stability of the banks (and especially the stability of the large bank), as well as examining the possibility of imposing price controls on services provided to customers. In addition, the regulator examines existing rules set (for example, the duty to obtain a license to operate a bank and the requirements therein to be met). These rules may have created the current market structure in which the one player has an advantage making them the dominant force. It is possible the examination will reveal that the solution to the risk is not new regulation, but a change in existing rules.

- It is also recommended to examine the adoption of international methods and standards. These alternatives make it possible to build on accumulated experience and ensure international compatibility.
- All regulatory methods and policy

tools should be directly related to the risk in question. It is important to ensure a clear logical connection between the nature of the intervention and the mechanism of occurrence of the risk.

Illustrative example: When dealing with a risk caused by too few players in the industry – the solution should be related to increasing the number of players in the field. The relevant regulatory methods and policy tools will relate to the removal of entry barriers or to the solution of the phenomenon that reduces the number of participants in the field. In such a situation, the use of the licenses tool will not help. A license is designed to ensure the level of competence of those who enter the market and not increase the number of players in it. In addition, a license constitutes a regulatory entry barrier to the market and will therefore only exacerbate the risk.

Sorting and selecting key alternatives

The criteria for selecting key alternatives are:

- Technical feasibility to implement the method or policy, including the possibility to supervise and

enforce. Is the method applicable and feasible? In examining the feasibility, the OECD guiding principle regarding policy-making can be applied: regulation should not be adopted if it cannot be applied and enforced consistently, transparently and proportionately;

- Proportionality between the problem and the solution. Does the risk justify the proposed method and the scope of the intervention? This criterion may also include an initial cost-benefit analysis of each of the alternatives;
- The influence of the regulatory method or policy tools on cross-sectoral interests: competition, small business, social issues, and environmental issues. Annex H contains supporting rules for identifying the risk of harm to competition. Annex I contains focused questions to identify concerns regarding harm to small businesses.
- Coherence with existing policy and decisions.
- Legal and political feasibility for the adoption of the method or policy.

After the sorting, 3 - 5 key alternatives

to the method should be chosen. These alternatives must be defined so that cost-benefit analysis can be carried out. In some cases, there won't be 3 - 5 alternatives, for example there may be a fundamental question with a yes/no binary answer.

Illustrative example: Bitcoin is a decentralized system for transferring value between two entities. Unlike traditional currencies, Bitcoin is not controlled by a central bank of a country but is based on computerized encryption mechanisms. When the Bank of Israel examines its position towards Bitcoin, a variety of policy questions may arise regarding the risks involved in the use of Bitcoin. One of the basic questions to be decided is what is the regulatory method and the general policy to be applied in respect of Bitcoin? There are two possible methods of defining and regulating financial assets: as currencies and as commodities. Therefore, at the stage of selecting the method, it will not be possible to examine five different alternatives.

At this stage, the key alternatives should also be analyzed in-depth to determine whether they provide a response to the risk or to the group of

risks being treated. The fact that there is widespread use of any rule or policy is not sufficient. The effectiveness of the proposed method should be based on data or through an analysis of the effect of the method on the causes of the risk or on the process of the risk occurring.

It is important that all key alternatives that undergo a cost-benefit analysis will be real alternatives and implementation is a viable option. There should be no futile alternatives ("scarecrows").

Cost-benefit analysis of alternatives and comparison between them

In order to choose the preferred regulatory method or policy tools, a cost-benefit analysis should be performed for each one. The benefit of each alternative is reflected in the reduction of the damage from the risk in one year. The cost of each alternative is the amount of resources it requires from the government and public. With regard to each alternative, the damage from the risk, the cost to the government and the cost to the public must be quantified. The alternative in which the total costs (the total social price) is the lowest is then chosen. In this manner, the government decision

maker minimizes the social loss and thereby promotes the principle of maximum aggregate welfare. The analysis is carried out using the following table:

	The expected annual damage from the risk	The cost of coping methods to the government	The cost of coping methods to the public	Total costs
Current situation				
Alternative A				
Alternative B				
Alternative C				

A comparison of alternatives must always include the current situation. The damage from the risk is known based on the risk analysis performed (taking into account historical data or on the expected probability of the risk being realized in a year and the severity of the risk's impact), as well as data on the amount of resources invested by the government in dealing with the risk (if any).

As with the Risk Assessment stage, cost-benefit analysis is based on data. At this stage, the quantification of the benefits and costs is done from an overview, as the individual content of the alternative has not yet been determined. Therefore, there is no need to strive for accurate quantification of each component, and an assessment is sufficient. However, it is important the assessment is not too crude, as a comparison of the alternatives must also be carried out.

Cost-benefit analysis is also a tool for budgetary management - it helps identify the extent of government resources required and whether they are justified in terms of the economic benefit of the policy.

Assessing the impact on the damage from the risk:

- The annual damage from the risk is composed of the two main variables of each risk: probability and severity. It is necessary to examine how the examined alternative is likely to affect the severity or probability of the risks' occurrence.
- Usually the severity of the damage is a professional variable. It is therefore advisable to consult with professionals from the field (supervised entities, government agencies, business organizations, researchers and lecturers, international organizations). The key question is - how does the rule reduce the severity of the damage?

Illustrative example: The obligation of wearing a helmet at construction sites is intended to mitigate the severity of the damage arising from risks in the field of work safety. A rule that requires wearing a helmet completely prevents damage to the head and reduces injury from additional risks. The assessment of the impact on the severity of the risk will be based on professional data on the protection that helmets provide in the event of a work accident.

- The probability of occurrence of

the risk is a more statistical variable, requiring assessment and working with quantitative data. Similar to the risk analysis stage, assessment of probability is based on two components: the number of risks in the year expected to occur and the volume of the relevant activity. Professionals (from the private sector, public sector, and from academia) can help formulate this assessment, as well as international organizations. Estimating the impact on probability depends on the type of change the method is designed to achieve. If the regulatory system reduces the probability of the risk occurring at night, and the risk only occurs at night –the potential impact of the method will be high. Even when dealing with risk that occurs at all hours of the day equally – the potential for impact is at most 50%. When it comes to processing risk, it is possible to assess what stage the regulatory process will affect and to what extent.

- Impact on the volume of activity associated with the risk creation. Government intervention may also

affect the scope of activity – increase or decrease it. The imposition of restrictions, obligations and costs will usually reduce the scope of activity. The distribution of benefits, training and the reduction of entry barriers will generally increase the scope of activity. This should be taken into account as a change in the volume of activity also affects the probability of occurrence of the risk (this figure is also included in the quantification of costs to the public. It is important to use it consistently).

- Consideration of the level of effectiveness and compliance. Every instruction or rule has a limited level of effectiveness – 100% compliance with the state's instructions is rare. It must be taken into account that the public will not comply with regulation and that policy tools will not be implemented in all cases – as such, their impact will be partial. It should never be assumed that the compliance rate will be 100%. As a rule of thumb, a compliance rate of 80% is a realistic estimate. It is recommended to determine the expected rate of

compliance in consideration of the relevant population (based on past data) and also according to the content of the regulation or the policy tools to be implemented.

- The change in the severity of the damage or the probability of its occurrence must be reduced, in accordance with the expected rate of compliance. For example, if helmet use is expected to reduce the severity of the damage by 60 (from 100 to 40) and the compliance rate is 80%, this means the reduction will only be 48 (from 100 to 52).
- After weighing the level of effectiveness of the regulatory method or the policy tools, the annual damage expectancy from the risk is calculated (μ):

$$D \cdot \frac{E}{A} = \mu$$

D (Damage) – the damage caused by a single occurrence of the risk

E (Events) – the number of events of the realization of the risk per year

A (Activity) – the volume of activity relevant to the year

μ – the annual damage expected from the risk

Creation or increase of other risks (risk trade-off). Regulatory intervention may cause a change in economic activity that increases or creates a different risk. If the alternative involves increasing other risks, they should be taken into account and the change in their annual damage must be summed up.

Illustrative example: In order to reduce pedestrians being injured by cyclists, cyclists are not allowed to ride on sidewalks. Although this ban significantly reduces the injuries to

pedestrians, it also creates a new risk. The ban removes the cyclists from the sidewalks to the roads. As a result, there are more traffic accidents between bicycles and vehicles on the road. Along with reducing the risks to pedestrians, the increase in risk to cyclists should also be taken into account.

Example: after the terrorist attacks on September 11th, 2001, the US government tightened the security check for passengers on commercial flights. The stricter inspection had direct costs: the additional manpower was financed by an increase in airline ticket prices and the waiting time before take-off was significantly prolonged. There was an unexpected result: because of the increased cost of tickets and delays at the airport, a percentage of travelers chose to travel by car instead of by plane (especially on relatively short routes like New York – Boston). The result was the shifting of travelers from air transport to land transport. Statistically a plane is a safer vehicle than a private vehicle.

The choice of many people to drive instead of flying created congestion on the roads and increased the number of fatalities and injuries in road accidents. Part of the price of tighter security checks on flights was increasing the number of traffic fatalities. The quantification of damage from the risk can also include other social benefits, along with the mitigation of other risks, provided that they are under the explicit authority of the decision maker. No external influences that are not at the core of the decision maker's authority may be included, so as not to distort the decision-making process by applying external considerations. These considerations may be taken into account as part of the following additional considerations.

Evaluation of the impact on government costs:

- Regulation or new government policy is another activity the government must carry out. Whether it is the establishment of a new mechanism (for example, the distribution of subsidies or the issuance of licenses), or involves knowledge management

(gathering reports or publishing information to the public), or supervision and enforcement. The scope of government resources involved in implementing the regulatory method or policy tools should be quantified.

- The main cost components of the government budget are: personnel (jobs), procurement budgets, IT budgets, and other related budgets. It is recommended to base the quantification of these costs on international experience (if any) or previous experience of government officials (for example, if a parallel office has set up a system for supervising activities of a similar nature). It is also recommended to evaluate the resources required together with the budget and administration personnel of the ministry or government authority.

Evaluation of impact on costs to the public:

- The cost to the public consists of a number of key components.
- Compliance costs must be quantified, these include acquiring equipment and supplies, as well as bureaucratic processes and mechanisms. In order to quantify compliance costs, all the duties that

will be imposed on the public (filling out forms, acquiring equipment, employing workers) and the cost components required (dedicating work time, paying employees and experts, slowing down the work rate) must be mapped. In addition, the waiting times for government approvals must be taken into account. When carrying out this mapping, a regulatory policy expert should assist. The Regulatory Burden Reduction Guide and the Regulatory Impact Assessment Guide published by the Prime Minister's Office are sources that should be referred to.

- The impact on business and social activity should be quantified. Government intervention may harm or reduce existing activity. Regulatory compliance costs may create entry barriers into a specific market or inhibit beneficial social activity. Social losses must be taken into account in the costs imposed on the public by each alternative.

After completing the table for the current situation and for each of the alternatives, the principle of maximum aggregate welfare can easily be compared. At this stage, the goal is to choose the alternative whose total social costs are the lowest. In light of

the comparison, a decision to change the alternative may occur.

Quantitative analysis is not the most important element. It is possible and desirable to take into account additional considerations and, if necessary, choose an alternative with higher social costs – if the additional considerations justify it. It is recommended to examine these considerations together with the ministries responsible for these issues. Below is an open list of additional considerations and effects.

- Social effects – the impact on weak social groups, education and employment.
- Human Rights – impact on basic rights such as the right to vote, to be elected, freedom of movement and the right to equality.
- Impact on competition – creating entry barriers and facilitating coordination among competitors. Annex H contains rules for identifying the risk of harm to competition.
- Impact on small businesses – small and new businesses are particularly vulnerable to the impact of government policy. Annex I contains focused questions for identifying concerns pertaining to

potential harm to small businesses.

- Environmental impacts. Although environmental impacts are often taken into consideration, sometimes it is advisable to consider them in a qualitative manner. Such impacts include harm to environmental resources (soil, water, air), harm to biological diversity and the creation of nuisances (noise, odor, heat, radiation and dispersion of materials).
- Dispersal of damage and dispersal of costs. Small scale damage can cause serious harm if the harm is concentrated on a small number of people. Similarly, the dispersion of costs associated with coping with the risk is also important. For example, a small group creates risk, but the entire public bears the costs of regulatory compliance designed to deal with the risk. In this case, a large group of citizens is forced to bear the costs of coping with a risk that is caused by a small group or a risk that only harms a small group.
- International considerations. Sometimes regulation and public policy affect the international obligations of the state. Government policy has a direct link to the international arena when dealing with import and export activities,

cross border transactions, and cross border human traffic. In such cases, one must ensure that policy tools do not conflict with international treaties or individual trade agreements.

On the basis of the principle of maximum aggregate welfare (the quantitative comparison) and using the additional considerations, an alternative should be chosen for a regulatory method or a policy tool for dealing with the risk. The exact content of this alternative will be defined in the next section.

After selecting the regulatory methods and policy tools, they should be presented to management for internal discussions. At this stage, it is recommended to hold a public discourse on how to implement the method and to understand the main issues in determining content.

The package of methods chosen, prioritization and allocation of resources must be reviewed. It is not enough to select the method that provides optimal response in terms of achieving the maximum aggregate welfare. The package of methods chosen should be examined to ensure they can be applied. After comparing the alternatives to the method and choosing the appropriate method for

treating and mitigating each risk, the option of executing all of them should be considered. Usually the decision maker will be required to prioritize the various methods. At this stage, each of the selected methods must be examined. Questions need to be asked and answered including; the extent to which each method is applicable, what its chances of success are and what resources are needed to implement each method for risks mitigation (budget, personnel, supervision and enforcement, legal jurisdiction and more). It is therefore necessary to examine the total resources of the government unit and to prioritize the chosen methods.

Examination of the content of the regulatory method and policy tools

After selecting the regulatory method and the policy tools, their detailed content must be determined. This chapter presents various questions designed to assist the decision maker in identifying the key issues and defining the various components of the regulation or policy they are developing.

General questions that are relevant to most cases will be presented. Following this, additional questions should be presented according to the

group to which the chosen method belongs (permits and licenses, binding rules or advanced regulation and non-regulatory tools). Each question expresses a content issue, for example – how often the license should be renewed. A number of possible content alternatives should be posed and compared. The comparison is both quantitative (cost-benefit calculation, according to the principle of maximum aggregate welfare) and qualitatively, in view of the range of considerations presented in the previous chapter. The scope of the examination depends on the extent to which each issue is material and significant. The more complex and influential the issue, the greater the comparison will be required for the formulation of the content.

While formulating the content, professional issues that are not necessarily within the purview of the decision maker will arise. It is important to hold a discourse with stakeholders and professionals in order to gather information and clarify the meaning of the various rules.

Illustrative example: The Ministry of Health wishes to establish a set of rules to ensure medication is kept at the permitted temperature during transport. It was decided to define which types of containers would be allowed to transport medication, in order to ensure uniform and continuous cooling. The Ministry of Health has no expertise in refrigeration equipment or the technical specifications of various containers. In order to determine the rules, a knowledge base must be established with the help of stakeholders and professionals in these fields.

General questions for developing the content of the regulation or policy

Does the proposed content contradict or overlap with other regulations or policies?

- To avoid contradictions and repetition, a check must be done to ensure the various content alternatives are not identical or contradictory to existing regulations or policies. As such, the government's involvement in the same area (by type of assets, type of risks and regulated activity)

must be examined.

- Such an examination also helps identify existing frameworks for anchoring the proposed regulation or policy. For example, it is possible that another department in the ministry regulates the same activity and has the tools to treat the risk involved.
- A "holistic vision" of government involvement in the relevant field will also assist in planning the supervision and enforcement procedures. If, for example, an entity already performing supervisory actions on the activity is identified, a decision may be taken granting that entity additional powers or tasks, as opposed to establishing a new supervisory unit.

Who will be subject to the regulation or policy?

- Defining the players who will be directly affected by regulation or policy is a key issue. In principle, the answer to this question is easy – those whose behavior one wishes to change in order to deal with the risks or potential risks.

- It is important to accurately define who is included in this definition – too broad an application will affect groups that are irrelevant, and a narrow application will impair the effectiveness of regulation or policy. A good definition will reduce the possibility of various players deliberately removing themselves from the regulation.

Example: The basic definition in the Income Tax Ordinance is that only individuals (or companies) must pay taxes to the Israeli tax authorities if they are residents of Israel.

Tax planning is designed to enable taxpayers to evade tax obligations. One such prominent method in tax planning is to establish a company in a tax haven (a foreign country where there is a preferred tax regime), while continuing business activity in the country of origin. In this case, the taxpayer evades the application of Israel's tax laws. In order to deal with this phenomenon, the Income Tax Ordinance includes the "Control and Management Test". This determines whether the control and management of the business of a foreign company is

executed from Israel. If it is found so, the company is deemed a resident of Israel and accordingly is required to pay taxes in Israel.

At this stage, it is also important to set exceptions to the definition – groups which the regulation or policy need not apply to, or groups where the application is partial. For example, it can be determined that burdensome regulatory directives need not apply to small businesses (see Annex I). Similarly, it is possible to determine in advance different tracks or different steps of application as part of the regulation or policy.

The decision on the application of the provisions also affects the content of the regulation or the policy to be determined. If the provisions are intended to apply to foreign corporations (for example: foreign investors) they must be written in English and in compliance with the regulations and policies of foreign countries. If it applies to individuals such as consumers, it should be very simple and clear.

In addition, it is advisable to define activities where the level of risk to the

public is low and can be exempt from policy or regulation. For example, it is customary to exclude personal imports for private use from the system of supervision over the import of goods. The reasoning was created in the risk assessment stage: personal import for private use is limited, as such the severity of the damage is limited to the importing person and their immediate surroundings. In addition, if it's limited to a person and their immediate person circle, there are no externalization problems.

In addition to the supervised entities or those directly affected, it is important to identify who will be indirectly affected by policy regulation. This must be done to take into account the effects on them or reduce residual negative effects on that group.

Impact on work processes and business processes – Use of process standards or performance standards?

The regulation or policy is designed to influence behavior ultimately mitigating the probability or severity of the risk. It is therefore necessary to examine how one wishes to change

the work processes and business activity.

There are two main ways to influence behavior: performance and design standards.⁽⁴⁾

Performance standards determine the required result. For example, the maximum level of bacteria in food or the maximum amount of time required to evacuate a building in an emergency; They do not define how the desired result will be achieved. In contrast, design standards determine how to act – what technology to use, what are the work processes, how to build a structure and what actions are to be taken in a business.

Design standards do not deal with the outcome and can be met (by performing the necessary processes or using the required equipment) without reaching the intended outcome of the regulation.

Performance standards define goals and objectives that must be met. They are usually based on measuring performance in achieving a desired outcome. The performance standard measures compliance with the

regulatory objective (a safe product is manufactured, tap water is clean, etc.). The manner of compliance with performance standards is flexible. Performance standards require meeting targets and do not define a binding way to do so. The strictness level of performance standards is based on a balance between the objective of risk mitigation and the practical and economic possibility of meeting the outcome. The objective of the performance standard is not to prevent damage, but an end characteristic and measurement of a process, product, or service.

Design standards require the use of a defined process, personnel, equipment or technology. To address risk, design standards require the public to comply with specific directives and do not require it to meet the desired results. The process approach of design standards is less flexible as it requires defined operations to be carried out in a specific manner. It is possible to add flexibility to design standards if the decision maker defines a wide range of permissible methods and technologies, but most design

standards include a single binding technical/process option.

Illustrative example: Fire safety

A design standard determines the required equipment (fire extinguisher on each floor) and the engineering content of the building (location of several emergency staircases, permitted and forbidden construction materials, window size, etc.).

The performance standard requires a design enabling all occupants to be evacuated within two minutes. It will allow the engineer to design the building in a way that meets the performance standard (location of the emergency staircase, type of fire equipment, etc.) in a manner that can be proved.

The main limitation of performance standards is that they require a measurable goal and compliance with the goal can be determined. For example, when wishing to determine a performance standard for being free of disease x, there must be a reliable and available test for pathogens of disease x.

The selection of design standards has three drawbacks that must be considered:

- First, the use of design standards

creates a high degree of government involvement in the content of business processes. This entails extensive involvement and the imposition of costs on the supervised entities. As a result, changing or updating the requirements imposes significant costs on the public due to factors such as replacing equipment or changing the work method (it is easier to adapt to changes in performance standards). Design standards require many resources from the government regulator: addressing numerous details, updating requirements and, of course, extensive supervision of each of the requirements.

- Second, design standards suppress innovation and prevent preparation for new problems. As the public and businesses are committed to the process defined, design standards do not enable them to adopt new solutions and technologies to cope with any unforeseen risks. Therefore, design standards prevent operational streamlining and improvement of safety levels. Performance standards enable the supervised sector to find a solution that encourages competition and innovation. In the world of performance standards, if in the

future a new risk arises, or new technology becomes available – the supervised entities will readjust accordingly in order to comply with the performance standard. The design process does enable this.

- Third, design standards may give preference to the technology or work method of a particular factor, thereby affixing the economy and providing a regulatory advantage to certain players.

Performance standards and design standards can be used both in a permits and licenses regime and in a norms regime.

Example: Licensing the driving of a motor vehicle

License with design standards: In order to obtain a driver's license, attending a minimum 28 practical lessons with a driving instructor is mandatory; License with performance standards: In order to obtain a driver's license, it is necessary to prove the control of the vehicle (practical exam).

Example: Binding norms for sanitation in public pools

Binding norm with design standards: In a public swimming pool, use of water filtering and disinfecting system is mandatory; Binding norm with performance standards: In a public pool no more than 10 general coliform bacteria in 100 ml of water is permitted.

It is important to clarify the distinction between design standards and performance standards and not to confuse them. For example, a regulator responsible for food safety is debating whether to demand the cleaning of a factory floor twice a day or to demand that the factory floor be clean and dry. Although the second requirement is formulated as a result, both options are a design standard aimed at reducing the risk of food contamination. The performance standard relates to compliance with the regulatory objectives: food safety and health (for example, the number of harmful bacteria in the finished product). When policy or regulation defines the means to achieve the regulatory objectives then it is a design standard rather than a performance standard.

When to choose design standards and when to choose performance standards?

- A choice between a design standard

4) The discussion relates primarily to regulatory requirements belonging to the permits and licenses group and the binding rules group. In addition, focusing on the process or outcome may also be relevant to non-regulatory.

and a performance standard dictates the manner in which the public's work processes and business processes are influenced. It is necessary to examine the existing processes and assess how they will change following the use of design standards or performance standards.

- As a rule, one should strive to use performance standards rather than design standards. Performance standards are effective when a desired outcome can be defined clearly and in a measurable manner (for the supervised entities and the supervising entities). Defining a binding result enables the government to focus on the goal of reducing risk. It transfers to the supervised entities the responsibility to mitigate the risk (in order to meet the performance standard).
- However, in some cases there is justification for using design standards. First and foremost, a design standard is efficient when handling small points in the process, when a critical impact on the formation of risk can be achieved. For example, a design standard requires cooks to wash their hands before entering a

restaurant's kitchen. Second, it may be used in cases of occasional players entering the process and are unable to acquire the knowledge and skill to manage the risk. Third, when there is overwhelming professional and public agreement that in a defined aspect there is no value in managerial, operational and technological flexibility. Fourth, when it is expensive and impractical to perform post-process control before the risk is realized. An example of this is in risk of contaminations in a restaurant. There is no laboratory test fast enough to test portions before serving, and therefore the performance standard has limited effectiveness. Fifth, sometimes a design standard will be more effective than a performance standard because of control and supervision aspects. For example, it may be simpler and cheaper (for the government and for supervised entities) that supervision is carried out in respect of the production process rather than the finished product. An example is if testing a product prior to shipment delays delivery or requires supervisors to arrive at a specific time window due to logistic constraints of the supervised entities.

- Choosing a design standard requires in-depth professional consultation with the supervised entities, experts, entrepreneurs and international companies in these fields. In addition, a regular mechanism for consultation and the update of design standards should be established, in order to change them or add additional process alternatives.
- Addressing a number of risks in a particular area will often require a number of steps. A combination of performance and design standards may be carried out in order to address the various risks.

To whom or to what will the regulation or public policy refer?

- Regulation (prohibitions, obligations and conditions) and public policy (instructions, benefits, requirements, etc.) can refer to elements of three main types: people, actions, and objects.

People – for example, to require an individual (who meets a certain definition) to take action or to refrain from taking it, to have certain skills, etc. Benefits can also be granted on this basis (for example, tax benefits for women). This group also includes groups and corporations.

Actions (or processes) – for example, to determine prohibited actions, actions that must be carried out, or defining how certain actions should be carried out.

Objects (including equipment and technology) – for example, to determine equipment that must be maintained, the level of maintenance of equipment, or prohibiting the use of defined objects.

- For each element of the regulatory content, it should be defined whether it applies to people, actions, or objects. The choice should be based on the risk assessment that has been carried out. An identification must be made as to whether risk mitigation will be achieved by focusing on people, action, or objects.

Illustrative example: In order to address the risk of accidents on construction sites, it is necessary to identify the cause of the risk or its severity.

If the accidents occur because of the recklessness of intoxicated workers – the focus should be on the human factor.

If the accidents occur due to carelessness when operating the crane – the focus should be on the regulation of activities.

If the accidents occur due to improper equipment or improperly maintained equipment – focus should be placed on the state of the equipment.

It is possible to formulate almost identical rules, the difference between them being the element to which they relate. This is of great importance. Untargeted treatment will not be effective. Supervision and enforcement will deal with objects that have been defined in the content of the regulation or policy. It is important to apply the regulation or policy to the element that will best mitigate the risk and to select an element on which efficient and effective supervision can be carried out.

Illustrative example: One must ensure that equipment is maintained at a satisfactory level. A rule focused on action can be determined: for example, maintenance must be performed once every six months. It is possible to determine a rule focused on equipment: the equipment must be maintained in proper working order.

Both rules are similar and have the same purpose, but they have different content. They impose different compliance costs on the public and the scope of the supervision and enforcement activities will be affected by this choice.

Focusing on treatment of risk and limiting intervention

- It is important that the regulatory or policy content be focused on the problem and not exceed the minimum required to address the risk. Any government involvement may impose costs – including unexpected costs and effects. The best way to minimize undesirable effects is to focus the content only on the treatment of risks. It is important to define exactly what the problem is (the mechanism of risk creation, reasons for the severity of the problem) and to ensure the government tool focuses only on them. It is easy to set sweeping rules, but they also impose restrictions on activities that do not contribute to risk.

Example: When driving in the dark, it is necessary to use the lighting installed in the vehicle. Many countries (including Israel) have identified that some power

lighting is required, but if the lighting is too powerful, it can temporarily blind other drivers. Therefore, the regulation states that every vehicle must have two headlights – one that illuminates using a high light and the other using a low light (Section 334 of the Israeli Traffic Regulations, 5721 – 1961). Over the years, advanced lighting systems have been developed, allowing for high-light illumination all the time. An automatic control system identifies other vehicles and selectively weakens the intensity of the light at the angle of the other vehicle (for example, <https://goo.gl/QQDyFk>).

Current regulation prevents the adoption of the new technology. If the regulation had focused on the problem (ensuring lighting and preventing glare), the adoption of new solutions would have been possible. Instead the decision was to use a detailed definition (there must be two separate headlights with different characteristics) which prevents the implementation of another solution.

What will the work process look like within the country? What elements will be involved?

It is necessary to examine and define

who is the authorized body responsible for the regulation or the new policy, what workers are required to operate the mechanism, what actions will be taken, etc.

In addition, it is necessary to examine what the work processes will look like. For example; what will be the process of issuing a license, who provides service to the public, which factors are involved in handling a query and whether assistance is required from supporting units (accountancy, legal office, laboratories, headquarter divisions, expert committees).

What will be the administrative control mechanisms. Who will manage the new government activity and who is responsible for checking it?

All the entities mapped in this process are required to devote resources to the implementation of the regulation or policy. It is important to assess how much time each need and whether additional resources are required. Increasing the scope of activity and areas of activity of a government unit will also require an increase in the resources available to it (mainly manpower).

Different methods and content

elements require a different scope of personnel. For example, a system of permits and licenses requires both office personnel to issue permits and manpower for supervision operations in the field. On the other hand, a regime of binding rules is more economical as it requires very little office manpower and mainly requires manpower for field supervision.

What will be the update mechanism?

How often should the regulatory content or policy be updated? If there are technical instructions (list of geographical locations, specifications of permitted materials) it may be necessary to design a mechanism that takes into account frequent updates.

Who is responsible for gathering information (on the impact of regulation or policy on the risk, on risks that have been realized, on changes in means of coping) and initiating updates of regulation or policy?

The Supervision and Enforcement System

Supervisory and enforcement tools must be adapted to every violation and element. The choice of the various tools should be based on the following considerations:

- Type of regulatory instruction – supervision over instructions to record information will be different from supervision over instructions on implementing a defined action. When it comes to information requirements – reporting can be demanded. There are instructions in which supervision requires expert examination: physical examination, information analysis or laboratory testing. When it comes to supervision over processes (design standards), the physical presence of the supervising bodies and the examination of compliance with the various instructions are usually required. When it comes to enforcement of performance standards, usually only the examination of the final result is required and there is no need for physical presence. Traditional supervision is based on physical arrival to the supervised entity. This supervising framework is still widely accepted and widespread, although it requires considerable resources. It is recommended to examine other methods of supervision.
- Severity of the risk – instructions designed to prevent serious risks

require closer supervision and stricter enforcement.

- Expected level of compliance and deterrence considerations – it is necessary to assess what the level of compliance will be and how necessary is it to deter the supervised entities? Effective deterrence requires appropriate use of supervision and enforcement tools, as well as ensuring that the combination provides sufficient deterrence (comparing the benefit of the violation with the expected sanction).
- Does the discovery of a violation require professional discretion and knowledge, or is it possible to perform a technical examination? Does the discovery of the violation justify specific equipment or individual testing? As a rule, testing methods that do not require expertise or special equipment, and can be quickly applied to large quantities of supervised entities are preferable.

Defining the supervisory powers required – professional supervision and investigative and evidence gathering powers in case of a violation. Is an authority required to conduct

a search, carry out an investigation, prohibit activity or give mandatory orders?

Another issue is who will carry out supervisory actions. Traditionally, the government has carried out its own supervising activities. However, the supervision system can be operated by other entities. The supervisory characteristics relate to the various supervisory activities: physical arrival and inspection, information requirements, professional examinations, document reviews, training and information, etc. The possible formats for supervision should be presented in a hierarchical structure according to the level of decentralization of the supervisory powers:

At the top of the pyramid –

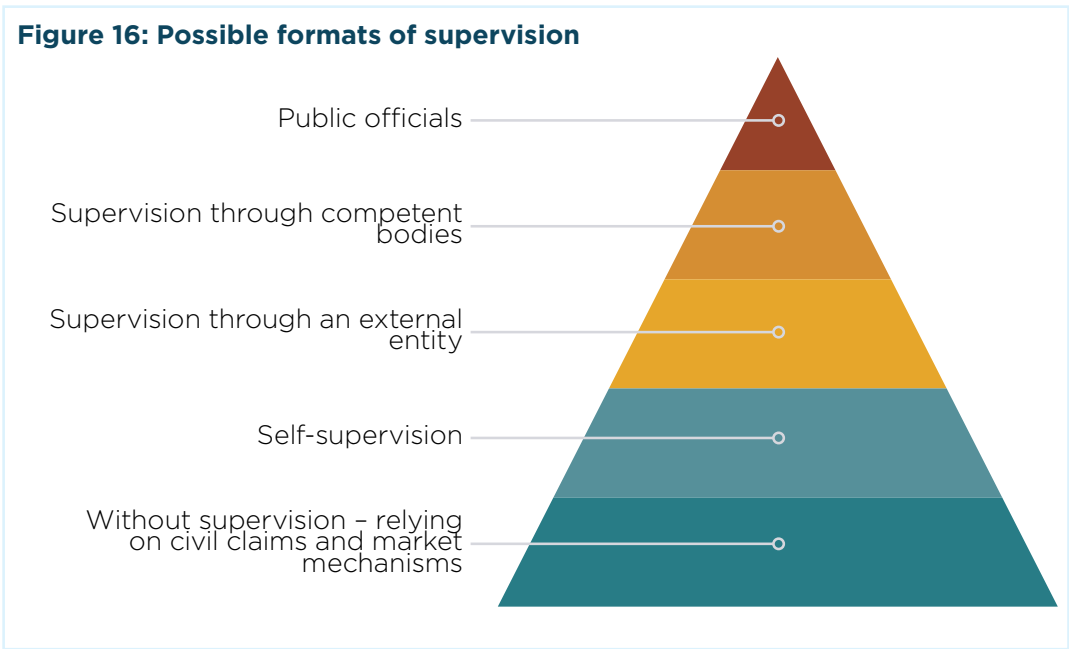
Traditional supervision by public officials.

At the second level –

Supervision through competent bodies (sometimes they form "recognized bodies"). In order to obtain supervisory authority, a process of certification is required. The process of granting certification is overseen by

the government. Once a supervisory body has proven compliance with government requirements, it is granted individual accreditation for specific supervisory actions (it can be said that the granting of supervision authority is carried out in the form of "permits and licenses", as advance

authorization is required in order to carry out supervisory activities). Since these bodies are individually authorized and subject to rules and criteria, they can be given relatively extensive powers and supervisory functions.



Example: Licensing agencies are authorized by the Ministry of Transport to perform a competence test for vehicles ("Test"). The Commissioner of Standardization in the Ministry of the Economy and Industry determines which laboratories are authorized to perform conformity tests for commodities which must be of a defined standard.

The supervision of the cultivation of organic produce is carried out by approval and control bodies authorized by the Ministry of Agriculture.

At the third level –

Supervision by an entity external to the supervised entity. The external entity is not required to undergo a certification process. Anyone who meets the general requirements can carry out this kind of supervision. The general requirements were not determined in connection with the supervision of a specific area but were determined in order to ascertain the general competence and independence of the external entity. This model allows for the recognition of a wide range of supervisory and control bodies that carry out supervisory activities. It does not require the creation and operation

of a certification mechanism (their recognition is carried out through a regime of binding rules, whereas the certification of authorized bodies is carried out in the form of permits and licenses). Most external entities must belong to a group which requires them to comply with other legal requirements, such as lawyers, accountants and accredited laboratories.

Example: The obligation that financial statements of companies must be audited by an external accountant (auditor). Any accountant who is independent of the audited entity may serve as an auditor. The accountant/auditor does not need prior examination or individual government consent.

The Ministry of Energy requires that equipment for work in an electrical facility be tested in a laboratory accredited by the Israel Laboratory Accreditation Authority, which has the appropriate means to test the equipment. The accreditation from ISRAC is general for the accreditation field and is not given only for tests required by the Ministry of Energy.

The Ministry of Transport is authorized to require vehicle importers present a certificate from the vehicle manufacturer regarding the vehicles' proper working order. In this case the vehicle manufacturer serves as an external supervisor of the importer.

The Ministry of Culture and Sport requires the presentation of a physician's approval as a condition for enrollment in a gym. Any physician is authorized to issue this certificate.

At the fourth level –

Self-supervision. As part of this method, the responsibility for supervision and monitoring rests directly with the supervised entities. They are responsible for checking for violations, gathering information and reporting to the regulator if necessary. This model is appropriate when supervising specific actions or soft norms that are difficult to monitor outside the organization. The advantage of this model is that the organization implements self-supervision activities, creating a structural supervision mechanism that encourages the assimilation of instructions in the organization's cultural code.

Example:

The appointment of a supervisor for the prevention of sexual harassment in the workplace. The board of directors of a public company must have an audit committee in which all the independent directors are members.

The requirement that laboratory tests be carried out for products, while allowing the use of the laboratories of the supervised entity.

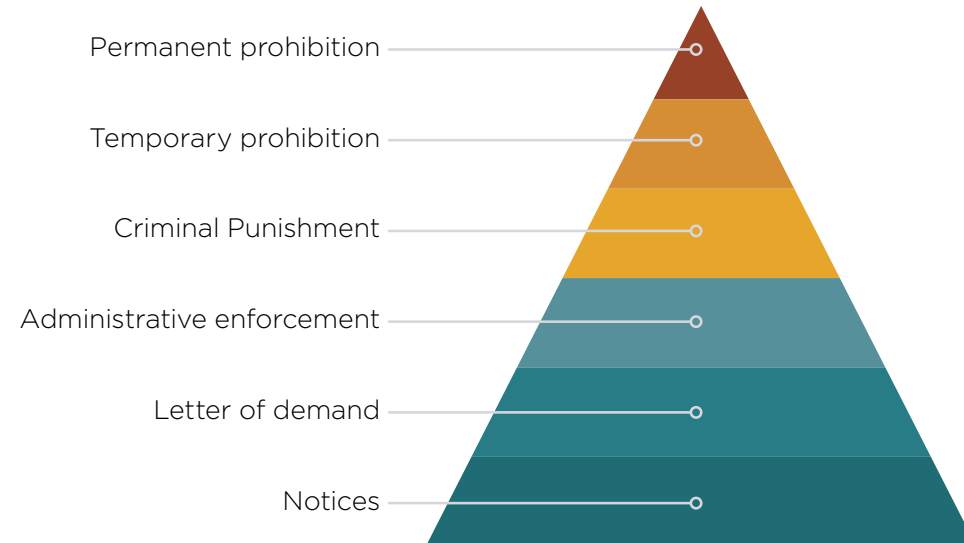
At the fifth and final level –

A decision not to carry out supervisory actions and to rely on civil claims and market mechanisms, as explained above.

An Enforcement and Punishment System –

What are the enforcement and punishment tracks for ensuring compliance with regulation or policy? Are these criminal sanctions? Is this administrative enforcement? Does one rely on civil enforcement through claims filed by the public? The enforcement and punishment tools include the following measures (which are arranged from the heaviest to the lightest in the following diagram):

Figure 17: The means of the enforcement pyramid



- **Permanent prohibition** – usually through the revocation of a license (administrative authority of the regulator). This constitutes severe harm to the supervised entity since they are subject to a permanent restriction on carrying out the supervised activity.
- **Temporary prohibition** – usually by suspending a license (administrative authority of the instance in a motion for the issue of an injunction).
- **Criminal Punishment** – Prosecution. There are relatively quick and lenient penal routes such as a fine and trial options.
- **Administrative enforcement** – administrative fines and financial sanctions.

- **Letter of demand** – issuing administrative instructions (to carry out an action or refrain from carrying out an action); Issuing a demand to rectify a violation within a specified period of time. Failure to comply with the demand will lead to more stringent enforcement measures.
- **Notices** – notice, warning, and reprimand.

Punishment by economic means (criminal fines, administrative fines and financial sanctions) is a central tool for enforcing regulation and public policy. However, it is important to know the limitations of these tools.

- First, economic enforcement has a different effect on different supervised entities. Individuals and small businesses can suffer severe harm as a result of fines, while large businesses will be less affected and may not be deterred by the same fines. Therefore, it is appropriate to make a distinction according to the size of the violating business and the economic profit derived from it.
- Second, economic enforcement does not necessarily reflect all the damage that the risk causes (for

example, damages to society, the environment, etc.).

- Third, the use of economic enforcement reduces and breaks down the violation to an economic price tag only. When enforcement is based only on economic tools, there is concern that the failure to comply with the sanction will be perceived by the supervised entities as an additional expense ("the cost of doing business") and not as a sanction resulting from breaking the law. This problem can be particularly severe when it comes to repeat players, who find it worthwhile to bear the costs of economic enforcement from time to time as part of their routine activities.

In addition to traditional enforcement tools, the publication of collected information can be used as an enforcement tool. The supervision and enforcement system can disseminate the information they collect to the public or interested parties. The publication can include positive or negative information. This is primarily relevant for supervised entities who invest in their reputation. The publication of information (for

example, rating of supervised entities) is particularly helpful when there are gaps in information and the public does not have enough information on the supervised entities. This tool is effective over time only if the supervisory system (which collects and produces the information) is reliable and consistent. An additional condition is that the supervisory activities are performed frequently enough so the information is up-to-date and relevant.

It is recommended to select a number of supervision and enforcement tools in advance. The supervision and enforcement system should have a mix of tools with a hierarchy of severity that can be adjusted according to the level of risk, level of compliance and difference among the supervised entities.

The level of effectiveness and deterrence the supervision and enforcement system creates is closely related to the degree of compliance with regulation or policy. An important principle to remember – deterrence can be strengthened by increasing the probability of being sanctioned (by strengthening the supervision system) or by increasing the intensity of the

sanction (reinforcing the punishment). As such, these variables have a mutual influence and effect on the level of deterrence.

Supervision and enforcement are also actions whose performance depends on manpower. In the content review stage, it is important to define how many resources are needed to supervise and enforce the policy – staff workers, professional inspectors, investigators, attorneys and legal advisors. These resources should be considered when comparing content alternatives.

In this context it is important to remember the guiding principle in **polycymaking**: regulation should not be adopted if it cannot be implemented and enforced consistently, transparently and proportionately. When the method of supervision and enforcement is inadequate, and resources are insufficient to produce effective supervision and enforcement – other alternatives must be examined.

Guiding questions for developing a regime of permits and licenses

Which factors require a license, and which actions require a permit? A regime of permits and licenses

determines that the legal default is a prohibition on the activity until government approval is obtained. Therefore, it is necessary to examine which factors and what actions will require advance approval and to limit the obligation to the extent possible. On the other hand, it is important to ensure that the rule is extremely clear and simple to implement.

How often is it necessary to renew the license?

- A license is granted to the licensee and allows them to perform various actions as long as the license is valid. In contrast, a permit is permission to carry out an action, and accordingly is generally limited to one use. It is therefore necessary to examine whether the goal is to supervise every action in advance, or the factors that carry out the actions. Choosing to use licenses over permits significantly reduces the burden on the public.
- It is necessary to examine how often a license should be renewed (how long is it valid for?). The answer to this depends on the frequency at which there are systemic factual or legal changes that require the license expires, and the licensee contact the issuer. It is important

to remember that the license is not a substitute for routine monitoring. The government can always conduct an initiated examination of the state of affairs, and if justified – change the terms of the license or revoke it.

Does the preliminary approval include additional instructions?

A requirement to obtain prior approval constitutes a “threshold condition” for carrying out the activity. In addition, provisions that must be followed after receiving the license may be mandatory. Therefore, it is necessary to examine whether the prior approval deals only with threshold requirements to enter the market or contains instructions regarding subsequent conduct?

What is the scope of discretion of the issuer of the preliminary approval?

A system of permits and licenses gives the government tight control over the market and creates a high level of friction between the regulator and the supervised entities (every entity is required to contact the regulator, stand for inspection and receive individual approval). Any entity that wishes to start operating in the market (and also someone who holds preliminary approval) will depend heavily on the

regulator. Due to the high level of control and the high level of friction, it is important to define the scope of discretion of the license issuer in exercising their authority: what are they entitled to examine, how and what information they may demand, how to appeal their discretion, etc. This is necessary to ensure certainty and consistency in the operation of this tool.

What does the process of receiving the preliminary approval look like? Receipt of preliminary approval requires contacting the regulator, carrying out a series of actions and culminates in receiving approval. This regime requires the operation of a bureaucratic process, including submission of documents, professional discussions, examinations, etc. It is important to anticipate all phases of the process and examine them from supervised entity’s view point.

Which test is performed for the purpose of issuing prior approval?

- When the trust in the supervised entities is low and the risks handled have a high severity level and high probability, a government official is elected to carry out an examination of compliance with the conditions for receiving preliminary approval.

- When the level of trust in the supervised entities is moderate or high, and when dealing with risks that do not have a high severity and probability level, it should require an affidavit of compliance with the requirements, together with an expert opinion (attorney, CPA, engineer) according to the relevant field. This mechanism provides assurance of compliance with the requirements, without the need for a government agency to review of any application.

- When the level of trust in the supervised entities is high, and in cases where the level of severity or probability is low, prior approval can be granted on the basis of an affidavit given by the applicant. In these cases, it is recommended to examine whether a system of prior approvals is required, and if it is possible to move on to a regime of binding rule.

Guiding questions for developing a regime of binding rules

Are the instructions simple and clear? A regime of binding rules consists mainly of imposing instructions, compliance and supervision. In order to make it easier for the

supervised entities to comply and for the government to supervise, it is important the instructions are clear and understandable. Vague rules raise questions as to how to implement them and, as a result, their enforcement may become more complicated since it is unclear what constitutes a violation. Accordingly, vague and complex instructions may reduce the public’s level of compliance.

What is the scope of the information required? Information is a valuable resource and therefore governments tend to expand information requirements. However, the value of information is also due to the fact that it is expensive to generate, process, collect and transfer. Any information requirement (reporting, publication, disclosure, documentation) must be examined. For example, is the information necessary to mitigate the risk? Information requirements are not intended to collect “general” data, but to address the risks identified in the Risk Assessment stage.

Intermediate hierarchy between permits and licenses and rules: Are the supervised entities required to demonstrate compliance proactively? The use of binding rules with a complementary supervising mechanism can facilitate the transition

from a strict regime of permits and licenses to a regime of binding rules. The use of a supplementary mechanism is intended to enable increased supervision, similar to a system of permits and licenses, without the need for the bureaucratic mechanism involved. The easiest and most effective mechanism is a combination of binding rules and a one-time reporting requirement on entry into the market. This mechanism is almost identical to the normal use of binding rules, but it informs the regulator of the entry of a new player into the field. Thus, the regulator knows all the players operating in the industry and knows where to carry out supervision and enforcement activities.

Guiding questions for developing a regime of advanced regulation and non-regulatory tools

This group includes a variety of policy tools, with a great deal of variance between them. A policy concerning the effect of risks through tax benefits is materially different from the granting of civil claim rights or the use of behavioral economics. The main rule of thumb is to use an expert from the relevant field to examine

suitable content alternatives for each tool. However, there are a number of questions that may be relevant to a significant portion of the tools in this group.

What are the governmental resources required? Many of the tools in this group are based on non-coercive government activity. Budgetary support, tax benefits and even a campaign to preserve the flow of information – all of which require proactive government action and allocation of budgets. Different content alternatives will have a significant impact on the scope of government resources required to implement the policy (for example, defining eligibility criteria for tax benefits or the manner of publication of the information campaign – via traditional media or the Internet).

Is the use of the tool liable to distort behavior, and if so – how? All the tools presented in this guide are designed to influence behavior in order to mitigate risks. However, advanced regulation and non-regulatory tools do not impose a defined action on the public but seek to direct the behavior voluntarily. In particular, use of these tools involves the danger of unplanned and undesirable influence on public behavior.

Illustrative example: A well-known anecdote tells of the period of colonial rule in India. The British sought to reduce the number of cobra snakes on the streets of Delhi. It was decided to give a reward for each dead cobra snake collected. At first it seemed that the new policy was achieving its goal, but later on the British discovered that many Indians recognized there was a "demand" for dead cobras and therefore began to grow and breed cobras in order to collect the reward. When the British became aware of this, they revoked the reward policy. As a result, the snake growers released the cobras from their breeding farms, and the city of Delhi was flooded with cobras.

The first mistake the British made, at the policy-making stage, was the failure to recognize that monetary compensation for cobras would not necessarily stimulate hunting of snakes but would result in their institutionalized growth.

The second mistake was the lack of understanding of the decision to revoke the reward policy on the behavior of snake breeders.

Does the policy have undesirable economic-competitive effects? Is government intervention liable to

improve the situation of the strongest players in the industry or indirectly harm small or new businesses. Even non-coercive intervention requires an examination of the effects on the dynamics among the market players. The selection of self-regulation tools requires an examination of the competitive effect, in order to ensure that the self-regulation mechanism is not used to coordinate between competitors or to enforce arrangements that harm competition.

Analysis of the content alternatives and selecting between them

Based on the questions, a number of content alternatives must be developed. A number of alternatives can be compared for each issue separately. Examples include; a comparison with respect to the question of the validity of the license, a comparison with respect to the question of the supervising system, a comparison of reporting obligations, etc.); or several sets of content alternatives developed that contain a number of issues for comparison. It will usually be easier and simpler to examine each issue separately (starting with the most important issue), and make sure that the different content elements are integrated.

The comparison of these alternatives should be carried out according to the stages presented above in the [method review section](#): A cost-benefit calculation must be made according to the principle of maximum aggregate welfare (effect of the risk, government costs, costs to the public) in order to choose the alternative with the lowest social costs:

	The expected annual damage from the risk	The cost of coping methods to the government	The cost of coping methods to the public	Total costs
Current situation				
Alternative A				
Alternative B				
Alternative C				

It is important to consider the effectiveness of each content alternative in influencing the risk mitigation. For example, the degree of impact on the probability, the degree of impact on the severity, and the percentage of cases that will be affected by the content alternative.

In addition, attention must be given to other considerations and influences – social, environmental, human rights, competition, damage and cost dispersion, small businesses and international considerations. Alternatives with minimal damage to these considerations should be preferred, and an attempt must be made to adjust the content in order to minimize the harm to them. These considerations are explained above in the method examination chapter. It is recommended to review Annexes G and H, which present rules for identifying harm to competition and small businesses.

Preventing redundancies and overlaps:

- Once the content development has been completed, it should be checked that the regulation or policy does not create redundancies. A redundancy will be expressed in excessive costs (to the government

or to the public), with low benefit to reducing risk expectancy.

- In principle, two methods of coping directed at the same risk should be avoided.
- For this purpose, the method as a whole and all the processes it creates must be examined. Redundancies in risk management can be internal (two instructions treating the same risk), or external (creating redundancies with regulation or other policies).

Recommendations for the stage of implementation of the selected regulation/policy

Performing a pilot trial. After completing the process and choosing the desired solution and content, a trial run of the regulation or policy should be implemented on a limited scale and for a fixed period – for example, by publishing a “temporary order”. The pilot allows for the empirical testing of the risk management, and for determining whether it is effective in dealing with the risk (and what are the government costs and costs to the public as a result). It is important to define in advance criteria and outcome indices for evaluating the trial run, and to establish a mechanism for collecting and analyzing data during the trial run period. A pilot run is especially important when it comes to regulation or policy with extensive social or economic impact.

Limiting validity and periodic examination. Even after the pilot run, it is recommended to limit in advance the validity of the selected regulation or policy. One can assume with certainty that in the future, circumstances will change – risks will arise, new risks will disappear or change, a new method (regulatory, business or technological) will be developed for dealing with the risks, etc. Unlike the budget, which is often examined, policies and regulations anchored in laws, regulations and procedures, are not examined once every specified period and are not required to be re-ratified. It is recommended to limit the validity of the decision. This will assist the system in the future, ensuring the system updates the risk management and will not rely on a process carried out in the distant past. The customary way to do this is through “sunset clauses” which presuppose that the law, regulation or procedure will be valid, for a defined period. Towards the end of the period, a renewed risk management process will be conducted, and based on the updated risk assessment, updated decisions will be made. This step strengthens the decision maker as it maintains the flexibility of the government system in the future (in view of the inherent difficulty in changing laws and regulations that have no effect).

CRISIS MANAGEMENT

To date, the process has dealt with early steps aimed at pro-actively reducing the probability of the risk occurring or mitigating its severity. In contrast, Crisis Management/Contingency Planning deals with risk management once it has been realized.

Preparing to respond to risks retrospectively is especially important when dealing with high severity risks. A successful crisis management system will prepare the system for risks mapped at the risk identification stage, but also for dealing with new risks and risks that have not been identified.

The main challenge of crisis management is that the system is required to act quickly, under conditions of stress and uncertainty. These are challenging conditions for making decisions. The target in crisis management is to reach as high a level of readiness as possible prior to any risk occurring. This enables decision-makers to determine what form of action is best and that all necessary resources will be ready. When there are clear guidelines for all functionaries and all equipment and resources are prepared, if the crisis occurs, everything will be ready for

implementation, thus reducing the risk of errors in the exercise of discretion.

Crisis management is a continuation of the process of assessment and early treatment of risks. It also complements them. Crisis management can be a major strategy for dealing with risks. The decision maker may choose a risk acceptance strategy (not to act in advance to mitigate the risk) and prepare to treat it retroactively. In addition, crisis management can be part of a risk mitigation strategy based on insurance methods. Crisis management enables the system to recover from the risk damage, but it requires dedicating resources in advance (planning, training, purchasing equipment) – similar to the purchase of an insurance policy. Therefore, crisis management is especially necessary for risks of a high level of severity and can be a major tool when severity is high, and probability low.

Crisis management entails early preparation (preparation of "contingency plans" and resources in times of need) and real-time response actions. Since this process is designed to provide a response to risks – it should be performed soon after the

Risk Assessment. Crisis management is designed to deal with damage and to ensure functional continuity in disaster situations – both of the government unit and other vital functions (such as public infrastructure or business).

A key component of crisis management is prior coordination with other entities and joint preliminary preparations. If the event materializes, the various bodies will be required to act in unison, while being familiar with the division of powers and functions between them. In Israel, the main emergency bodies include the National Emergency Authority, the Home Front Command, the Emergency Economy Department, the hospitals, the National Fire and Rescue Authority and Magen David Adom (Israel's official ambulance service, blood service and disaster rescue agency).

The key stages of treating the risk that has occurred, should be the key points of the crisis management contingency plan. As with any significant decision, it is preferable to examine the steps that will be taken after comparing alternatives and conducting a cost-benefit analysis.

In addition to the three risk treatment stages, it is important to prepare

the regulation and public policy so there are appropriate powers and methods of action for operation during the crisis. Preparation may include the option of making policy or regulation more stringent (taking extra caution in implementing the rules or operating the supervision and enforcement system in times of crisis) and facilitating policy or regulation (removing barriers to improve crisis management and recovery).

Illustrative example: Authority to ease regulatory or policy requirements

During an outbreak, vaccines and medicines are needed. In preparing for such an event, it was decided to determine in advance that the director of the unit has the authority to allow accelerated licensing of drugs and vaccines without undergoing all the usual tests and procedures. This temporary measure will be activated in times of crisis, in order to enable crisis management and recovery.

Illustrative example: Authority to make regulatory or policy requirements more stringent

At times of a suspected unusual security or safety incident involving civil aviation, it is necessary to isolate the dangerous aircrafts and evacuate all aircrafts from Israeli airspace. Therefore, it must be determined in advance that the director of the Airports Authority has the power to order all civilian aircrafts to land immediately. This provision expands the powers of the government to limit activities that involve or are associated with risk. It is not intended for routine use, but only in exceptional emergency situations.

Immediate response: stabilization of the situation and urgent treatment

Lifesaving operations – search and rescue, evacuation and medical treatment. It is necessary to define what actions should be carried out, how, who is responsible for executing them, and to allocate resources in advance for these actions.

Example: "Magen David Adom" is the State of Israel's national rescue organization for emergency medicine. The existence and activity of the organization is intended to save lives after a risk has been realized.

Delineating the source of the damage and preventing the expansion of the damage's effect. The planning must include how to treat the cause of damage in order to contain its reach. For example, in the case of an epidemic, it is necessary to locate patients and carriers who may infect their environment. In addition, a plan must be prepared to contain the effects of the damage on the protected assets. In the example of an epidemic this would include vaccination of the population within a defined radius around the risk epicenter.

Communicating the event. Guidelines on how to act and provide information about the risk should be prepared in advance. Communication channels and the way the messages are conveyed (written, visual, through a senior official interview) must be considered. It is important the guidelines and explanations to the public be adapted to emergencies in a clear, simple and immediate way.

Example: In 2016, the National Emergency Authority began preparations for the possible occurrence of a tsunami event. The preparations included the preparation of escape routes from the beaches and the deployment of warning and guidance signs. These actions are intended to enable a better response to the risk that has been realized. They are not intended to reduce the probability of a tsunami occurring or to mitigate the severity of the impact on the coast.

Actions required to prevent further injury. Preparation for any further action required to prevent additional damage.

Example: Traffic accident

In addition to the regulations concerning the reduction of the probability of road accidents (obligation to maintain distance) and the mitigation of the severity of the damage (obligation to wear a seat belt), the government is also involved in the preparation of crises.

Lifesaving operations – MDA ambulances and rescue organizations

are ready to treat the wounded and rescue casualties.

Preventing the expansion of the impact of the damage – the Traffic Department of the Israel Police is responsible for reporting traffic hazards and preventing obstacles to traffic resulting from traffic accidents.

Communicating the event – the Traffic Department of the Israel Police is responsible for reporting traffic hazards and providing information on accidents and blocked roads.

Secondary response: activation of essential functions

Reactivation of essential functions. Contingency plans should include instructions for specific actions for each critical unit in the system. These instructions are intended to prevent the disruption of essential services and infrastructures.

Example: The Israel Electric Corporation is preparing to operate power plants following rare events such as accidents and natural disasters.

Example: The Traffic Department of the Israel Police removes vehicles that were involved in traffic accidents to

the side of the road, in order to enable the road to fulfill its function – transportation infrastructure.

Emergency facilities and alternative work methods. Special facilities should be prepared for use in emergencies, as well as work methods and equipment for emergency situations. An example of this is the provision of flashlights in the event of a power failure. These facilities and methods will be used as an alternative to facilities and methods used in the routine – for example, in the event of a crisis, accelerated procurement processes can be performed without a tender committee. They are also necessary to serve emergency activities for the treatment of risk (for example, installing generators to supply electricity to a war room established in the field). Pre-planning of the deployment and operation of alternative infrastructure (for example, communications infrastructure) may be required at times.

Provision of essential services. Risk may disrupt the provision of essential services to the public or a defined group. Preparations must be made in advance for the supply of electricity, water, fuel, food, medical services,

medical equipment and other essential services. At times it is necessary for emergency supplies to be prepared and maintained.

Conclusion of the event: Rehabilitation and return to routine

At this stage, actions will be taken to restore activity to routine. This process may take several weeks. Since this process also requires dedication of resources and attention, it is advisable to prepare for it in advance. The conclusion of the event is defined as a situation in which the relevant activity returns to normal function, as it was prior to the occurrence of the risk.

At the conclusion of the event it is important to conduct a professional enquiry of the event, led by the managerial level. The enquiry is integrated into the organization's supervision and control system (detailed in following chapters). The main role of the enquiry is to identify and treat faulty processes in order to prevent the realization of future risks. It is important to conduct an enquiry regarding rules and work processes, rather than focus on the responsibility of one factor or another.

CRISIS MANAGEMENT

Risk communication includes the transfer of information from the government decision maker to the relevant stakeholders and the public. The explanations in this chapter are suggestions and not a substitute for effective and productive working systems already in place with the government spokesperson's office. The instructions of the spokespersons must be on a case-by-case basis.

The decision maker's goal is to manage the risk optimally according to the principle of maximum aggregate welfare. This is done by influencing public behavior. To ensure maximum effectiveness, it is important to provide the public (or relevant groups within it) with information about the risks and the government policy for addressing them.

Risk communication is carried out throughout the risk management process: from the Risk Assessment stage, through the Risk Treatment stage and during the implementation of the selected policy.

Risk communication has three main objectives:

- To create public awareness of risks;
- To instruct the public on how to act in order to avoid or mitigate risk;
- To present and explain government policy – what risks it addresses (and how) and what risks it does not address.

Since communication with the public can influence its conduct, risk communication can serve as an advanced regulation method for risk management. For example, if the public believes a significant risk is a negligible risk, it is possible that the transfer of information and explanations will change the public's perception and conduct on the subject. For both individuals and organizations, information is a fundamental element in the decision-making processes. Proper use of this tool can have a material effect on the public's exposure to risks.

Example: The public mistakenly thinks that an airplane is a dangerous vehicle and that private vehicles are a safer mode of transportation. In practice, the situation is reversed – an airplane is a much safer vehicle. The public misunderstands the probability of occurrence of the various risks. The consequence is the public fears the use of airplanes, resulting in excessive use of cars – a behavior that increases the risk to the traveling public.

Illustrative example: Parents think children get lice from playing in a sandbox. In actuality, lice are mainly transferred through contact with dogs and cats. In this case, the parenting community misunderstands the circumstances in which there is high exposure to risk. As a result, they may be ill-prepared to prevent and treat the risk.

It is recommended to check retrospectively whether the risk communication has achieved its purpose. The accepted process for this is through surveys / questionnaires, in which the public is asked how it

understands the risks and the risk management policy.

Communicating the risk assessment

Communicating the risk assessment presents the insights and conclusions the decision maker has arrived at from Risk Assessment (risks identification, risks evaluation and risks analysis) to the public and relevant stakeholders. The goal is to equip the public with relevant information that the decision maker has collected and analyzed.

Communicating the risk assessment has two main principles: communicating the message clearly and providing clear guidelines for action. In order to help the public to act correctly in face of the risk, it is important the communication of the risk assessment includes practical instruction on how the public should act.

In the first stage, it is necessary to clarify the public's position:

- How familiar is the public with the risk?
- What experience does the public have in exposure to the risk?

- How does the public assess the severity of the damage from the risk and the probability of its realization?
- What other characteristics does the public attribute to the risk?

This information will determine whether there is a gap between the public's perception of the risk and the actual risk and enable more effective planning for the risk communication stage:

- Does the public's misunderstanding of the risk cause the public to act recklessly?
- What information about the risk must be provided to influence public behavior?
- Who is the information intended for? The target audience must be accurately defined.
- In what way must the information be disseminated in order for it to be received and understood by the public and its target audience? (I.e., length of the message; how information is transmitted: visual

message, warning, training, publication of press release).

- When should the information be distributed and how often should an update be released?
- How should the public act? What actions to avoid, what actions are vital, etc.

Communicating the risk management policy

- After completing the risk management phase, it is important to present and explain to the public the chosen regulation or policy and why it was selected so they can fully comply with it. Communicating the chosen policy also gives it public legitimacy and expresses principles of reasoning and transparency.
- Successful communication of the selected policy will include an explanation of the process carried out, the considerations taken into account and the reasons for the decision made. A summary of the outcomes of the process can be used to communicate the selected policy.

- There is value in giving the public information about the decision to treat or not treat certain risks. These decisions reflect government policy and clarify to the public which risks will be addressed and which not through government involvement.
- Another case of communicating risks during policy execution is if the risk materializes. The guiding principle in these situations is to provide the public with clear and concise information that can be easily processed and used in an emergency situation if the risk occurs. Complex, lengthy messages or messages phrased in professional language, will not be understood, and of no or little benefit to the public.
- When the risk is realized, the manner in which the government responds is significant. There are a variety of possible responses to a risk that has been realized. These include refraining from providing information until the completion of the enquiry, through the gradual provision of information culminating in an immediate notification acknowledging responsibility for the occurrence. As part of the broad concept of risk management, it is proposed to formulate the message to the public according to the following principles:

It is important to provide information to the public, as long

Communication in the course of policy execution

as the information is reliable, and its publication will not damage crisis management operations.

Since there are no zero-risk situations, it is important to clarify that the risk occurrence is not necessarily the result of a government failure to completely prevent risks, but that the risk exists as part of the activity of the parties involved. Even during an event, it must be remembered and noted that the occurrence of a risk does not mean the policy or regulation has failed.

At the time of the risk's actualization and shortly thereafter, there is a tendency to look for those responsible for the event. It is important to conduct the review in a responsible manner and not publish unfounded assumptions. This principle applies both when the blame is directed at the government and when it is directed at private entities. A debrief is an internal professional tool and is also a control mechanism against the publication of unfounded information.

Initial messages about the

realization of the risk will include the following components:

Basic information: description of the event (what happened, place, date); An explanation of its effect (to the extent that there is information), and actions taken (by the government and by private entities) and their outcome.

Instructions to the public: provision of contact details of the person in charge of managing the event; Inviting the public to provide information and report unusual events; Instructions and directions for the public (to stay away from a certain area, to take precautionary measures, not to consume a product manufactured on a particular date, etc.).

Future actions: an explanation that the incident is under investigation; reference to the various options and alternatives to action facing the decision maker according to the outcome of the investigation.

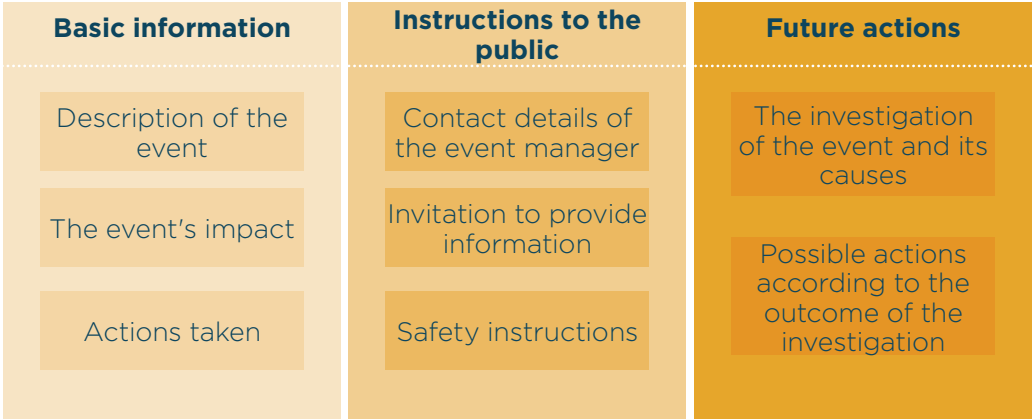


Figure 18: Key components of a media message about a risk that has been realized

Illustrative example: A press release following the explosion of a natural gas pipeline

"The incident that took place yesterday is considered one of the greatest disasters that has occurred in Israel in recent years. The explosion took place shortly after 23:00 near the Great Crater. As a result of the explosion two people were seriously injured and five people were lightly injured. The current data in our possession indicates that the explosion occurred as a result of an accident and not the result of deliberate action. The Ministry is examining whether the energy company has maintained the gas pipeline according to the provisions of the law. It is currently forbidden to drive on route 225. For further information and to report unusual events, contact the Director of the Safety Department. Suspicion of negligence by the employees of the pumping station is being examined as prior to the incident, an increase in the gas flow rate was observed. The results of the enquiry will be published upon its completion. If it is found that the policy and procedures are to be updated – they will be amended; if it is found that the explosion was caused by negligence or violation of the law, those responsible will be prosecuted."

Illustrative example: A press release following a drowning at sea.

"According to the information in our possession, the deceased entered the sea yesterday at an unknown time and apparently was pulled under the water and subsequently drowned. At this stage the circumstances of the case are being examined.

It is not yet known whether the deceased entered the sea at a time when there were rescue services on the beach and whether the deceased knew how to swim. After examining the full circumstances, the question of whether regulation of bathing sites (declared beaches, rescue services) will be examined and whether there is a need to clarify procedures or make changes in policy. The Authority reminds the public that bathing in a beach without rescue services is prohibited and dangerous. The list of declared beaches is published on the Ministry of Interior's website."

MONITORING AND EVALUATION

The implementation of the risk management policy should be accompanied by monitoring and evaluation for two main reasons:

- To monitor whether the risks have changed;
- To ascertain whether risk management is effective and to identify whether changes are required.

Monitoring and evaluation processes should be established to assess whether risks continue to exist, whether new risks exist, whether the risks identified have changed (especially in the aspects of probability and severity), and whether significant changes have occurred requiring an update of priorities. In addition, the overall risk management process should be subject to a periodic review to ensure it continues to be appropriate and efficient. Risk monitoring and the evaluation of the results of the process are different from each other. They do not replace each other.

A monitoring and evaluation system should include:

- An annual examination of all

components of risk management (strategies, methods, content, crisis management, risk communication);

- The risks identified undergo periodical examination (at least once every three years) – including up-to-date information collection. This can be done with the help of an external factor;
- Collection and monitoring of information to warn of the formation of new risks and changes in identified risks, in respect of a change requiring a new examination of the priorities or risk treatment policy.

Guidelines for monitoring and evaluating the risk management policy:

- Information should be collected on the risk management mechanisms (number of processes performed, number of applications for licenses submitted, scope of supervisory actions, rate of violations seized, application handling period);
- Information should be collected about events that occurred. This includes risks that have been

realized or risks that have almost been realized. Reporting of unusual events and risks that have almost occurred is vital to the system in order to learn about possible scenarios for the realization of the risk. Documentation and reporting of these cases is critical in order to investigate the reasons for the occurrence of risks and to assimilate insights into the system. An investigation of these events will focus on processes rather than functionaries. The goal is not to blame but to analyze and improve the process;

- Guiding questions to examine whether the risk management policy is effective in mitigating the risk:

Has the number of risks that have occurred changed? Has the relative prevalence of the occurring risks changed? (Number of events out of total activity).

Has the severity of the events that have occurred changed? Has the overall severity (total damage to the economy per year) changed? (It is important to ensure a dynamic

analysis and take into account changes and background trends).

Examination of the basic components: Is the scope of the risk management policy appropriate? (Too narrow, too broad). Is the method and content selected appropriate? Does the supervision and enforcement system meet its objectives?

What are the costs that the risk treatment policy actually imposes on the government budget and the public? (Direct compliance costs and indirect costs such as impact on supervised operations).

- Did the policy affect other interests? (Competition, international trade, fundamental rights, etc.).

Guidelines for monitoring and evaluating external changes:

- Has the profile of the identified risks changed? (With respect to each risk – is the data detailed in the Risk Register still current?).
- Have some of the risks identified disappeared and no longer exist? (For example, due to the cessation

of the activity involved).

- Did new risks appear? (If they are significant it is important to perform their evaluation and analysis – and if necessary update the strategy and priorities).
- Is there a change in the means of coping with the risks? (Development of new technologies or methods, changes in the legal situation). If changes have been identified – it must be ensured that government policy blocks the adoption of the new means of coping. Recommendations for an institutional monitoring and evaluation system:
- It is important that a government unit has a computerized information system;
- It is advisable to appoint a person responsible for collecting and analyzing the data. They will report the findings to the decision maker or the unit manager;
- A risk board that receives information on all the risks the government unit manages can

should be established in order to analyze the information and conduct periodic discussions with the decision makers.

Monitoring and evaluating risks and regulatory or policy effects requires resources and consistent handling of data collection, documentation, and analysis. It is important to allocate resources in advance. Collecting information during routine times requires little resources, whereas recovering and collecting information in retrospect requires extensive resources.

Consistent monitoring and evaluation / adaptation of risk management are essential to ensure that risk management continues to be effective and relevant over time. The reality is dynamic and without an update of regulation or policy, risk management will be devoid of content.

CHAPTER C – RISK MANAGEMENT IN SUPERVISION AND ENFORCEMENT

GENERAL

Risk management in supervision and enforcement is the continuation of risk management in developing regulation and public policy. The process of developing regulation and public policy defines the general framework, which consists of binding requirements and sanctions in respect of their violation. This chapter deals with the supervision and enforcement system. This system determines how to use the policy-defined tools to ensure compliance with the requirements - for example, decisions regarding what actions are to be taken, vis-à-vis whom, and with what frequency.

Once the risks have been identified and the methods for treating them have been determined, it is necessary to decide how to ensure compliance with the regulation or the policy established, in order to mitigate risks. The meta-goal of the supervision and enforcement system is to encourage and ascertain regulatory or policy compliance.

Noncompliance may arise from the supervised entities' lack of incentive or inability to comply with the requirements or ignorance of their existence. Therefore, the supervision system includes traditional actions

such as data collection, inspections and enforcement (imposition of sanctions), alongside actions for guiding the supervised entities on how to meet the prescribed requirements.

The role of the regulator and the responsibility of the supervised entities. According to OECD guidance, it should be made clear - in law, in procedures and in practice - that the primary responsibility for complying with regulation and not taking inordinate risks lies with the supervised entities. The role of regulators and inspectors is to assist and promote compliance, but they are not responsible for regulatory compliance by the supervised entities or for preventing risks, as they are not the supervised entities. The regulator's job is to set rules, guide the supervised entities and penalize in cases of noncompliance. The supervised entities should be given responsibility for compliance with the requirements and for the risks they pose to the public.

If risk management is performed at the regulation or policy-setting stage, the risks tackled will already be defined and characterized and the supervision and enforcement system

development process will be simpler. If risk management is not performed at the regulation or policy-setting stage, developing the supervision and enforcement system will necessitate the collection of data and evaluation of the different risks and violations.

Risk management in supervision and enforcement is essentially similar to risk management in the development of regulation and public policy. Both are based on an attribute analysis of the problem (noncompliance or public risk), adaptation of government handling measures to the problem's attributes, and prioritization of problems by degree of importance. Planning the supervision and enforcement system is based on the criteria of probability and severity: the probability of a violation being committed and the contribution of the requirement to damage mitigation. Nevertheless, risk management in supervision and enforcement is simpler, as it should be a routine work process used by supervision and enforcement units on a daily basis (supervision and enforcement are routinely performed actions, as distinct from policy-setting).

This following model presents a guide

for the planning and operation of a supervision and enforcement system. The model is based on the mapping of the attributes of the supervised entities and requirements and on the rating of each pairing of a requirement with the supervised entities. The model is suitable a considerable portion of the supervision and enforcement tasks. However, some situations may require a different model. For example, there may be cases where the more appropriate approach would be to rate each supervised entity (business, buildings, shipments, etc.) by a dedicated set of criteria.

The model in this Guide is a baseline. It is advisable to use a computerized system for the management of supervision and enforcement actions. Computerized systems have predictive capability, based on machine learning, statistical analysis and professional information. Automation and digitization of the supervision system will help to weight a multitude of data, ensure consistency, uniformity and facilitate the inspectors' work.

The risk management process in supervision and enforcement consists of three key parts: defining the supervision objectives, establishing

the supervision system, and execution and monitoring. The structure of the process is described in the following diagram.



Figure 19: Risk Management Process in Supervision and Enforcement

The first part - Defining the Supervision Objectives

This involves mapping the elements of the supervision and enforcement system and consists of the following stages:

Mapping the requirements - Identifying all the requirements and analyzing the effect of each requirement on reducing the relevant risk's likelihood and severity.

Mapping the supervised entities - Identifying the entities that are subject to the regulation and characterizing them according to size of supervised group, expected compliance level and major reason for noncompliance.

The second part - Establishing the Supervision System

The decision maker relies on the information collected about each requirement in order to select and apply suitable policy tools:

Allocating supervision and enforcement resources - In this stage, the supervision system and the allocation of supervision resources among the various requirements and different supervised entities should be planned. An automated

tool for planning the supervision and enforcement system can be downloaded through the following link: <http://bit.ly/EnforcementToolEng>.

Planning the supervision and enforcement activities - Which supervision tools will be used, at what frequency, and how will the inspections be carried out. Also, work methods with the aim of increasing compliance with requirements must be defined.

The third part - Execution and Monitoring

This includes tools for operating the supervision and enforcement system:

Execution of supervision and enforcement activities - Including instructions on the operation of the supervision and enforcement system, with a view to creating a reliable, deterrent, compliance-encouraging and exact system.

Monitoring and evaluation - Collecting information on the supervision and enforcement system's operational process, examining the actual effect on the risk profile of the supervised entities, drawing lessons and adapting the system to changes.

MAPPING THE REQUIREMENTS

As already noted, supervision and enforcement should increase regulatory or public policy compliance. Compliance is the fulfillment by supervised entity with the government requirement. Accordingly, the first stages of the process include mapping the requirements and the supervised entities. Before starting to evaluate the different requirements, it is advisable to screen out those requirements that will not be handled by the supervision and enforcement system:

Requirements in which effective non-government enforcement mechanisms already exist.

As mentioned in Chapter A, the two key mechanisms are civil lawsuits and market forces. Wherever these mechanisms operate and guide the behavior of supervised entities, the need for government supervision and enforcement is lower.

Negligible risk.

This refers to a low-rated risk in terms of severity and probability. If the requirement is intended to mitigate a risk that is negligible compared to other risks on the agenda, the correct approach may be to not allocate supervisions resources to it. It is

important to bear in mind that such requirements could be enforced by non-government mechanisms.

Overlapping requirements.

When there are several requirements in place aimed at mitigating the same risk, it may be decided that those requirements with a low contribution or limited enforcement capacity will not be subject to enforcement action initiated by the supervision and enforcement system.

The supervised entities have both a high capacity and high motivation to comply with the requirements.

In such cases, the same interests are shared by the supervised entities and the decision maker and the need for supervision and enforcement is low.

After the initial screening, a list should be made of all the requirements. Each requirement should be assessed for how much it decreases the probability and severity of the risk. A "requirement" can be an isolated or a set of requirements directed at a single supervised entity for dealing with a single risk.

Some requirements affect both the

probability and severity, while others affect only one of the two variables.

Illustrative example: A requirement obligating motorcyclists to wear a helmet reduces the severity of damage in an accident but does not diminish the probability of an accident occurring. The requirement mandating a driver to signal before changing lanes or making a turn reduces the probability of an accident but does not diminish the severity of damage resulting from an accident.

The guiding question - assuming full compliance with the requirements - is to what extent each requirement contributes to risk mitigation (influence on risk probability and severity).

Probability:

To what extent does full compliance with a requirement influence the probability of risk occurrence?

Severity:

To what extent will damage severity be reduced by full compliance with a requirement?

It is important not to confuse the risks' probability and severity, with

the influence exerted by a particular requirement on the probability and severity of the risk.

Illustrative example: The expected damage of risk A is 70 and that of risk B is 40. Two requirements have been prescribed for the treatment of the risks: The first requirement will reduce risk A from 70 to 50, and the other will reduce risk B from 40 to 10. Only one of the requirements may be selected for the performance of supervision and enforcement. In this case, supervision and enforcement tools should be applied to the second requirement reducing risk B by a value of 30 (from 40 to 10). Although the expected value of risk B is lower than A, the requirement regarding risk B will better protect the public and better reduce risk exposure.

A score is assigned to the influence of each requirement on probability and to the influence of each requirement on damage severity. The score is based on a scale of 1 (slight to negligible influence) to 5 (great influence), with the two respective influences of each requirement subsequently placed in the requirements matrix. The matrix is a 5*5 table structure with two coordinates: probability and severity.

Illustrative example: The requirement to wear a helmet while riding a motorcycle reduces the damage severity in an accident to a greater degree than the requirement to wear a helmet when riding a bicycle. In this example, we would score the requirement in respect of a motorcycle as a 4 and the requirement in respect of a bicycle as a 2.

The cells of the requirements matrix represent the product of the scores (the influence on probability and severity) for each requirement. Thus, a requirement that was assigned scores of 3 and 2 will be awarded an overall importance ranking of 6; and a requirement that was assigned scores of 5 and 4 will be awarded an overall importance ranking of 20. The overall importance ranking will subsequently be used to prioritize supervision actions.

		Probability				
		1	2	3	4	5
Severity	1	1	2	3	4	5
	2	2	4	6	8	10
	3	3	6	9	12	15
	4	4	8	12	16	20
	5	5	10	15	20	25

Figure 20: Requirements Matrix

Illustrative Example: Mapping of Requirements			
Requirement	Influence on Severity	Influence on Probability	Requirement Importance Ranking
Use of protective equipment	5	4	20
Installation of alert system	1	3	3

The requirement importance ranking (the result of the influence on probability and severity) is used for ranking and prioritizing the different requirements. The requirement importance ranking must take into account the risk that is being treated (how significant the risk is) and the effect of the requirement on the risk.

It is possible to accurately calculate the effect of each requirement on the expected value of damage using an estimate of the reduction in damage probability and severity, thereby arriving at defined economic values. However, this concerns a complex and imprecise calculation that must be made for a large number of requirements. In many cases the value may be determined based on the requirements matrix presented here.

MAPPING OF THE SUPERVISED ENTITIES

This stage requires mapping the entities that will be subject to the supervision and enforcement system. First, it is necessary to identify all the supervised entities that are subject to the relevant regulation or policy. Generally, the mapping relates to categories of supervised entities (children, adults, self-employed, salaried employees, local manufacturers, importers, exporters, etc.). If it concerns a small and closed group of supervised entities, it is possible to make a specific identification.

The aim at this point is to draw up a list of groups of supervised entities and their characteristics. A group of supervised entities should include supervised entities of similar characteristics. Following are guiding questions for defining groups of supervised entities:

- What is the involvement of the supervised entity in the risk's creation (risk creator, overseer of the risk creator, an entity whose role is to prevent realization of the risk)?
- Are the supervised entities private individuals or companies / businesses?
- Division by defined professions and roles (in the workplace:

manager and employees; on a trip: tour guide, trip participants and a security guard; in a hospital: physician, nurse and patient).

- Where and when do they carry out the risk-associated activity?
- Scope of activity.
- Does this involve regular or one-time supervised entities (does this concern a rare or one-time meeting of the supervised entity with the requirement or does it concern a recurring or routine meeting)?
- What is the capacity of the supervised entities to comply with requirements? (Knowledge, skill, experience and tangible resources needed to comply with requirements)?
- What is the motivation of the supervised entities to comply with the requirements (is compliance with the requirements consistent with the interests of the supervised entities; do the requirements fit with the processes which the supervised entities have assumed voluntarily such as internal work procedures)?

Subsequently, the following three questions should be answered with

regard to each supervised entity group and each requirement:

- What is the size of the group of supervised entities compared to other mapped groups? The group size is scored from 1 (smallest group) to 5 (largest group).
- What is the expected non-compliance level for a requirement? In the absence of supervision, what percentage of the supervised entities is expected to violate the requirement? The non-compliance level (the violations) is expressed as a percentage. As such, 100% signifies that no supervised entity complies with the requirement and 0% means there are no violations and all supervised entities comply with the requirement.
- In the absence of any data or experience that would enable assessment of the non-compliance level, sampling of inspection actions may be carried out on a limited number of supervised entities as a basis for making a preliminary assessment. This can be updated at a later stage.
- What is the key reason for non-compliance with the requirement? There are five major reasons for supervised entities' non-

compliance with the requirements:

1. Ignorance of the requirements;
2. Misunderstanding of the requirements;
3. Distrust of the regulation or policy;
4. Inability to comply with the requirement (including inability to bear the costs entailed in compliance);
5. Deliberate criminality.

These three attributes – group size, non-compliance level and reason for non-compliance – will be used to decide on the allocation of supervision resources.

The requirement non-compliance level is a key criterion, as supervision and enforcement are meant to boost compliance. Where compliance level is high and the need for supervision and enforcement is low, resources may be directed to supervised entity groups with a lower compliance level. To assess the probability of violation, it is important to assess the expected compliance level of each supervised entity group with regard to each requirement. The compliance level may be assessed based on past compliance, research studies and

Illustrative Example: Mapping of Supervised Entities

Supervised Entity Group	Group Size Ranking	Expected Violation Percentage	Noncompliance Reason
Businesses	3	80%	Deliberate criminality
Private homes	5	45%	Ignorance
Public buildings	2	20%	Misunderstanding

general information on the supervised entity group. It is important to validate and update this information on a regular basis throughout the implementation of supervision and enforcement actions. If information is available, the impact of the supervision and enforcement action in improving compliance with the requirement should be evaluated.

At the end of this stage each of the

requirements is paired up with each of the supervised entity groups to form "requirement-supervised entity group" couples. This yields a list of supervision tasks - a table summarizing all the information on the various risks (see Annex J). Each task (row) in the list of supervision tasks (pairing of requirement and supervised entity) is assigned an overall score. This is calculated based on the data presented above:



Figure 21: Overall Score Calculation

The list of supervision tasks is arranged by descending order of the overall score.

Illustrative Example: List of Supervision Tasks						
Requirement	Supervised Entity Group	Requirement Importance Ranking	Group Size Ranking	Expected Violation Rate	Noncompliance Reason	Overall Score
Use of protective equipment	Businesses	20	3	80%	Deliberate criminality	48
Use of protective equipment	Private homes	20	5	45%	Ignorance	45
Use of protective equipment	Public buildings	20	2	20%	Misunderstanding	8
Installation of alert system	Businesses	3	3	80%	Deliberate criminality	7.2
Installation of alert system	Private homes	3	5	45%	Ignorance	6.75
Installation of alert system	Public buildings	3	2	20%	Misunderstanding	1.2

DISTRIBUTION OF SUPERVISION AND ENFORCEMENT RESOURCES

Distribution of Supervision Resources by the 50-30-10 Model

Mapping the requirements and supervised entities is followed by a determination of how the system will operate and how the various resources will be allocated (work time of inspectors, budget, technology equipment for identifying violations, training activities, etc.). The primary method for planning the supervision system is based on the list of supervision tasks and the 50-30-10 Model.

An automated tool for planning the system of supervision and enforcement, can be downloaded through the following [link: http://bit.ly/EnforcementToolEng](http://bit.ly/EnforcementToolEng)

Planning the supervision system is based on the overall score of each row in the list of supervision tasks. A row with a high total score will be considered a top priority in the supervision and enforcement system due to the effect on the risk, the supervised entity group size and the group’s disposition not to comply with requirements.

The list of pairings is divided into three equal groups according to the number

of tasks (rows) in each group.

A list of 30 rows is divided into three groups of 10 rows each). The upper group, comprising the highest overall scores, is defined as the red category; the middle group - the yellow category; and the third and lower group is the green category.

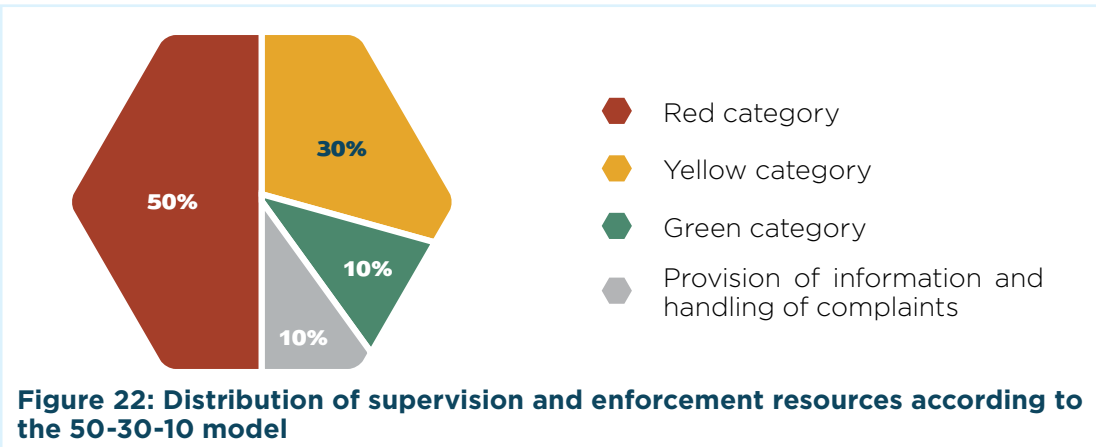
The distribution of the supervision and enforcement resources is based on the 50-30-10 Model:

50% of the resources will be allocated to the red category (the upper third of the supervision tasks);

30% of the resources will be allocated to the yellow category (the middle third of the supervision tasks);

10% of the resources will be allocated to the green category (the lower third of the supervision tasks);

A further 10% will be allocated for horizontal supervision tasks involving the provision of information to the supervised entities and for response to complaints, the handling of which goes beyond the regular handling of the three categories.



This distribution system allows for pro rata allocation of the resources (inspectors, investigators, laboratory tests, budget and any other resource) among the various tasks by their level of importance.

According to the 50-30-10 Model, half the supervision resources are dedicated to tasks in the red category, 30% to tasks in the yellow category, and 10% to tasks in the green category.

Illustrative Example: Effectuation of 50-30-10 Model							
Requirement	Supervised Entity Group	Requirement Importance Ranking	Group Size Ranking	Expected Violation Rate	Noncompliance Reason	Overall Score	Category
Use of protective equipment	Businesses	20	3	80%	Deliberate criminality	48	Red
Use of protective equipment	Private homes	20	5	45%	Ignorance	45	Red
Use of protective equipment	Public buildings	20	2	20%	Misunderstanding	8	Yellow
Installation of alert system	Businesses	3	3	80%	Deliberate criminality	7.2	Yellow
Installation of alert system	Private homes	3	5	45%	Ignorance	6.75	Green
Installation of alert system	Public buildings	3	2	20%	Misunderstanding	1.2	Green

The division of the list into categories provides a simple formula for the allocation of supervision resources based on the contribution of each requirement to risk mitigation and on the need for supervision actions (requirement compliance level without supervision actions). The 50-30-10 Model reflects the principles of risk management in supervision and enforcement. The scope of supervision actions is suited to the risk attributes, the requirement, and the compliance level of the supervised entities.

The allocation of resources within each category need not be uniform. On the contrary, it should be based on information, reflect suitability to the risk level, encourage compliance and promote deterrence. It is recommended that the supervision task list be used to help in the prioritization of resources within each category.

Resources may be transferred from the green to the yellow category, and from the green and yellow categories to the red category, if there is no need for the resources allocated to them to begin with. This model recommends an advance allocation of 10% of the resources for an information provision and complaint handling infrastructure. This is a basic amount of resources

for the benefit of two key types of supervision actions for which there is often no pre-planned allocation of resources (these details are elaborated upon shortly).

Infrastructure for providing information

The information infrastructure should facilitate the communication between the supervision and enforcement system and the supervised entities. Examples of which include; preparing a checklist with key requirements, establishing guidance and instruction systems for the supervised entities, developing instruction material and information for the supervision staff, and identifying information gaps among the supervised entities that could result in violations.

Handling of complaints

The handling of complaints consists of actions in response to a complaint lodged directly to the supervision system. Though complaint actions are not planned, it is important to allocate resources in advance for implementation of designated actions. Resources should be allocated for the establishment of an information provision and complaint handling infrastructure; the stated 10% allocation of resources is the

recommended minimum threshold.

Additionally, the resources distributed to the three main categories (red, yellow, green) may also be directed to the provision of information and handling of complaints regarding the requirements included in the respective category.

The 50-30-10 Model is based on the premise that the supervised entities comprise known, key groups, in respect of which the supervision actions are carried out. Nevertheless, reality teaches us that unusual, rare or exceptional supervised entities are identified after the allocation of resources. An example could be an isolated shipment with unique attributes or a rare product that arrives on the market occasionally. Such supervised entities or events are not always identifiable in advance, but it is necessary to decide what resources should be allocated for their supervision and enforcement. As a rule, special supervised entities are included in the red category and are subject to close supervision due to insufficient information about them. This generally involves only a small number of cases, and therefore, there is no concern that considerable resources will be diverted away from the other actions included under

the red category. If the available information indicates that the supervised entities or requirements concerned are not of great importance (based on the compliance level, effect on risk, etc.), they may also be included in a different category.

Planning the Supervision System to Suit Each Supervised Entity

An alternative method to the supervision task list and the 50-30- 10 Model is customization of the system. Customization is necessary when a small number of supervised entities are involved and both risk and complexity levels are high. If the population of supervised entities is small and consists of recurrent players, customization enables a more precise definition of the components of the supervision system. Therefore, this method is less suitable when the requirements are directed toward the general public.

Planning a supervision system based on the individual characteristics of each supervised entity requires a system for scoring each entity according to a fixed set of evaluation criteria to be applied to all entities. The criteria and scoring method must be customized to the pertinent field and The scoring system should be based

on the following key criteria:

Criterion A – Risk Attributes

Type of activity / equipment / people involved.

For example, using electrical equipment is more dangerous than using non-electrical equipment. Pasteurized food is safer than unpasteurized food.

Nature of activity being performed. Does the activity profile of the supervised entity include risk-increasing attributes or risk-mitigating attributes?

A short production line, use of vacuum packaging for food and use of insulating materials are risk-mitigating attributes. Long shift work, driving at night, rapid employee turnover that harms experience and skill levels are risk-increasing attributes.

The public that is exposed to risk.

An evaluation should be made of the susceptibility of the population exposed to the risk (e.g. babies versus adults) and the size of the public exposed to risk (number of consumers that purchased products from the supervised entity's business or the number of people who are likely to be injured as a result of an accident),

and the number of events expected to occur during a given year.

Criterion B – Precautions and Internal Control of the Supervised Entity

Establishing principles for approved workflows or for approved tools and precautions can guide the supervised entities on operational procedures. It also helps the supervision system to consistently inspect the actions initiated by the supervised entities in order to comply with the requirements.

Workflows –

Do the supervised entities' workflow include precautions designed to diminish the probability of risk realization? For example, does the

workflow include safety inspections of the product.

Tangible tools and precautions –

Does the supervised entity use tools or precautions that diminish risk probability or severity, for example the use of helmets, use of automation to identify defects in products, use of non-flammable materials, etc.

Internal control and enforcement systems for boosting compliance with requirements –

Such systems carry out control procedures on workflows and tools used by the supervised entity to ensure requirement compliance and risk mitigation. If the supervised

entity has an internal control mechanism or internal enforcement system, these mechanisms may be assigned a score (mechanism components, violation detection capability and internal enforcement capacity).

Example: The Israel Securities Authority grants reliefs to supervised entities that operate an internal enforcement program designed to detect and prevent violations of securities laws. An internal enforcement program includes mapping regulatory requirements, performing a requirement compliance survey, well-ordered procedures, an effective enforcement system, handling failures and learning lessons. An internal enforcement program implements a culture of law abidance at the supervised entity.

Example: The HACCP (Hazard Analysis and Critical Control Points) system is designed to prevent hazards through process analysis, potential hazard identification, and determination of critical control points. HACCP systems are widespread in the food safety field. One of the key stages in the system is defining critical values: determining a measurable range of values within which a product is defined as safe.

This matter is regulated under ISO 22000, a quality standard for food safety management systems.

Criterion C – Current Compliance Level

The compliance level can also be evaluated in relation to requirements related to or associated with the field of occupation. It is not only based on compliance with the requirements for which the supervision system under consideration is intended. It is important that compliance level rating relate to the number, as well as nature, of the violations. This enables differentiation between technical/marginal violations and material violations. Violation of a single material requirement could be more serious than ten minor violations in technical matters.

Items for inspection should be determined for each criterion, with a defined score assigned to each item. A table should be prepared specifying the scope or frequency of each of the supervision actions, based on the score assigned to each supervised entity.



Example: The British government issued a document setting out the rules for the supervision of food industry regulation: Food Law Code of Practice. This document includes a method for establishing a supervision and enforcement system to oversee requirements concerning the maintenance of food hygiene, in accordance with the principles of the HACCP system. The method includes a system for determining the frequency of the supervision actions carried out in respect of each supervised entity. The riskier the supervised entity, the higher the score it will receive. Following is the table of supervision frequency prescribed in said document:

Category	Score	Minimum Inspection Frequency
A	92 or more	At least once in 6 months
B	72-91	At least once in 12 months
C	42-71	At least once in 18 months
D	31-41	At least once in 24 months
E	0-30	Will not be included in the supervision program described, and will be subject to an alternative, more lenient program, in which framework they will be inspected at least once in 36 months.

Complaint-Based Supervision

Up to this point two methods have been presented for establishing a pre-planned proactive supervision system. In addition, supervision actions may be carried out based on complaints. This means that some supervision actions will be executed based on complaints regarding violations that have occurred, and not based on advance planning that attempts to predict where violations will be committed.

Example: A considerable portion of police enforcement action is based on complaints filed either by people arriving at the police station or telephonically via the Israel Police 100 emergency number. The Israel Police differentiates between police-initiated cases (cases opened at the initiative of the Police, without a complainant) and other cases. In 2015, 82% of all cases were opened following a complaint, whereas 18% of cases opened were police-initiated.

Example: It is prohibited to chop down mature trees without a permit from the Forestry Commissioner. The Forestry Commissions’ supervision and enforcement system is based on complaints lodged with the Forestry Commissioner and local

Authorities by members of the public. Reports on an attempt to chop down a tree are received from passer-byes and people living in the vicinity of the incident, on this basis part of the supervision and enforcement actions are executed.

The key advantage of complaint-based supervision is that the supervision and enforcement system focuses resources on handling violations that have occurred and not on searching for violations that may or may not occur. A traditional supervision system assumes some of the supervised entities will be found violating the requirements, while some will be found complying with the requirements. A complaint-based supervision system directs part of the resources to tasks with a high probability of finding only supervised entities that are violating the requirements.

A further advantage of complaint-based supervision is its focus on violations that create public risk or damage. Traditional supervision is likely to focus on technical or negligible violations that do not significantly influence the public risk exposure. Complaint-based supervision is directed at handling violations that create public risk, damage or disturbance, since otherwise no

complaint would have been filed. Complaint-based supervision screens out violations that have no complaint filed against them – technical violations or violations of formal requirements that do not bother the public.

Employing complaint-based supervision is particularly appropriate and effective when it concerns a public that is aware of the requirements and can actively report violations, and when public violations are credible (a minimal number of false complaints and consistent and uniform reporting).

This supervision format has several limitations and drawbacks that must be taken into account. First, complaint-based supervision is likely to be biased due to reliance on external complaints. The bias could arise from false complaints and from a lack of uniformity in the method of filing the complaints (non-reporting populations, cases that go unreported, etc.). Second, there are violations that necessitate proactive government supervision in order to discover and expose them (for example, at the criminal level: offenses of possession and receipt of stolen property, offenses of extortion and blackmail, offenses of drug use, etc.). Such offenses are generally not discovered through complaints received from the public. Third, under this mechanism,

supervision is exercised only after the violation has been discovered. Therefore, it is suitable only in cases where the violation is retroactively identifiable (an ongoing violation or a violation with a retroactively identifiable result), and unsuitable for violations that must be identified in real-time. Fourth, complaint-based supervision is appropriate for handling violations of binding norms, but inappropriate for supervision and enforcement of permits and licenses.

Owing to these limitations, it is generally difficult to base the supervision system solely on complaints. As such, under suitable circumstances it is recommended to incorporate this method as part of the supervision and enforcement system.

Under the 50-30-10 Model, at least 10% of total resources should be earmarked, among other things, for handling complaints received by the supervision system. It is important to be aware that complaint response and handling requires a similar level of supervision resources as is needed for proactive action. Hence, planning and advance allocation of resources are essential to enable supervisory activity of this type.

PLANNING SUPERVISION AND ENFORCEMENT ACTIONS

There remains two additional issues to define in order to complete the setup of the supervision and enforcement system: what supervision tools are to be used (content of meeting with the supervised entity) and how the supervision actions should be executed.

When planning the supervision actions, it is important to consult with various parties that can provide different views and perspectives. It is important to consult with both headquarter and field personnel within the supervision system, who are familiar with the actual situation on the ground and with the circumstances characteristic of the relevant area. It is also recommended to seek advice from regulation/policy setters. The process should also be based on an international comparison of supervision and enforcement methods, as well as accumulated global insights and experience.

Selection of Supervision Tools

Traditional supervision tools include; a request for information, physical arrival at the supervised entity's premises and performance of inspections, summons for the supervised entity to appear before an inspector for an inspection,

performance of a professional inspection (expert, laboratory, garage), supervision by a private body (accreditation of supervisory and control bodies – and exercise of meta-supervision over them) and inspection of documents and examinations. There are also advanced supervision tools aimed at promoting compliance with requirements: instruction activities, explanations on regulation or policy, guidelines on how to meet the requirements, detailed proposals for improvement of the supervised entity's performance, disseminating information relating to the requirements to the public, in order to encourage public demand that the supervised entities meet the requirements, and more.

While the supervision system is in operation, it is advisable to explore the use of alternative tools as necessary, and not be limited to the use of a single tool, namely: exercising control over compliance with requirements; conducting instruction sessions for supervised entities (how and why to comply with requirements); and communicating the risk and its treatment measures to the supervised entities and the public.

Reason for noncompliance

Since the supervision and enforcement activities are intended to promote compliance with the requirements, the key component for selecting the supervision tool is its suitability in addressing reasons for noncompliance. When noncompliance stems from ignorance or misunderstanding, supervision must focus on increasing the awareness of the supervised entities and explaining the requirements. If noncompliance is the result of an inability to comply, supervision should focus on the skill or resources that the supervised entity lacks in order

to comply. This may include review of the requirements and the need to update them. Where noncompliance is due to disaffection with the policy, action should be taken to boost identification with the requirements and their objectives. This should be carried out in conjunction with the moderate application of traditional supervision and enforcement tools. In the event noncompliance is the result of deliberate criminality, traditional supervisory action must be executed to ensure that enforcement and punishment provide a deterrent for the supervised entity.

It should be emphasized that problems of ignorance, misunderstanding or inability to comply will not be solved by punishment and deterrence. It is important to bear in mind that traditional supervision (inspection and punishment) is ineffective for a substantial percentage of violations and supervised entities. As such, it is important to invest resources in direction and guidance in order to encourage compliance. This is particularly important where professional requirements in complex business areas are concerned. Violations stemming from ignorance, misunderstanding and disaffection

with the requirements are common among the public. These violations should also be resolved through the application of advanced supervision tools.

Example: The Israel National Road Safety Authority and the Israel Anti-Drugs and Alcohol Authority do not conduct inspections for the purpose of detecting violations and enforcing the law. Their ramified activities focus on studies, surveys, dissemination of information, training, instruction and coordination of government and non-government activities. The definition of their role stems from an understanding that the prevailing driving culture and widespread use of drugs are a social phenomenon that cannot be attributed solely to deliberate violation of the law.

Motivation and compliance ability

Another way to match the supervision tools to the supervised entities and their attributes is by categorizing the supervised entities by motivation level and ability to comply with the requirements (knowledge, understanding, technical capability). Small businesses and new business enterprises are often characterized

by their low compliance capacity and problems with knowledge or understanding of the requirements. Recurrent players and large businesses with strong public visibility (brands, high media exposure, large market share) are likely to be characterized by high motivation and a high compliance capacity, as many are reputation focused. Determining the placement of supervised entity groups on the following diagram will help clarify the mix of supervisory measures to be employed to each group.

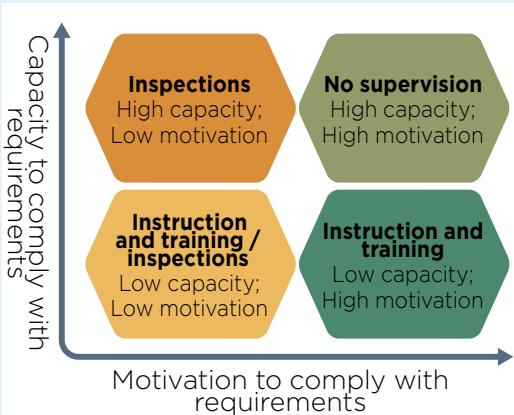


Figure 23: Guiding Rules for Supervision Type Selection According to Noncompliance Reason

Selecting the Supervision Format

Surprise inspection and scheduled inspection

Based on the supervised entities' attributes and type of supervision activities selected, a mix of surprise inspections and scheduled inspections must also be selected. Surprise inspections are aimed at checking the supervised entity when it is unprepared for an inspection. This is vital in evaluating the entity's level of compliance with day-to-day requirements without special preparation. The purpose of the scheduled inspection is to give the supervised entity time to get organized and increase its compliance with requirements. Although, traditionally, government supervision has been based on surprise inspections, a scheduled inspection significantly affects the supervised entities' compliance level and also mobilizes the supervised entities for the benefit of a joint effort with the supervision system. These are two complementary tools, both contribute to the compliance level, according to the violation reason and supervised entities' attributes.

The supervised element can be persons, objects or actions.

Supervision of persons and objects can be done ex-ante, ex-post or in real time. Supervision is generally done in real time, in order to identify the violation at the time of its occurrence. Accordingly, it is necessary to plan when and where effective supervision actions will be executed.

Rules must be established regarding the frequency of inspections and the supervision activities to be carried out with respect to each requirement and supervised entity group. The rules must be based on the latest information (which is gathered on a regular basis), and the frequency must be updated accordingly. The frequency of the supervision activities should be based primarily on the historical compliance level of the supervised entity, the type of violations the supervised entity may have committed in the past (technical, minor, serious), the reason for the violation (does it stem from a misunderstanding of the requirements or was it a deliberate violation). It must also be determined how resources are to be allocated for the various supervision actions. It is important to also take into consideration the benefit gained from increasing or decreasing the frequency of inspections. Visits and supervision actions need not be carried out at a higher frequency than

the rate of changes at the supervised entities.

Illustrative example: Supervision to check the production line at a pharmaceutical plant meets the requirements. If a pharmaceutical plant renovates and upgrades its production line once every five years on average, there is no need to conduct a monthly inspection, nor even a yearly inspection.

Illustrative example: Supervision of public bus journeys aimed at identifying passengers that did not purchase a ticket. The minimum usage unit on a bus is a ride between two successive bus stops. Thus, there is no need to make two inspections during a ride between two bus stops. Additionally, if the common bus-ride duration of passengers (in general or ticketless passengers) is known, then the supervision frequency can be set accordingly. If 95% of bus passengers take rides of at least ten stops, there is no need to make an inspection between every stop. This is particularly apt when the typical bus ride duration of ticketless passengers is known. In such a case, a few focused inspections should be carried out on bus lines and bus stops that are most prone to this risk.

It is necessary to define a minimum sampling unit to be examined in each inspection. The supervised element is inspected to ascertain that it meets the requirements. The sampling unit should be predetermined – for example, should one or ten products be examined? It must also be decided what sampling method will be employed: how to prepare a representative and credible sample, how to take physical samples, and what knowledge is needed to perform the sampling operation. These are essential questions that affect the quality of the supervision system and the supervision staff's scope of work. As a rule, it is best to select a simple sampling method and the minimal sampling size needed to establish the credibility of the supervision system. Sampling or inspections that are not necessary should be avoided.

Supervision activities should be carried out both vis-à-vis the risk creators and the entities that are likely to detect and prevent the risk. Therefore, it is necessary to explore the possibility of early violation detection. This will identify the entities that could be influenced in a way that will boost compliance and prevent violations:

Early detection by the supervised entities.

If the supervised entities are able to detect the violation early and handle it, then the supervision system (and the incentives it creates) should be built in a way that encourages them to discover and remedy the violation and, if possible, also to report it to the supervision system. Enlisting the supervised entities in a manner that makes them part of the supervision system is an efficient and effective tactic. This way it is possible to replace a considerable part of the government inspections with self-supervision of the supervised entities, while the government supervision system focuses on performing meta-supervision. The degree of technological advancement of the self-supervision mechanisms and the extent of reliance on them depends on the professional level of the supervised entities, their motivation to take an active part and their ability to institutionalize an effective supervision system.

Early detection by the public.

Sometimes the public can identify a violation and handle it before the risk is realized. The potential injured parties in this case of risk could be the public or other entities such as service providers and recurrent players (lawyers, accountants,

investment managers, physicians, customs agents, engineers, internal auditors, other stakeholders). These entities can serve as gatekeepers that prevent the realization of the risk or as whistleblowers that report the violations.

Early detection by the group exposed to risk.

Occasionally, the group that is exposed to risk can identify the violations or the risk and act to avoid it. In such cases, it is both possible and desirable to shunt resources from the supervision of risk creators to other supervision tasks. Resources can also be focused on actions that boost awareness or impart a skill in the risk-exposed group. The possibility to do this depends on the sophistication level of the risk-exposed group, i.e. to what extent its members are involved and possess the skill and experience enabling them to act on their own as de facto inspectors and damage preventers.

Predetermination of considerations for not performing supervision activities. It is not imperative to handle every supervised entity and every "observation." The inclusion of a supervised entity and requirement in the red category

does not necessarily justify the implementation of supervision actions due to the presence of defined attributes. Risk management means matching the means/measures to the circumstances. As such, risk management-based supervision is not carried out uniformly and rigidly, where there is no such justification.

Consolidation of Supervision Activities

In establishing the supervision system, it is important to take into account the bureaucratic actions deriving from the creation of the system (issuance of approvals/ notices of violation, taking laboratory samples, etc.). Likewise, it should be established whether there are supervision actions that can be joined with other government actions that are being carried out in connection with the same supervised entities. This may spare the need for especially carrying out supervision.

Moreover, there should be a focus on supervision actions that address a broader set of requirements. There are supervision actions that enable monitoring several requirements. For example: when arriving at the production plant to conduct an inspection, it is possible to examine both the products themselves and

the plant workers' familiarity with the requirements. On the other hand, if in an inspection is conducted at the product shipping stage, there may be no plant workers present and thus their knowledge cannot be evaluated.

Generally, every government entity has its own separate supervision system (the "silo" effect, meaning a narrow, inward-looking attitude). Therefore, every supervised entity is subject to a number of separate supervision and enforcement systems (for example – a plant is subject to the supervision of the Health Ministry, Labor Ministry, Environmental Protection Ministry, etc.). As a result, the government system devotes overlapping resources to the execution of supervision activities. Such situations involve a waste of resources for all entities: double costs for the supervised entity (devoting time to several inspections), double government supervision actions, and splitting of information on the supervised entity among several entities. Consolidation or combination of supervision and enforcement actions leads to a savings of much needed resources and raises the effectiveness of the supervision system.

Hence, the aim should be to unify clusters of supervision actions and

execute them as a consolidated whole. Consolidated supervision activity includes the conduct of inspections, checks, examinations and instruction sessions in respect of a range of requirements and areas. This can be done by setting up mechanisms for coordination and collaboration among the supervision units operating in allied areas and vis-à-vis similar supervised entity groups (e.g., supervision of the engineering design of power plants is carried out by both the Energy Ministry and the Labor Ministry). Such collaborations augment the capabilities of both supervision systems and enhance their capacity for control, instruction and enforcement vis-à-vis the supervised entities.

Illustrative example: Environmental regulation of chemical plants includes provisions regarding wastewater, emission of air pollutants, water purification and treatment of hazardous materials. Each of the four topics include different requirements relating to a different activity in the plant. Supervision of each topic within a separate inspection framework, by a separate supervision unit, results in a waste of resources of the supervision system (four different inspectors arriving at the plant to conduct an inspection). Moreover, each inspector will be exposed only to the hits associated with the specific topic they are supervising and will not see the complete picture and the mutual effects of compliance or noncompliance with the various requirements. Integrated inspection enables streamlining the supervision system, strengthens the supervising entity and achieves greater effectiveness.

An advanced way to consolidate supervision activities is to establish an integrated inspection system with structural unification. The framework of this includes a unit set up for performing integrated inspection on behalf of several authorities and government ministries.

Example: In 2017, Malta enacted the Coordination of Government Inspections Act.

Pursuant to the law, the Inspections Coordination Office was set up. According to the law, the purpose of the move was two-fold - the reduction of costs imposed on the supervised entities and the promotion of transparency and effectiveness of the supervision actions. The law, inter alia, empowers the Inspections Coordination Office to instruct the government authorities how to consolidate their supervision resources, issue binding standards regarding the manner of carrying out the supervision actions and set up a common database for the different supervisory authorities. The law applies to 23 regulatory entities in various fields, such as: Commission for the Rights of Persons with Disability, Regulator for Energy and Water Services, Tourism Authority, Competition and Consumer Affairs Authority, and Plant Health Directorate.

Defining Measurable Targets and Customizing the Supervision System

In planning the supervision system, it is important to predetermine the success indices for the supervision activities,

so that in the following weeks or months it will be possible to evaluate the activities in order to improve them and make them more suitable for implementation. The pivotal question is whether the supervision action has impacted compliance. It is important to ensure that the indices will be quantifiable and that it will be possible to gather information on them. For an elaboration on this topic, see the subchapter "Monitoring and Evaluation."

Whereas policy and regulation are relatively rigid frameworks, the supervision and enforcement system must be flexible and adapted to the needs and circumstances. It is recommended to predetermine a routine for evaluating data and updating the work procedures of the system to fit the changing circumstances. For example, it is reasonable to assume the compliance level of the supervised entities will be influenced by the supervision actions and change frequently.



EXECUTION OF SUPERVISION AND ENFORCEMENT ACTIONS

Uniformity and Consistency in Operating the Supervision System

A common problem among supervision systems is the lack of uniformity and consistency in the inspectors' work. For example, when two inspectors interpret the requirements differently, they end up presenting different requirements to the supervised entities. This impairs the ability to achieve the regulation or policy objectives (since some of the inspectors are deviating from the policy). This gives rise to variability among the different supervised entities. Furthermore, that same supervised entity is likely to run into different requirements, according to the demands of the inspector who examined it. A lack of uniformity and consistency lowers the compliance level of the supervised entities. From their perspective the requirements are inconsistent, giving them no incentive to comply with them. Non-uniformity and inconsistency make it difficult for the supervised entities to comply with the provisions, leaving the inspectors with the task of enforcing them effectively.

Typical causes for the lack of uniformity and inconsistency are: multiple

provisions (a long and detailed list of requirements); vagueness regarding the substance of the requirements; absence of written documents with a description of the requirements.

One of the most effective tools for contending with these problems is the issuance of a checklist by the supervision system. The checklist must include all the requirements demanded of the supervised entities and clearly emphasize the core requirements – the principal and most important requirements for risk mitigation. The inspections will be carried out according to the checklist, and the latter will serve as the implementation document of the regulation or policy.

The checklist must be a closed list. It should not be possible to introduce further requirements beyond the original document. Creating the checklist also helps the supervision system to know whether the requirement list is too long and detailed (which makes it difficult for the supervised entities to comply). If it is, it may be decided to cut down requirements or simplify them, in order to encourage compliance. Use of a checklist helps manage

the supervision system (ensures consistency and uniformity among the inspectors, enables measurement) and also lets the supervised entities know which provisions are the most important for risk-mitigation.

Example: The authorities in Lithuania make extensive use of checklists for purposes of supervision and enforcement. The checklist sets out clearly all the provisions and requirements the supervised entities must meet, with the supervision actions carried out only in respect of the checklist's content. Lithuania's experience demonstrates the use of checklists solves one of the chief problems of the supervision systems: inconsistent and non-uniform inspection. Moreover, the checklist is useful when the inspectors are unable to analyze the risks themselves. Checklist-based inspection limits the discretion of the inspectors and favors consistency over flexibility. Use of the checklist helps clarify the requirements, encourages supervised entities to comply with the provisions and improves the supervision system.

It is useful to issue a checklist for each type of supervision action: inspection when receiving a business license,

annual inspection, inspection of large businesses, inspection of high-risk businesses, etc.

It is important to update the checklist on a regular basis with the aid of the field workers. The checklist should be issued along with guidelines, principles and tools for complying with requirements, as a complete whole aimed at promoting compliance with requirements. These materials can serve as a basis for an instruction system that helps the entities supervise themselves and verify compliance with requirements.

Certainty and Clarity for Supervised Entities

In addition to uniformity and consistency in the operation of the supervision system, it is important to ensure that the supervised parties understand what they are required to do. This involves a fourfold problem, namely: multiplicity of requirements; dispersion of requirements over numerous documents (regulations, procedures, guidelines and directives); requirements which are not set down in writing; and use of complex legal or professional language.

These factors make it difficult for all

supervised parties to understand the requirements and comply with them. This is especially so for new, one-time or small players.

The guiding principle is that the rules and processes of the supervision and enforcement system must be clear and public. The duties of the supervised entity and the duties and responsibilities of the government units must be made public. In particular, it is important to explain how the supervised entity will be examined and investigated. There is no need to "reveal all the cards" and examination methods, but the supervised entity must know exactly what is expected of it and how it should prove compliance with the requirements.

The supervision system must be as transparent as possible. The system will more effectively encourage compliance and deter violations if the supervised entities are aware of the mechanism that underlies the supervision and enforcement actions. For example, if the supervision and enforcement system is based on the compliance record of the supervised entities, this is important knowledge that will promote compliance. On the other hand, the detailed supervision

and enforcement plan must remain confidential and privileged, since it comprises sensitive intelligence information.

Where the requirements or processes are complex, it is recommended to prepare a guide or manual for the supervised entities that explains how to meet the requirements, what steps are advisable to take, how the requirements interrelate, what solutions exist for the treatment of risks, etc. To ensure its usefulness, such a guide must be prepared as a joint project between the supervision system (headquarters and field personnel) and the supervised entities.

Example: In the United Kingdom, the Health and Safety Executive (HSE) operates a project called "Health and Safety Made Simple," which issues clear and simple guidelines on health and safety at the workplace. The project, among other things, explains the regulatory requirements and how the supervised business can identify risks to workers. It also distributes posters and informational material to businesses.

Example: The British program "Safer Food, Better Business" (SFBB) is designed to help small and medium-sized businesses comply with the complex regulatory requirements in the food industry. The program strives to improve the safety level in the industry and help supervised entities know whether they are operating in accordance with the requirements. In the framework of the program, guides were issued for various businesses in the food industry: restaurants, catering businesses, etc.

The guides contain step-by-step explanations, clear illustrations and examples of correct and incorrect ways of performing actions. The guides were translated into the most widely spoken languages among workers in the United Kingdom's guest accommodation and catering sectors.

The guides can also set out approved implementation methods that were examined in the past by the supervision system and found useful. The publication of approved methods makes it clear to the supervised entities that anyone implementing the approved method is operating in accordance with the requirements. This makes compliance easier for the supervised entities and encourages

the use of effective risk treatment measures. As part of the mechanism of approved implementation methods, one can also remunerate supervised entities that take steps to improve compliance and mitigate risks – for example, reducing the number of supervision actions when the supervised entity implements risk treatment measures or operates an internal enforcement program in order to ensure compliance with the requirements.

Enhanced certainty and clarity regarding supervision processes: To increase the level of certainty of the supervised entities, it is customary to issue a Service Level Agreement (SLA). This document details; how the supervision unit "contracts" with the supervised entities, how information is provided in an efficient manner (answers to questions, instructions, solutions to problems, etc.), within how much time a supervised entity's communication is handled, what are the supervision processes, how the fine-imposition mechanisms operates and how to appeal a decision.

Since uncertainty and lack of clarity harm the supervised entities' compliance capacity, efficient handling of the questions posed

by them contributes greatly to the compliance level. A rapid response to questions and problems of supervised entities helps both to comply with the requirements and to understand the supervision system. It also builds trust in the system. It is advisable to document the questions and the answers provided. When an important or recurring question is encountered, it should be written up and published with an official response. Additionally, maintaining a follow-up of questions of the supervised entities is an excellent tool for identifying gaps in their understanding of the system. If a certain issue raises numerous questions or disagreements, it obviously requires clarification.

Example: The Israel Securities Authority (ISA) publishes on its website answers to general questions posed by supervised entities, as well as responses to preliminary inquiries made on concrete issues. The competent answers clarify the ISA's position on various issues. The responses to the preliminary inquiries are published without identifying the party that submitted the inquiry, but they are detailed enough to elucidate to the supervised parties how the ISA enforces the law and the requirements.

It is highly recommended to set up a telephone support center, or appoint a public service officer in the regulatory unit, to serve as an authority who can be approached by business owners for information on compliance with regulatory requirements.

Supervision actions (such as inspections) reveal various gaps between the requirements and actual compliance by the supervised entities. Therefore, at the conclusion of a supervision action it is important to advise the supervised entity (orally and in writing) on the matters requiring improvement. It is also desirable to give instructions on how this should be done. It is advisable to identify and focus on critical issues, given the limited resources of the supervision system and the supervised entities.

Additionally, it is recommended to adopt a certainty and stability code, according to which changes in the requirements and their interpretation will be made biannually at set times – for example, in the middle and end of the year. This allows supervised entities a period of at least half a year without regulatory changes, thereby increasing clarity and certainty and improving their capacity to adapt their activity to the requirements. Under this

code, any change in the requirements will become effective only after each supervised entity has been given prior notice and allowed a period of at least three and a half months in which to organize and adapt to the new requirements.

Deterrent and Proportionate Enforcement

In the preface to the Guide, guidelines were set forth on supervision, enforcement and deterrence. The following explains how to operate the supervision and enforcement system in order to achieve maximum deterrence.

Communication and explanation of the enforcement system.

When operating the enforcement system (imposition of administrative fines, filing of criminal indictments), it is important to explain the risk and the nature of the violation. It is recommended to relate to the following four parameters: (1) the nature of the risk (severity, annual number of cases, annual damage to the national economy); (2) the nature of the violation and its influence on the realization of the risk; (3) the benefit accruing to the violating entity from the violation; (4) the degree of

probability that violating entities will be caught. It is especially important to explain this to the court, in order to be able to request the imposition of a sanction that will make the violation unprofitable for supervised entities.

Enforcement and punishment adjusted for risk and compliance

The sanction for a violation must be suited to the character of the violating entity, to the nature of the requirement and to the circumstances of the violation (extent of the risk's effect, number of violations, gain from committing the violation, and number of actions performed by the supervised entity). The punitive system must take into account the balance between ensuring effective deterrence and encouraging future compliance.

A harsher punishment should be meted out for recurring violations by the same supervised entity (even if they are relatively minor violations). Punishment should also be more stringent in the case of an isolated violation with far-reaching risk implications. The fact that the multiple violations are minor in nature, or the fact that the violation is a one-time occurrence, will be taken into account

when determining the scope of the supervision, but not when deciding on the severity of the punishment.

There are supervised entities that may respond better to an administrative sanction (restriction of activity, suspension of an activity permit) combined with counseling on appropriate conduct, whereas other supervised entities may be deterred only by the imposition of a criminal penalty.

One should take into account the size and financial robustness of each violating entity. It should ensure effective deterrence, but not cause the collapse of small supervised entities.

It is also recommended to apply the principle, which has already been adopted in several countries, according to which sanctions should not be imposed on a business in its first year of operation. This allows the business time in which to learn the requirements and adapt its activity accordingly.

Compliance-enhancing enforcement – "compliance program"

- A highly effective method for correcting defects in compliance with regulatory requirements is

the "compliance program". When the head of supervision and enforcement identifies a violation of a material requirement, they allow the supervised entity to choose between one of two alternatives: punishment (usually the payment of a fine) or entry into a compliance program.

- If the supervised entity chooses the compliance program, it is given instructions to help it improve its compliance with the requirements and correct the defects that were found. Additionally, the supervised entity must prepare a detailed work plan including binding times for the correction of each defect, as well as a detailed budgetary plan for implementing the program. The program is submitted to the

supervision authorities and is contingent on their approval. If the supervision authorities approve the program, it becomes binding on the supervised entity. Any deviation from, or violation of the compliance program, results in a more severe punishment than the regular punishment (usually the payment of a fine at double the amount of the original fine).

- Many countries use this compliance program model for boosting compliance of supervised entities. It creates an incentive for supervised entities to improve their compliance (avoidance of the payment of a fine and entry on a structured track for the correction of defects), within a framework of close supervision and increased deterrence.

MONITORING AND EVALUATION

The monitoring and evaluation process described in Chapter B on risk management in the development of regulation and policy requires the collection of data. This helps ensure the regulation/policy is achieving its objectives and adapting to changes. The same data collection requirements also apply to the supervision and enforcement system. To avoid repeating the description of the topic in the previous chapter, this subchapter presents only issues that are specific to monitoring and evaluation of the supervision and enforcement system.

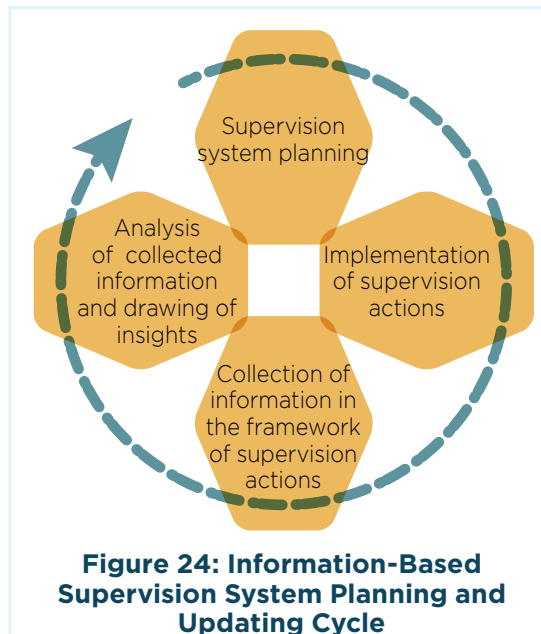
Monitoring and evaluation of the supervision system is intended to help the decision maker manage the system effectively and ascertain that they are achieving their objectives. Each year, the decision maker must be able to justify that the supervision and enforcement method applied based on current data that supports the risk analysis is achieving its objectives. This will show that the monitoring and evaluation are being performed correctly.

Unlike risk management in the development of regulation and public policy – in the supervision



and enforcement stage, risk management is performed on a daily basis, as part of the operating routine. Supervision actions can be regarded as activities dealing with information, including data collection (does the supervised entity comply with the requirements) and provision of information (encouragement of compliance, informational activities,

instruction). The monitoring and evaluation stage takes place as a matter of routine and is an integral part of planning the supervision activity. Supervision actions enable the collection of information and data used for monitoring and evaluation. This information should then be used to update the supervision and enforcement program.



Information should be collected on the following topics:

Up-to-date information on the supervision and enforcement system:

- First, basic data should be collected: How many supervision and enforcement actions are performed, how many actions were planned, and what percentage of these actions was carried out?
- It is necessary to know the amount of resources dedicated to the supervision and enforcement actions. Supervision actions may require government resources (working time of inspectors, budget) as well as private resources (setting aside time for the inspection, payment for laboratory tests, etc.). Collecting information about the private resources necessitated by the supervision actions requires communication with the public.
- How are the supervision resources provided by the government invested (distribution by: identity and groups of supervised entities, types of supervision actions, which elements are subjected to

supervision, where and when)?

- How are the enforcement resources provided by the government invested (investigations and claims – regarding which requirements and violations these actions are performed, and on which supervised entities)?

Current overview:

- Have there been changes in the risks (reduction/increase in the probability or severity of each risk)? Have there been changes in the risk treatment means? If so, how does this affect the supervision system planning and operation?
- Profile update for various supervised entities: Risk level, compliance level, reason for nonconformance, etc.
- Update of identity of supervised entities and of activity format: Who are the relevant supervised entities, where and when a supervised entity may create a risk, how a violation of the requirements can be identified, what is the level of compliance with the requirements for each supervised entity group.
- Has there been a change in

nongovernmental mechanisms, such as civil lawsuits and market forces, that can replace the need for government supervision actions?

Collection of information on the impact of the supervision and enforcement system

Has there been a change in the number of violations or in the level of compliance with the requirements? The compliance level most likely will change over time (which is actually the purpose of the supervision and enforcement system). It important to monitor this parameter and update it accordingly.

- Is it possible to identify violation patterns or compliance patterns (regarding defined supervised entity groups, all or any category of the elements subjected to supervision, and types of supervision actions)?
- Were some of the violations caused by ignorance, misapprehension or inability to comply with the requirements? If so, were they resolved, and how?
- Are the supervision and enforcement tools suited to the needs of the supervision and

enforcement system?

- Do the supervision and enforcement tools create sufficient deterrence among supervised entities that violate the requirements out of deliberate criminality?

An effective method for monitoring the changes over time is through the definition of key risk indicators (KRI).

Key risk indicators are quantifiable and measurable variables on which information can be collected and which can be compared over time. KRIs allow the decision maker to measure the compliance level over a period of time. Possible KRIs are: number of identified violations, number of complaints received, number of compliance programs set, number of indictments filed/fines imposed, number of realized risks (accidents that occurred, infections that were imported from other countries, people that became ill), etc.

Comparing the KRIs to earlier periods provides an indication of whether the supervision system is achieving its objectives. When studying these indicators, the number of supervised entities and actions that were added or subtracted with respect to the

population of supervised entities should be taken into account. This is so background changes will not distort the measurement.

Illustrative example: Initially, 500 businesses operated in the non-bank credit market, with an average of 80 violations discovered each year. Subsequently, the market structure changed to include 1,000 businesses, and the number of violations rose to 90. At first glance, it appears the compliance level fell; but when taking into account the number of supervised entities and the actions performed, it becomes obvious there was a relative decrease in the number of violations.

Guidance on the collection and use of information:

- In principle, data should not be used that is directly affected by a government unit's actions (number of supervision actions, number of indictments filed). This data describes government actions rather than the compliance level of the supervised entities. Moreover, these are "dependent variables" that can be influenced by government actions. Relying

on them could create a negative incentive or distort the inspectors' activity in the field.

- It is important to define in advance data that can be measured consistently and effectively, and to ascertain that such data is accessible and can be collected.
- It is advisable to use information technology to allow focusing on the risks. Manual data collection wastes resources and is subject to bias and error. Furthermore, methodical collection by means of information technology enables big-data collection.
- It is recommended to cooperate with related government units and with government units that deal with the same supervised entities. Different government agencies deal with the same risks or the same risk creators, in different stages and from different aspects (e.g., the Ministry of Health has data on medical treatments provided by the HMOs and the hospitals. Since many government ministries are involved in safeguarding human health and life, information that is collected by the Ministry of

Health can be used by decision makers in other ministries). It is recommended to share and disseminate information through shared IT systems that are available to several government units.

- It is recommended to consider employing an external party to perform data collection and processing. The use of an external party frees up time for the supervision and enforcement units to perform their core tasks. It also removes the concern that information will be collected selectively or be affected by the inspectors' behavioral patterns.
- It is important to collect basic information on entities that are not subject to supervision and enforcement actions. The government cannot and need not investigate everyone, but only entities that pose a danger. As a result, supervision actions themselves provide the government only with partial information about the supervised entities, relating mainly to dangerous entities. This naturally creates a biased picture of the reality. It should be noted that

the supervision and enforcement system collects information mostly on serious cases, and that the information contained in the system does not reflect the situation throughout the market. Hence, care must be taken to collect information from all the relevant entities. Another alternative is to perform a market-wide survey (based on statistical criteria) once every few years. Such a survey can serve as a "reality check" regarding the situation in the entire market.

- Apart from monitoring supervision enforcement actions, it is important to review and investigate events involving risks that were realized. Lessons can be learned from these events about the usefulness of complying with the requirements (e.g., the extent to which compliance with a requirement reduced the severity of the damage) and about the attributes of non-compliant supervised entities.
- There are two key changes that could impact the future operation of the enforcement and supervision system. One is a change in the supervised entities'

compliance level as a result of the supervision and enforcement. The other is a change in the extent of the damage due to violations of the requirements (violations which previously were perceived as dangerous, and violations which previously were perceived as insignificant). Such changes necessitate a redistribution of the supervision resources and a change in the type of tools to be applied in each case.

- After reviewing the information, the decision maker may decide to reallocate the resources among the different actions and elements. Broader powers of search and investigation may be applied and the hierarchy of sanctions available to the enforcement system could be expanded. Another option taken may be that of supervising a previously unsupervised population sector or to discontinue supervision of a previously supervised population sector (e.g., when there is an alternative mechanism that makes government supervision redundant).

BIBLIOGRAPHY
AND REFERENCES
FOR FURTHER
READING

אבי טבח "אכיפת חוק אופטימלית ואי-שוויון כלכלי" משפט ועסקים ט, 347 (2008). זמין ב:
<http://portal.idc.ac.il/he/lawreview/volumes/volume09/documents/tabbach.pdf>

גיא מור "הנשר הגדול בשוק ההון - דיני עשיית עושר ולא במשפט ככלי לאכיפת הפרות כלכליות" תאגידים
י(3), 273 (2013). זמין ב:
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2439653

הארגון לשיתוף פעולה ולפיתוח כלכלי (OECD) "ארגו כלים להערכת השפעותיה של הרגולציה על התחרות
- כך 1: עקרונות" (גרסה 2.0, 2011).

הארגון לשיתוף פעולה ולפיתוח כלכלי (OECD) "ארגו כלים להערכת השפעותיה של הרגולציה על התחרות
- כך 2: הדרכה" (גרסה 2.0, 2011).

זוהר גושן "בעיית הנציג כתיאוריה מאחדת לדיני התאגידים" עורכים: אהרון ברק, נפתלי ליפשיץ, אוריאל פרוקצ'ה,
מרדכי ראבילו חיבור משפטי, ספר זיכרון לגואלטיה פרוקצ'ה 239 (1996).

משרד האוצר משרד התחבורה והבטיחות בדרכים "נוהל פר"ת 2012" (2012). זמין ב:
<http://media.mot.gov.il/PDF/HofesMeyda/NOALP/Prat2012.pdf>

משרד ראש הממשלה "מדריך ממשלתי - הערכת השפעות רגולציה - מדריך RIA" (2015).

משרד ראש הממשלה "מדריך הפחתת הנטל הרגולטורי" (2015).

נעם זוסמן ודורון סייג: התחלקות הסיוע בשכר דירה בין שוכרים למשכירים: המקרה של סטודנטים במרכז
ירושלים", בנק ישראל - חטיבת המחקר (2015).

ניר בקר "ערך חיי אדם סטטיסטי" הוגש למשרד להגנת הסביבה (2012).

עומר זליבנסקי ויובל פלדמן "שיטות התנהגותיות לטיוב עבודת הממשל" המכון הישראלי לדמוקרטיה - כנס
אלי הורוביץ לכלכלה וחברה (טיוטה, 2017). זמין ב:
<https://tinyurl.com/EHC17c>

קבוצת פארטו "מתודולוגיה לחישוב רווחי מזהמים" (עורכים: ניר בקר ואורן פרז), הוגש למשרד להגנת הסביבה
(2012).

AS/NZS ISO 31000, Risk management - Principles and guidelines (2009).

Australia, Queensland Government, A Guide to Risk Management (2011).

Australia, the Australian Government Guide to Regulation (2014).

Australia, Risk Management Framework Policy (2014).

COSO, Enterprise Risk Management - Integrating with Strategy and Performance (2017).
url: <http://www.coso.org/IC-IntegratedFramework-summary.htm>

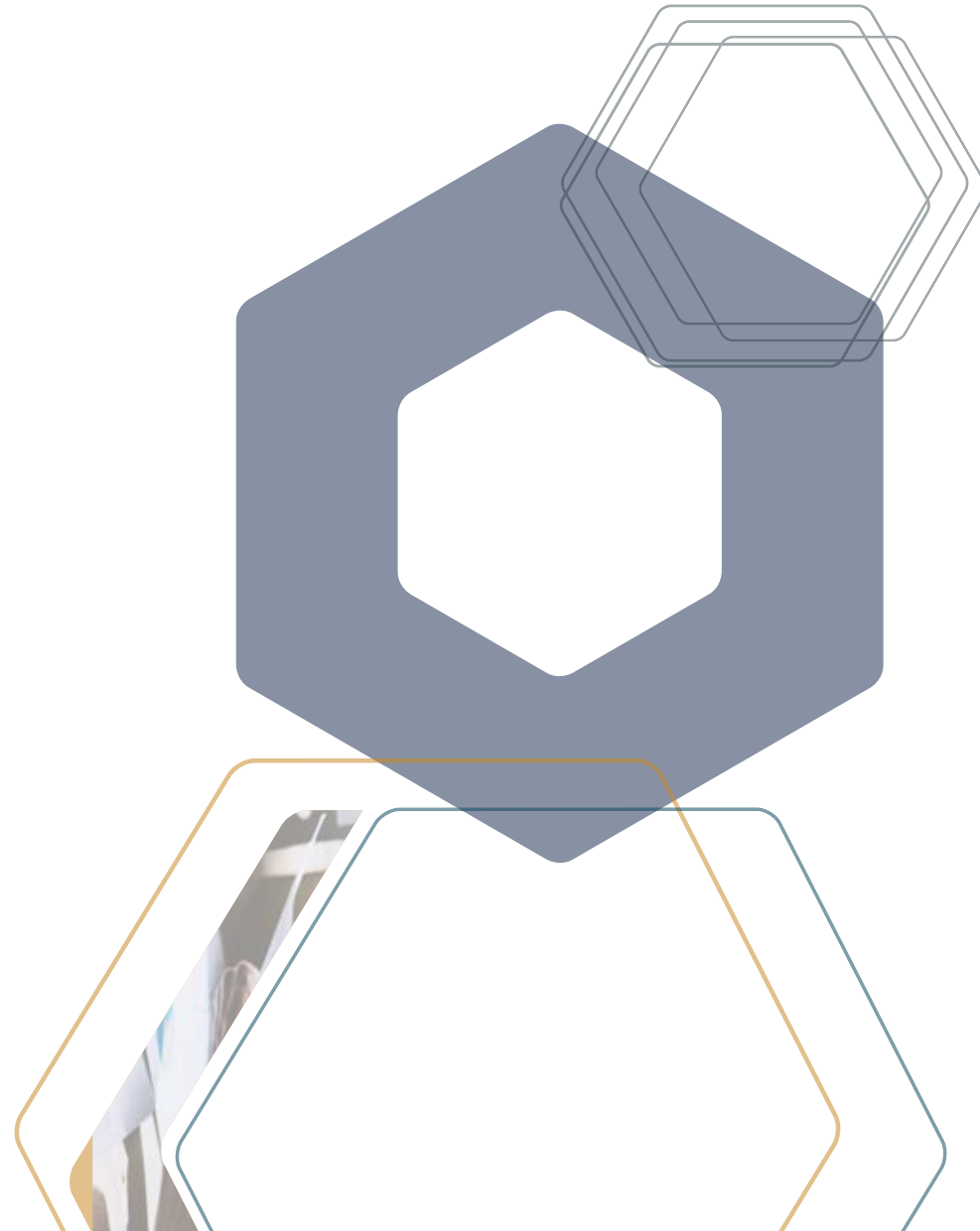
European Commission, Smart Regulation in the European Union (2010). url: <http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=COM:2010:0543:FIN:EN:PDF>.

European Commission, Report From The Commission To The Council And The European
Parliament - Minimizing regulatory burden for SMEs: Adapting EU regulation to the needs
of micro-enterprises, COM(2011) 803 (23.11.2011). url: [http://ec.europa.eu/smart-regulation/
better_regulation/documents/minimizing_burden_sme_en.pdf](http://ec.europa.eu/smart-regulation/better_regulation/documents/minimizing_burden_sme_en.pdf)

European Commission, Communication From The Commission To The European
Parliament, The Council, The European Economic And Social Committee And The Committee
Of The Regions - Smart Regulation - Responding To The Needs Of Small And Medium -
Sized Enterprises, COM(2013) 122 (7.3.2013). url: [http://eur-lex.europa.eu/LexUriServ/
LexUriServ.do?uri=COM:2013:0122:FIN:EN:PDF](http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=COM:2013:0122:FIN:EN:PDF)

European Commission, Commission Staff Working Document - Better Regulation
Guidelines (2015). url: [http://ec.europa.eu/smart-regulation/guidelines/docs/swd_br_
guidelines_en.pdf](http://ec.europa.eu/smart-regulation/guidelines/docs/swd_br_guidelines_en.pdf)

European Commission, Better Regulation Toolbox (2015). url: [http://ec.europa.eu/smart-
regulation/guidelines/docs/br_toolbox_en.pdf](http://ec.europa.eu/smart-regulation/guidelines/docs/br_toolbox_en.pdf)
France: Autorité de la concurrence, Guide for Competition Impact Assessment of Draft
Legislation (2012).



Gary Stanley Becker, Crime and Punishment: An Economic Approach, 76 J. POLIT. ECON. 169 (1968).

Heinz-Peter Berg, Risk Management: Procedures, Methods and Experiences, RT&A 2(17) Vol. 1 (2010).

IEC/ISO 31010, Risk management – Risk assessment techniques (2009).

ISO 31000, Risk management – Principles and guidelines (2009).

ISO/IEC 17020, Conformity assessment -- Requirements for the operation of various types of bodies performing inspection (2012).

IRGC, Risk Governance: Towards an Integrative Approach (2006).

IRGC, Risk Governance Deficits: An analysis and illustration of the most common deficits in risk governance (2009).

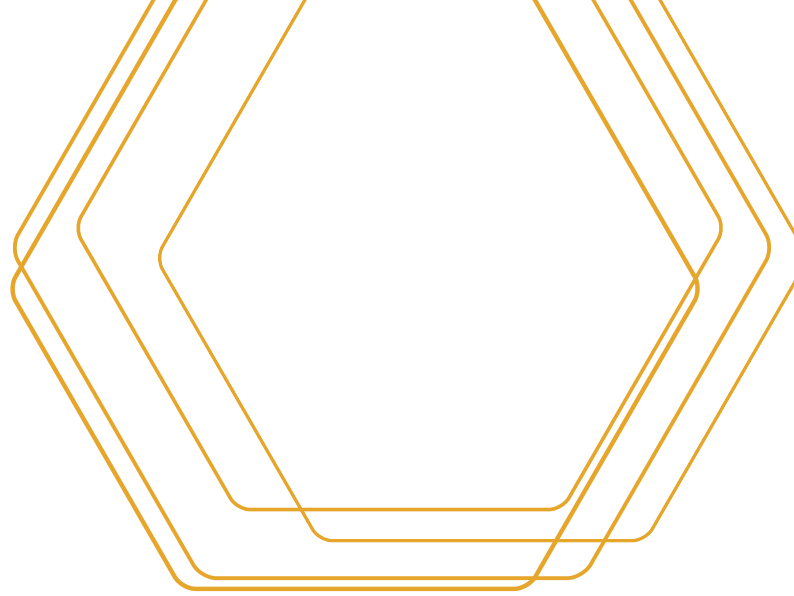
Jachia Lorenza and Valentin Nikonov, Applying risk management concepts in the design of legislation, in organizational and regulatory issues of public-private cooperation for trade facilitation (2011).

Jachia Lorenza and Valentin Nikonov, Effective regulatory processes for crisis management: an analysis of codified crisis management in Europe, in governing disasters: the challenges of emergency risk regulation (2011).

Louis Anthony Cox Jr., What's Wrong with Risk Matrices?, 28(2) Risk Anal 497 (2008).
Mercatus Center and Sherzod Abdukadirov, The Unintended Consequences of Safety Regulation (2013).

Nederland, Actal, Actoren en Factoren: Over Regeldruk en het Samenspel Van Beleid en Toezicht (2015).

New Zealand, Ministry of Economic Development, Code of Good Regulatory Practice



(2006). url: http://www.med.govt.nz/templates/MultipageDocumentTOC_____22149.aspx
New Zealand, Ministry for Primary Industries, Biosecurity: Risk Analysis Procedures (2006). url: <https://www.mpi.govt.nz/document-vault/2031>

Barack Obama, Toward a 21st Century Regulatory System, Wall Street Journal (18.1.2011). url: <http://online.wsj.com/article/SB10001424052748703396604576088272112103698.html#mjQuickSave>

OECD, Regulatory Impact Analysis – Best Practices in OECD Countries (1997).

OECD, Reducing the Risk of Policy Failure: Challenges for Regulatory Compliance (2000).

OECD, Regulatory Policies in OECD Countries: from Interventionism to Regulatory Governance (2002).

OECD, Building an Institutional Framework for Regulatory Impact Analysis: Guidance for Policy Makers (2008).

OECD, Risk and Regulatory Policy: Improving the Governance of Risk (2010).

OECD, Better Regulation in Europe – Executive Summaries (2010).

OECD, Reform of Regulatory Enforcement and Inspections in OECD Countries (2012).

OECD, Inspection Reforms: Why, How, and With What Results (2012).

OECD, Regulatory Enforcement and Inspections, OECD Best Practice Principles for Regulatory Policy (2014). url: http://www.keepeek.com/Digital-Asset-Management/oecd/governance/regulatory-enforcement-and-inspections_9789264208117-en

OECD, OECD Regulatory Compliance Cost Assessment Guidance (2014). url: <http://dx.doi.org/10.1787/9789264209657-en>

OECD, Competition Assessment Toolkit - Volume I: Principles (Version 3.0, 2016).

OECD, Competition Assessment Toolkit - Volume II: Guidance (Version 3.0, 2015).

OECD, Competition Assessment Toolkit - Volume III: Operational Manual (Version 3.0, 2015).

OECD, Rana Roy and Nils Axel Braathen, The Rising Cost of Ambient Air Pollution thus far in the 21st Century: Results from the BRIICS and the OECD Countries, OECD Environment Working Papers, No. 124, (2017). url: <http://dx.doi.org/10.1787/d1b2b844-en>

Richard H. Thaler and Cass R. Sunstein, Nudge: Improving Decisions about Health, Wealth, and Happiness (2008).

Stanley Kaplan and John B. Garrick, On the Quantitative Definition of Risk, Risk Analysis, 1: 11-27 (1981).

The Institute of Risk Management, A Risk Management Standard (2002).

The Institute of Risk Management, Bribery Risk Guide (2016).

United Kingdom, Health & Safety at work, Reducing Risks: Protecting People - HSE's decision making process (2001). url: <http://www.hse.gov.uk/risk/theory/r2p2.pdf>

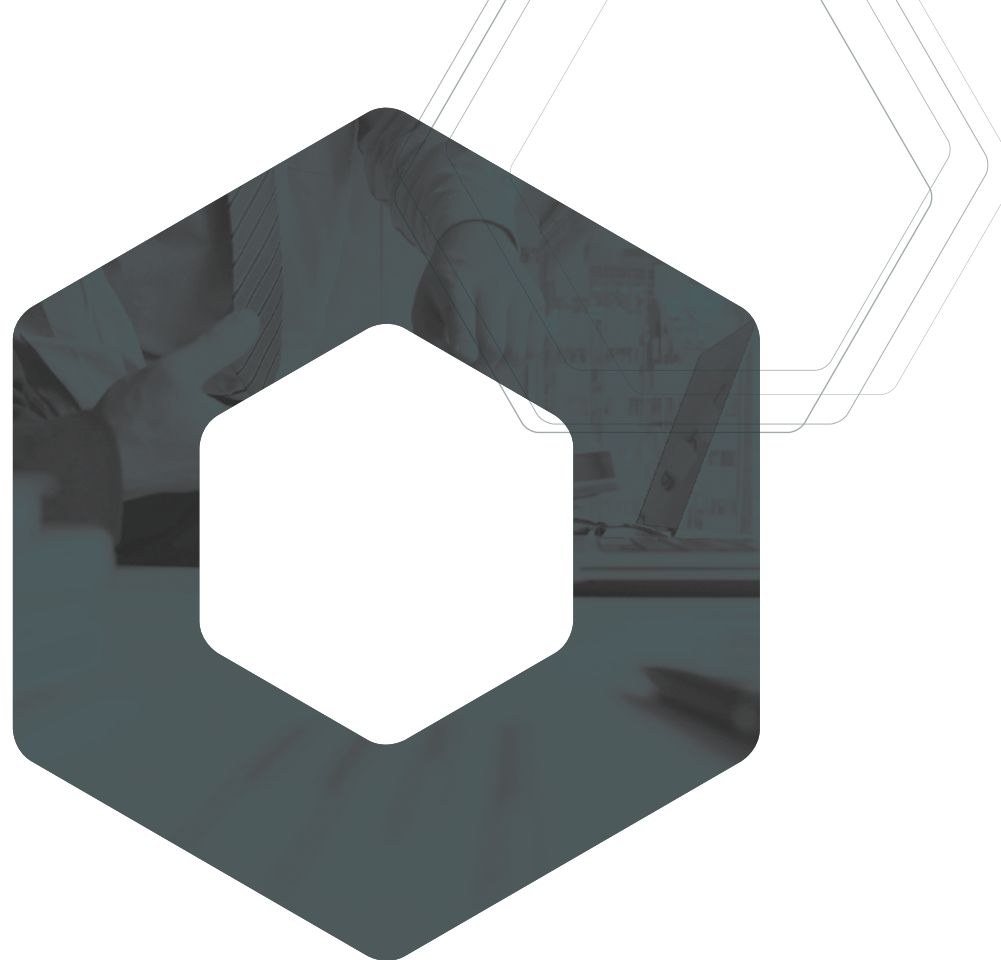
United Kingdom Treasury, The Orange Book: Management of Risk - Principles and Concepts (2004).

United Kingdom Treasury, Managing risks to the public: appraisal guidance (2005).

United Kingdom Treasury (Philip Hampton), Reducing administrative burdens: effective inspection and enforcement (2005).

United Kingdom, Better Regulation Task Force, Regulation - Less is More - Reducing Burdens, Improving Outcomes (2005).

United Kingdom Treasury, Thinking About Risk - Managing Your Risk Appetite: A Practitioner's Guide (2006). url: <http://webarchive.nationalarchives.gov>



[uk/20130107005526/http://www.hm-treasury.gov.uk/d/tar_practitioners_guide.pdf](http://www.hm-treasury.gov.uk/d/tar_practitioners_guide.pdf)

United Kingdom, Better Regulation Executive, Regulatory Justice: Making Sanctions Effective - Final Report (2006).

United Kingdom, Regulatory Enforcement and Sanctions Act 2008.

United Kingdom, RRAC, Response with responsibility, Policy-making for public risk in the 21st Century (2009). url: <http://www.berr.gov.uk/files/file51459.pdf>

United Kingdom, RRAC, An introduction to public risk (2009).

United Kingdom, Department for Environment, Food & Rural Affairs, Economics and Farm Animal Welfare (2011). url: http://www.fao.org/fileadmin/user_upload/animalwelfare/Report_on_Economics_and_Farm_Animal_Welfare.pdf

United Kingdom, Competition and Markets Authority, Competition Impact Assessment, Part 1: Overview (2015)

United Kingdom, Competition and Markets Authority, Competition Impact Assessment, Part 2: Guidelines (2015)

United Kingdom Treasury, The Green Book: Appraisal and Evaluation in Central Government (2016). url: <https://www.gov.uk/government/publications/the-green-book-appraisal-and-evaluation-in-central-government>

United Kingdom Treasury, Management of Risk in Government (2017). url: https://www.gov.uk/government/uploads/system/uploads/attachment_data/file/584363/170110_Framework_for_Management_of_Risk_in_Govt__final_.pdf

United Kingdom, Food Standards Agency, Food Law Code of Practice (England) (2017). url: https://www.food.gov.uk/sites/default/files/food_law_code_of_practice_2017.pdf

United Kingdom, Food Standards Agency, Safer food, Better Business. url: <https://www.food.gov.uk/business-industry/sfbb#toc-2>

United Nation Economic Commission for Europe, What Happened In Uganda? Declining HIV Prevalence, Behavior Change, and the National Response. Janice A. Hogle, ed. (2002). url: www.usaid.gov/our_work/global_health/aids/Countries/africa/uganda_report.pdf

United Nation, World Health Organization and Food and Agriculture Organization, Codex Alimentarius, Working Principles for Risk Analysis for Food Safety for Application by Governments (2007).

United Nation Economic Commission for Europe, Recommendation on Risk Management in Regulatory Systems (2011). url: <http://www.unece.org/trade/wp6/recommendations/recommendations.html>.

United Nation Economic Commission for Europe, Recommendation on Crisis Management in Regulatory Systems (2011). url: <http://www.unece.org/trade/wp6/recommendations/recommendations.html>.

United Nation Economic Commission for Europe, Risk Management in Regulatory Framework: Towards a Better Management of Risks (2012). url: http://www.unece.org/fileadmin/DAM/trade/Publications/WP6_ECE_TRADE_390.pdf

United Nation, Guidelines for Assessing the Compliance of Regulatory and General Administrative Acts with Competition Rules (2013).

United Nations, Economic Commission for Europe (Steering Committee on Trade Capacity and Standards, Working Party on Regulatory Cooperation and Standardization Policies), Draft Recommendation S on “Applying predictive risk management tools for targeted market surveillance” (21.9.2016).

United Nations, Economic Commission for Europe (Steering Committee on Trade Capacity and Standards, Working Party on Regulatory Cooperation and Standardization Policies), Report of Working Party on Regulatory Cooperation and Standardization Policies on its twenty-sixth session (2.12.2016).

United States Executive Order 12866 - Regulatory Planning and Review (1993).



United States Congress Office of Technology Assessment, Environmental Policy Tools: A User's Guide (1995).

United States, Small Business Regulatory Enforcement Fairness Act (SBREFA), P.L. 104-121, 1996.

United States, Office of Management and Budget and Office of Science and Technology Policy, Updated Principles for Risk Analysis (19.9.2007).

United States Chief Financial Officers Council & Performance Improvement Council, Playbook: Enterprise Risk Management for the U.S. Federal Government (2016).

Van Der Hofstede, Mogelijkheden Voor Lastenluwe(re) Handhaving: Onderzoek Naar De Oorzaken Van Ervaren Handhavingslasten Door Ondernemers (2015). url: <http://www.actal.nl/wp-content/uploads/060-Ministerie-van-EZ-Bijlage-1-bij-brief-lastenluer-Handhaving-DLV-rapport-definitief-v1.1.pdf>

World Economic Forum, Rethinking Risk Management in Financial Services: Practices from other domains (2010).

Risk management process for [Complete]: Asset Registry

The key types of risks to be protected against are: [Complete] [For example: Risks to lives due to a criminal or security incident]

Serial Number	Asset Name	Description	Priority	Remarks

Risk management process for [Complete]: Summary of the Current Situation

The main types of assets to be protected are: [Complete]

[For example: the health of consumers of soya milk, occupational safety in steel plants]

The main types of risks to be protected against are: [Complete]

[For example: introduction of fake diamonds into the market; use of insider information in companies traded on the stock exchange by legal consultants]

Description of the existing activity [Complete]

[For example: the number of companies active in the market, the work processes, key players, activities with the greatest risk exposure]

Key processes / key players [modify as necessary] [Complete]

[Use of charts is recommended: flowcharts, organizational layout, charts, decision trees, etc.]

The existing legal and regulatory framework [Complete]

[For example: laws applicable to the activity or the relevant players, government intervention through licenses, tax benefits, supervision and enforcement activities executed, etc.]

ANNEX C: MAPPING RELEVANT STAKEHOLDERS

Risk management process for [Complete]: Mapping Stakeholders

1. Stakeholders in the inner-organizational circle:

2. Stakeholders in the government-wide circle:

3. Stakeholders from among the public:

- a. Parties injured by the risk:
- b. Causes or contributors to the risk:
- c. Associated with the place, time, or framework in which the risks occurred:
- d. May be impacted by changes in the regulation or public policy:
- e. Representative organizations (employees, employers, NGOs):
- f. Experts

ANNEX D: INSTRUCTIONS FOR CONDUCTING A PUBLIC DISCOURSE: RECEIPT OF INFORMATION, PARTICIPATION, AND CONSULTATION

Before embarking on a public discourse, four questions need to be answered:

What are the main questions that need to be clarified?

What are the questions concerning the risks? What are the questions that need to be clarified about possible directions for action? Do they concern quantitative data? Is it a qualitative assessment? Are they questions about a professional content world?

What is the purpose of the discourse?

With respect to each issue – is information lacking? Does existing information need to be verified and validated? Is there a need for ideas for solutions? Is there a need to learn from the previous experience of stakeholders or to listen to their opinions about an idea that is being considered for implementation etc? Once the issues for discourse and the purpose of the consultation have been defined, then the next steps can be taken.

Who should the discourse be held with?

A brief mapping of all the relevant stakeholders must be undertaken. It must also be determined whether it is a good idea to have them meet

together or to hold separate meetings for different groups (the public, business proprietors, importers, exporters, employers, employees, experts, etc.).

It is always a good idea to conduct a separate in-house process, that precedes the external process, in order to prepare for the public discourse. The intra-governmental partners should also participate in the meetings with the public. This way, they will be able to gain an impression from what is said and respond and participate themselves.

To the extent that regular forums are held (of participants, supervised entities, advisory committees), a dedicated discussion with them should be considered; their help can also be enlisted to reach relevant stakeholders.

How will the discourse be integrated in the process?

At what stages should the discourse take place, at what depth at each stage, and what are the appropriate tools in each stage (web survey, discussion, in-depth interview).

Considerations in planning a discourse meeting:**Meeting venue and date**

A meeting venue that is accessible to most of the participants should be arranged. Consider geographic area, parking, access by public transportation, and accessibility for the disabled. In the appropriate instances, consider going to the participants and being hosted their location. Also, consider the quality of the meeting venue in terms of atmosphere, acoustics, etc. These variables can impact the quality of the meetings.

Check the date in advance. Make sure

it does not fall on a religious holiday and that the time is convenient for most of the participants. Also consider the characteristics of the participants: if they are parents, make sure that schools are open on that day. If they are members of a specific industry, make sure that the timing fits in with the constraints of the industry (tourism companies are very busy in the summer and over public holidays; the food industry is very busy before major holidays).

Managing the meeting

Define in advance who will open the

meeting and how it will be conducted (a round of viewpoints, splitting of the time between the questions, who gives the right to speak). It is important to be aware of and to balance the various interests and the various groups. It is also vital that weaker groups have the ability to express their position (small businesses, entities that are not incorporated, private individuals, and participants who have difficulty expressing their position). Define in advance the role that you fulfill: moderator, listening only, and whether you will use another person to help you manage the meeting.

Points of emphasis for the contents of the discourse meeting:**Upon the initiation of any discourse, it is important that expectations are coordinated among the participants.**

Explain the aim and significance of the discourse. In what context it is being conducted, and at what stage of the risk management process it is incorporated. Explain who will use the information given and what influence they have. It is important to define the topics for discussion, and to define which topics are not open for discussion.

Check the participants' expectations.

Verify that everyone is on the same

page regarding the aims of the meeting and its possible impact. Take into account that some of the participants might want to talk about topics that were not planned for inclusion in the discussion. Listen with an open mind to suggestions. Suggestions don't have to be accepted, but it is important to understand these suggestions reflect the motivations of the participants in the discourse. It is preferable to include them in the discussion to the extent possible.

Summarize briefly at the end of the meeting.

Summarize the main points raised, and in particular, the things that the meeting highlighted that were new. Mention again what happens from this point on – is this a one-time meeting? Is a follow-up meeting expected? Will a summary be circulated? How will the participants ultimately know that what they said was taken into account?

It is always a good idea to circulate to the participants a summary of the main points raised in the discourse, and express appreciation for their participation in the meeting and the information that they provided.

The quantity of the meetings

In situations in which separate discussions are held with multiple groups, decide whether to make a concerted effort and hold a series of meetings on the same day or to separate between them. If the latter, there is time to learn lessons from one meeting to the next.

In cases that appear to require more than one meeting with stakeholders or with a specific group, consider coordinating expectations in advance and preparing accordingly. Multiple meetings will probably be needed in the following instances: in the case of a highly complex subject or a very broad

issue; time is required to thoroughly acquaint the participants with the issues; the discourse is designed to solve a difficult problem; the discourse is designed to formulate a new policy approach or a new way of working.

Use of questionnaires and surveys

Questionnaires and surveys are readily available useful tools. They require relatively limited inputs and enable a large number of people to be reached. The tool is especially useful when seeking to gather well defined information or ideas, and there is no need to clarify or work through the issue together.

Principles for the use of surveys and questionnaires:

- Be sure to keep the questionnaire short.
- Distribute the questionnaire in as diverse a manner as possible.
- Formulate short and clear-cut questions – simple sentences, one topic per question.
- For each question, check whether the answer is really needed to advance the risk management process.
- If an open-ended response is permitted, restrict the length of the answers (words/pages) in order to prevent being overwhelmed.
- Always leave an open field in which respondents can write things that they want to raise. E-Gov has developed a survey tool that is readily available and easy to use.

To use the survey tool, look through the E-Gov catalog of services: <http://catalog.govpro.gov.il>

Enlist the support of an external operator specializing in participation processes

Alongside the methods of participation mentioned in the previous page, a discourse may be needed that necessitates greater resources (for example: due to the complexity of the subject matter, the number of stakeholders, consultation with experts, holding of a larger number of meetings). In such situations, the support of operators of participation services can be enlisted via a special framework tender: Central Tender No. 2015-23: "Planning and Execution of Participation Processes in Government Ministries". This tender enables Government Ministries to hold a discourse using an external operator with expertise in participation processes.

The tender applies to all instances of the discourse, be it intra-ministerial or inter-ministerial, participation of experts, participation of organizations, or public participation. It should include a wide variety of participation services, including physical participation services (such as focus groups, roundtables and

conferences), online participation services, and supplementary services (content reviews, etc.).

Instructions for using the tender were established in the provisions of Mandatory Tender Regulations 16.9.0.3: "Planning and Execution of Participation Processes in the Work of Government Ministries": <http://mof.gov.il/takam/Pages/horaot.aspx?k=16.9.0.3>

In Israel, the support of the public participation team in the Governance and Social Affairs Department in the Prime Minister's Office can be enlisted for all matters pertaining to the planning of the consultation and participation steps: shituf@pmo.gov.il

This Annex was prepared on the basis of the Guide to Social Procurement, prepared by the Department Governance and Social Affairs Department in the Prime Minister's Office (July 2017): http://il1.activepoint.com/Government_tenders_Guide2017



ANNEX E: RISK REGISTRY

Risk management process for [Complete]: Risk Registry

The key types of risks to be protected against are: [Complete] [For example: Risks to lives due to a criminal or security incident]

Serial Number	Risk name	Description of risk scenario	Asset type	The possibility of detecting the risk	Necessary conditions	Normative cause of the risk

Various qualitative influences (society, environment, economics, international trade)	Description of the harm	Damage from one occurrence of the risk	Annual probability of occurrence of the risk	Degree of certainty in quantifying the probability and the damage from the risk (Low, medium, high)	The annualized loss expectancy

ANNEX F: COMPARISON OF REGULATORY METHODS AND POLICY TOOLS ACCORDING TO THE DEGREE OF GOVERNMENT INTERVENTION

The inverted pyramid presents groups of regulatory methods. The pyramid is arranged in order of degree of government intervention. The greater the degree of intervention, the greater the resources required – government and private.

Greater intervention does not guarantee a more effective solution to the risks. Permits and licenses are the most burdensome regulatory method with the greatest degree of intervention, but additional methods are effective in addressing other risks. Moreover, there are many risks that this treatment method is not suited for.

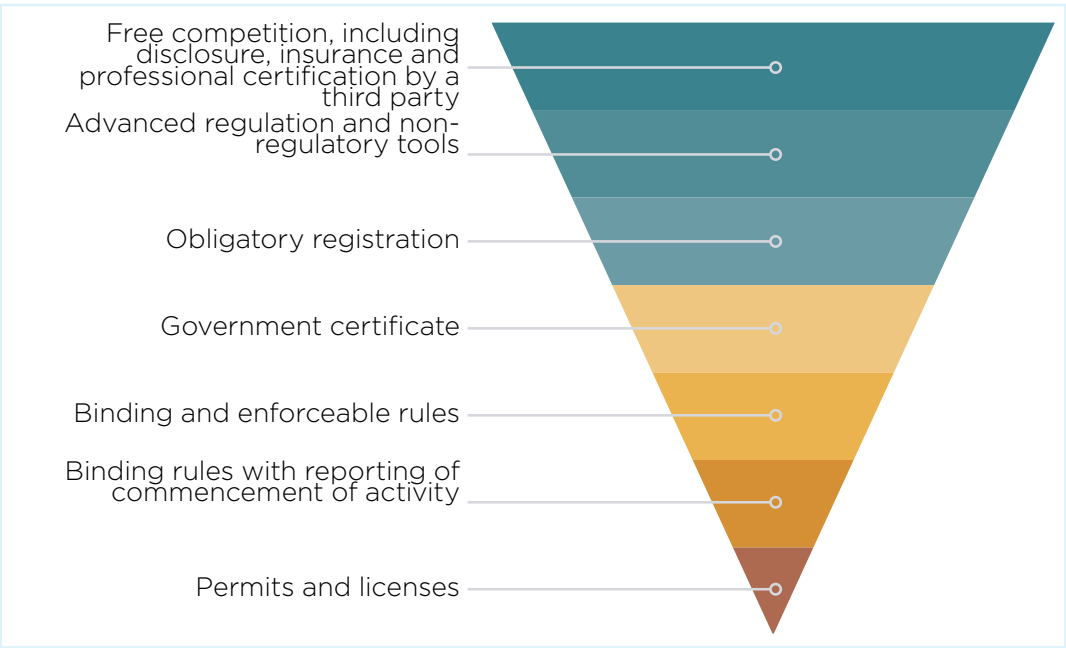


Figure 14: The inverted pyramid – regulatory methods and policy tools according to the degree of government intervention

It is preferable to start looking for solutions based on the methods at the top of the pyramid. These incur little government intervention.

The following is a summary description of the various methods specified in figure 14.

Free competition, including due diligence, insurance and professional certification by a third party

The terms of free competition provide a broad choice and create an incentive for cautious behavior. Sometimes a prior government policy has resulted in the creation or amplification of a risk. Therefore, the option of free competition should be considered, as well as examining whether previous government intervention adversely impacts this and prevents the market from operating safely and efficiently. Systems of free competition can include additional voluntary components:

Voluntary disclosure of information.

Problems of information flow and coordination can be resolved by the relevant players taking the initiative and providing information.

Professional qualification by a third party.

The players in a specific field can approach the certification body at their own initiative in order for the body to examine them and grant a certificate of competency. The certificate does not grant rights but informs the public that the certificate holder complies with standards examined by a third party. ISO standards are an example.

Insurance and guarantees.

The use of insurance and guarantees enables players to manage their risk by transferring that risk. As such, if the risk is realized, the injured parties can be compensated. The advantage of using insurance is that the insurance premium prices the risk of the insured. Consequently, the price of the premium encourages the insured to act with greater care. An example of this can be found in drivers' third-party insurance (insurance that covers property damage caused to a third party).

Advanced regulation and nonregulatory tools

These methods and tools are designed to change behavior without enforcing compliance by government measures. Advanced regulations include economic incentives, employing of

behavioral economics, and enacting grounds for a civil lawsuit.

Against the backdrop of the discussion on risk, it is important to remember that existing legislation may already address the risk. To begin with, the general law of torts enables the injured party to sue the party that caused the damage. Another example is the Israeli Consumer Protection Law and Standard Contracts Law, which provide consumers with a mantle of protection from suppliers who might harm them. A third example is commercial torts law – designed to prevent commercial losses through the stealing of trade secrets, infringement of intellectual property, passing off, or false use of product names.

Obligatory registration

Supervised entities can be required to furnish the government with details about themselves and the supervised activity (name, address, description of the proposed services) before commencement of the activity. A requirement to register ensures players operating in the field must identify themselves. This simple requirement can deter problematic players from executing an activity as their details

are known and they are traceable.

Government certificate

Using this method, the government issues a certificate stating that a person has specific competencies; however, there is nothing to prevent a person who does not possess the government certificate from engaging in that field. The use of a specific title is limited only to those who possess the government issued certificate but working in the specific field is also possible without a certificate. An example of this is accountants: in many countries, any person may engage in accountancy, but only those who received government certification may add the title "certified public" to their name.

Binding and enforceable rules

This concerns binding and enforceable rules, the violation of which are punishable; rules such as: prohibition of an activity, duty to act, or conditions regarding the manner of executing the activity. Binding rules may be in the realm of information: duty of disclosure, obligation to publish, duty to report, or the obligation to record and save information. Binding rules may impose an obligation on an individual to

adopt an internal risk management system – by means of an external standard (for example: ISO 31000 Risk Management), obligation to perform control activities or obligation to have insurance. Rules require the operation of a governmental supervision and enforcement system.

Binding rules with reporting of commencement of activity

A higher level of binding rules includes the duty to report commencement of activity. This requirement means the government knows who the active players in the field that are subject to the binding rules. All players must report intent to commencement activity prior to implementation of their activity.

This method constitutes a combination of obligatory registration and binding rules. It is an intermediate level before imposing an obligation to obtain permits and licenses.

Permits and licenses

Permits and licenses are the most stringent and comprehensive method of intervention that a government can adopt. Using them, the government prohibits execution of an act or activity in a field until the party is granted

a permit or license. This method greatly compromises the choices of the individual and requires extensive government and private resources in order to issue individual authorizations. Therefore, it is recommended the government decision-maker considers this a last resort after all the other methods have been reviewed.

ANNEX G: GUIDELINES FOR CHOOSING REGULATORY METHODS AND POLICY TOOLS

The rules presented in this Annex are designed to help adapt the best possible solution to a public problem. This means the solution is effective in resolving the problem and efficient in terms of the resources that it incurs. The annex is divided into five sections.

The first section presents various rules of thumb [according to the risk characteristics](#). Among others, rules are presented for supervision of market entry as opposed to supervision of routine activity; the need to collect information on the supervised entities; handling of information-related market failures; handling of concentrated markets and the normative cause of the risk.

The second section presents [regulatory methods and policy tools](#) according to their suitability in different situations. The tools are divided up according to their competitive impact.

The third section presents rules regarding [the information and the public comprehension of the risk](#). The better the public understands the risk, the more effective measures it can take to deal with it regardless of regulations or government policy.

The fourth section presents guidelines

for using advanced regulation and non-regulatory tools. This is a set of less well-known methods, and as such, an explanation is provided about when to use them. In addition, implementations are presented for improving and strengthening the civil lawsuit apparatus, use of behavioral economics, and resolution of problems of coordinated action between multiple parties.

The fifth section sets out the advantages and the disadvantages of the key regulatory methods and policy tools: permits and licenses, binding rules and advanced regulation and non-regulatory tools.

Rules of thumb according to the risk characteristics

Treating of risks that emerge during the commencement of activity versus risks occurring during routine activity:

Permits and licenses are suitable when a preliminary check is required before the execution of an activity, while binding rules are imposed on the public throughout the period of execution of the activity (after entry into the market).

The use of permits and licenses is particularly suitable if faced with a high probability risk of great severity, and when the risk results from gaps in competency (knowledge, skills, experience) or from gaps in the equipment (defective or unsuitable equipment).

Permit and license-type tools necessitate extensive resources – both from the government and the supervised entities. Consequently, limiting the use of them is recommended.

In the event of serious risks, both permits and licenses and rules can be used.

Supervision of people and objects as opposed to advance supervision of activities:

When those creating the risk are returning players or when a one-time check is required prior to entry into the market – the appropriate tool is licenses for every supervised entity (people or objects).

When those creating the risk are one-time players and an individual check is required before every activity – the appropriate tool is a permit prior to activity. Because a permit requires an

individual check before each activity, this rule may place a heavy burden both on the supervising agency issuing the permits and on the supervised entities. To illustrate the difference between the license burden and the permit burden, contrast the number of drivers in Israel with the amount of times a year a driver crosses a junction; or the number of importers in Israel as opposed to the number of imported shipments to Israel each year. Permits are particularly suitable in the case of a market that has few activities, where each one creates a high risk. In view of the heavy burden involved in conducting an individual check before every activity, it is advisable to examine converting the regime of permits and licenses into a combination of binding rules and the duty to report before executing the activity creating the risk. In this way, the decision-maker defines a binding requirement and even receives information in advance about the dangerous activities. This way, the decision-maker can choose on which ones to conduct supervision activities.

Need for information on the supervised entities in order to address the risk:

If there is a need for information about the supervised entities in order to monitor the risk, duty to report

is the most appropriate tool. When the supervision and enforcement system needs information on the identity of the supervised entities or on their characteristics, there is no need to establish a regime of licenses, permits, etc.. It is often possible to convert a regime of permits and licenses into binding rules (design standards or performance standards) in combination with a duty to report commencement of activity or entry into the market.

When information is required both by the supervised entity and the public, the duty to report can be expanded and obligatory registration can be defined. Obligatory registration includes collection of information on the supervised entities in a database ("register") that is publicly accessible, and the conditions defined in respect thereof (for example: Registrar of Companies, Pawns Registrar, Contractors Registrar, and Databases Registrar).

Handling with information-related market failings:

To resolve a problem of gaps in information between parties, use should be made of duty to disclose or obligation to publish. Another option that can make information publicly

accessible is to impose a duty to report and an obligation to publish information to the public (for example: to publish data about the financial strength of insurance companies or to publish the safety rankings of the supervised entities).

When the public lacks information about the risks or ways of dealing with them, use can be made of campaigns, training, providing information and communication of the risks or the means of dealing with them.

Handling of concentrated markets:

In dealing with concentrated markets, it is preferable not to use tools such as permits and licenses because they create entry barriers and exacerbate the structural problem. In the case of concentrated markets, use of binding rules or advanced regulations is preferable.

Nascent protection is one of the government tools for easing market entry barriers and encouraging competition. This is a time-based tool that grants an exemption from or the easing of regulatory requirements for small or new players. It enables them to enter the market and to establish themselves among the existing players. Nascent protection is a medium-strength intervention

method. The significance of the nascent protection is that some of the policy or regulations' provisions are not imposed on small or new players. Accordingly, the risk management policy will not apply to them and its effectiveness will be diminished. Consequently, it is appropriate when the policy or regulations are not intended to address serious, high probability risk.

In the case of a highly concentrated market without any prospects for increased competition, price controls can be considered. This tool though can be expected to perpetuate the market structure and preserve it as a concentrated market, rather than as a competitive one. In addition, price controls will in all likelihood set a high price and prevent prices from declining.

For additional considerations and rules for identifying possible harming of competition, see Annex H.

Distinction according to the normative cause of the risk:

Force Majeure. If the risk is due to events beyond human control, society is limited in its ability to affect the risk's realization. In such cases, the strategy should be to focus on reducing the severity of the losses from the risk and

on crisis management. In some cases, it is also possible to influence the probability of the risk being realized. It is important to distinguish between rare force majeure events (such as natural disasters) and common events (such as an outbreak of disease).

Market failure. A major portion of public risk belongs to this category.

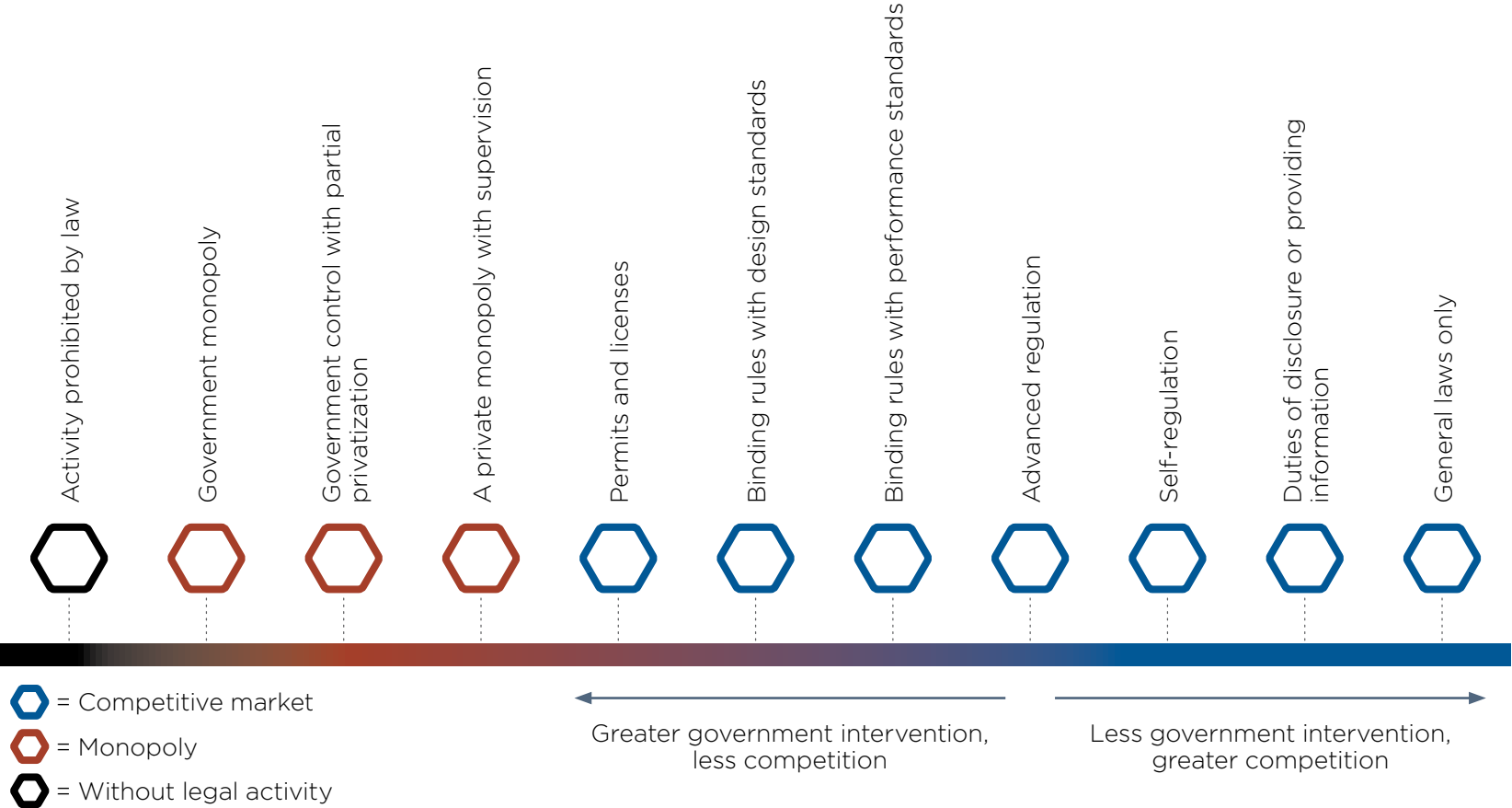
- The biggest group of market failures includes externalities (or "external influences"). An externality occurs when a person performs an action and the price of the action falls on others. Pollution is a classic example of an externality as the risk is dispersed and is not internalized by the polluter. In the case of externalities, the executor of the activity does not bear all the risk and the costs incurred. Consequently, they might execute the activity excessively without taking the appropriate precautions. The purpose of the regulations or the public policy is to cause the creator of the risk to internalize the implications of their action. When this is not feasible, the aim will be to prevent them from creating excessive risk (and effectively cause the creator of the risk to act as if they had internalized the implications of their actions).

The main tool to this end is not regulatory - civil lawsuits enable individuals to claim compensation from whoever caused them damage. As such, one should examine the civil lawsuit mechanism and whether it could mitigate the risk. When the externalities cause heavy damage, adding a regulatory layer can be considered, as well as establishing provisions obliging the creator of the damage to take various precautions.

- In the event of a market failure resulting from a problem in the flow of information (partial or asymmetric information), the solution should focus on improving the flow of information. Sometimes advanced regulations and nonregulatory tools can be used. For example: in the event of an information gap between a supplier and a consumer a presumption can be enacted. The presumption is a legal rule according to which a certain fact can be assumed, without the need to prove it, in favor of the party at a disadvantage due to the fact they possess inferior information. Such a rule may enable the weak party (in terms of the information in its possession) to sue the strong party even in

Impact of regulatory methods and policy tools on market competition.

The following figure presents the regulatory methods and the policy tools on a continuum according to their competitive impact. It can be seen that aggressive government intervention will, as a rule, weaken competition.



Does the public understand the risk in a way that enables it to be wary of it?

If the public underestimates the risk, tools should be considered to help them take appropriate protective measures. When the public understands the risks they are exposed to, they tend to take the necessary precautions. Such solutions could be aimed at providing information about the risk or increasing awareness about it. If the problem lies in the way the public assesses the risk, non-regulatory tools designed to inform should be considered.

Similarly, if the public does not understand the risk factors correctly, use of informative tools that clarify the situation should be considered. For example, if the public mistakenly thinks that the damage from a given risk can be mitigated by using tools or specific equipment, it will invest resources in solutions that are not applicable. It is important to make the risk characteristics (its causes, when and where it is realized, and which players are involved in creating the risk), clear to the public. Here too, it is a matter of information and understanding the risk, as such, the first thing to do is to consider tools that improve the

transfer of information to the public.

In this context, it is a good idea to examine just how complex the issue is, and the capacity of the relevant audience to understand it. When large businesses and knowledge-intensive companies suffer from knowledge gaps, it is easy to provide them with the information and the implications in respect of the risk. Conversely, when the information about the risk is abstract, complex, or requires professional knowledge to understand it, (especially when the information is intended for the public and not for professionals), providing information may not be an effective alternative. In such cases other tools should be examined, for example, mandating the use of protective measures against the risk. As stated in the section on "Risk Communication", the choice of mandatory regulation and the reasons behind it must also be explained to the public.

Guidelines for using advanced regulation and non-regulatory tools

Advanced regulations are particularly well-suited to treating risk that is not severe. When the risk is not typically severe or highly probable, and there are no serious compliance problems

advance regulations maybe more effective than traditional regulations. Since most of the public are law-abiding and the major portion of risks are not severe, advanced regulation is a solution for a wide range of situations. It provides simple solutions without the need to establish a supervision and enforcement system, where the origin of the risks is in a normative population. Consequently, advanced regulation provides an economical solution in terms of resources. It enable the decision-maker to concentrate on imposing restrictions focused on the severe risks and on instances of wrongdoing.

Advanced regulation may be a solution when deciding between a risk acceptance strategy and a risk mitigation strategy. In those instances, use can be made of advanced regulation, and even combined with preparations for crisis management. Advanced regulation may be added as another layer, as a tool that encourages compliance by providing positive reinforcement (for example: traffic police who hand out positive tickets for careful and safe driving).

When the aim of the policy is to grant a specific group financial aid or an

economic incentive, it is preferable to do so directly and as simply as possible. A complicated policy with indirect payments and benefits may miss their goal and create undesirable results. For example: a study by the Bank of Israel and the Central Bureau of Statistics examined the impact of rental assistance grants provided by the Jerusalem Municipality to students who live in the city center. According to the study, the grants led to a rise in rents in the city center. The implication: part of the student assistance was passed on to the homeowners. Furthermore, the grants led to a rise in rent for those renting apartments who had not received a grant.

Improving and strengthening the mechanism of civil lawsuits: Civil lawsuits are the most widely used tool for restraining risky behavior: they enable compensation for the injured party and serve to deter the creator of the risk.

- In the case of minor damage that does not require full, costly, and lengthy legal proceedings, a claim can be filed with the Small Claims Court. This is a simple, expedited procedure, so even someone not represented by a lawyer can sue for damage.

- In order to enable a civil lawsuit apparatus, a defined right must be enacted so that the injured party can sue in respect of it. The legal remedy may be financial compensation, or a restrictive order prohibiting the creator of the risk from acting.
- If there is difficulty in proving substantial financial damage, exemplary damages can be awarded. This mechanism enables the victim to receive compensation without proving damage. It also defines the maximum compensation that can be awarded without proof of damage.
- Sometimes, the concern is that because the damage is spread, the injured parties will not have the incentive to sue. This occurs when damage is dispersed across a wide population and each of the injured parties only absorbs a small portion of the damage. The solution in these cases may be to grant the option of instituting a class action on behalf of the injured group. This mechanism enables one of the injured parties to serve as a representative plaintiff and institute the class action on behalf of all injured parties. The rule in Israeli law is you have to opt out

i.e. by default, all members of the group are represented, but they can take active steps not to be included in it. This mechanism enables a wide group of injured parties who have sustained dispersed damage to institute a unified action (generally a large one) that justifies legal proceedings and creates a considerable deterrent for the creator of the damage.

- **In the event of evidentiary difficulties in proving the damage or the causal connection between the unacceptable behavior and the damage, evidentiary rules can be enacted.** One of the most commonly used rules includes presumptions and provisions about transferring the burden of proof. As a rule, the plaintiff must persuade the court that its argument is more reasonable than that of the defendant. Presumption is an evidentiary tool stipulating the burden of persuasion lies with the defendant. Presumptions can bridge inequalities in access to evidence and can simplify the legal proceedings. For example: Section 39 of the Israel Torts Ordinance stipulates that in a fire damage claim, the defendant (as the owner of the real estate from which

the fire emanated), must prove that the fire was not caused by negligence on their part (and is not the responsibility of the plaintiff). Similarly, the Israeli Standard Contracts Law stipulates a list of conditions that are presumed to be depriving and therefore invalid. By virtue of the presumption, the plaintiff is not required to prove that those conditions are depriving. Transfer of the burden of proof by presumption can resolve problems of evidentiary difficulties and enable injured parties to file a civil lawsuit against the risk creators.

Guiding behavior by means of behavioral economics:

- Economic theory analyzes human behavior on the basis of the assumption that human beings make rational decisions (i.e. define their goals and make decisions that advance these goals in the best possible manner). Behavioral economics is a branch that investigates systematic deviation from the making of "rational" decisions. Such deviations are also called bias.
- **Behavioral economics offers a tool set** that enables government to influence the behavior of individuals.

Behavioral economics can be used to support the decision-making process of the individual, without obligating the individual to do or avoid doing something.

- **One behavioral method is boosting.** This method focuses on irrational decision-making as a result of incomplete information and an inability to process and analyze it. This method encourages giving the public the tools to improve their decision-making process. An example is imparting knowledge and an understanding of financial concepts in order to improve the financial decisions made by the public.
- **Another method is called nudge.** Nudge is designed to shape the decision-making environment of individuals in order to influence their decision simply and inexpensively, without limiting their freedom of choice. Not only can the government use nudge to influence the decision-making process of the individual, so can commercial entities. In the commercial sector, an example of this is making consumers agree to accept advertisements as their default setting. It is therefore

sometimes necessary to "amend" existing use of nudge that produces an undesirable result. Examples of nudge are: simplifying complex decisions (especially in the case of uncommon decisions that the individual has little experience of); mapping and comparing alternatives (a concrete and clear-cut description of the alternatives and their implications, clear-cut presentation of comparable criteria) and defining a default (people tend not to deviate from the status quo, hence the default has great potency).

A third method is eliminating behavioral bias (debiasing). This method focuses on the root causes of problems of bounded rationality, such as partial information and cognitive limitations in processing and analyzing information. Behavioral debiasing is achieved by identifying the factors contributing to the bias and then modifying those factors in order to weaken them (solving problems of a lack of information, or an environment that contributes to distorted decision-making) or providing information about those factors in order to help the public avoid those biases.

Resolving problems of coordinated action:

- Sometimes risks stem from an inability to act in coordination or in sync. In such instances, the public has several options, but no unequivocal preference for a specific one; however, it is important the public act in a concerted manner according to the same rule or method.
- The preferred way of dealing with problems of coordination is to employ informal (nonbinding) standards. These standards do not impose a binding standard on the market, but they help all players converge on a uniform method. Together with the players, the government provides the market with the foundations for coordination and may also encourage voluntary adoption of a higher standard. One example is the layout of the keys on a computer keyboard: both manufacturers and consumers have an interest in the existence of a defined recognized standard that everyone can work according to. The generally accepted layout is QWERTY. In the United States an alternative layout called Dvorak is also recognized.

- When problems of coordination create significant risks (in terms of severity and probability), use can also be made of binding rules. An example is the rule to drive on the right side of the road. Driving on the right does not have any real advantage over driving on the left, but to ensure public safety, a binding and uniform rule must be established that everyone abides by.

Advantages and disadvantages according to groups of regulatory methods and policy tools**Permits and licenses:****Advantages**

In this regime, execution of the supervised activity is prohibited without advance receipt of an individual license or permit from the regulator. The main advantage of this group is the tight control afforded the government over the supervised activity. Furthermore, the obligation to operate under a government permit or license enables the State to revoke the permit or license (in the event of a breach), and thereby terminate the activity of the offending party.

Disadvantages

The typical disadvantage of the permits and licenses group lies in government intervention in the supervised activity. The government has to authorize every supervised entity or supervised activity (as the case may be). These tools create considerable friction between the government and the public, as every activity requires an individual permit or license. Operation of these tools generally requires a bureaucratic apparatus – submission of documents, checks before and after, and issuing of the permits and licenses. All these impose a regulatory burden on the public (expressed in direct costs, indirect costs and market entry barriers), and necessitate allocation of substantial government resources in order to manage the bureaucratic apparatus that supports this method. In addition, as with all government intervention, permits and licenses impose compliance costs on the public.

Binding rules:**Advantages**

The advantages of binding rules are that they can be defined to regulate or monitor activity without the need for active State intervention. In this regime, the State establishes rules and

enforces them, but the public does not have to engage with the State or obtain a permit or license. Binding rules are enforceable, and the government resources required are exclusively for supervision and enforcement purposes.

Disadvantages

As in the permits and licenses group, binding rules constitute traditional regulation including obligations and restrictions. Compliance with obligations carries an economic price (purchase of equipment, execution of processes). The restrictions and prohibitions may partially block or prevent beneficial activities. Binding rules are based on supervision and enforcement during the execution of the activity; consequently, government is unable to inspect compliance with the requirements in advance.

Advanced regulation and non-regulatory tools:**Advantages**

The advantages of this group are the flexibility and suitability of the tools, the implementation of which is not enshrined in legal provisions mandated by the State. Individuals are free to choose, and implementation

is custom tailored on a case-by-case basis. Tools in this group impose very few restrictions on the freedom of the individual and on socioeconomic activities. In addition, information related tools can be used to resolve a wide variety of failures by educating and empowering the relevant parties.

Disadvantages

Use of the tools in this group diminishes government control compared to traditional regulation. In addition, guiding behavior with soft power, such as incentive systems may have indirect, hidden, and unpredictable results. Furthermore, some of these tools necessitate government resources – from providing grants and tax benefits, to educational campaigns and making the information publicly accessible.

ANNEX H: RULES FOR IDENTIFYING POSSIBLE HARM TO COMPETITION

Regulation, and government policy in particular, may remove barriers to competition or strengthen them. Preventing harm to competition is essential in order to reduce prices, bring about higher quality, more variety in products and services, and contribute to productivity.

To examine the impact of regulation or public policy on competition, it is first necessary to identify which markets the decision will impact. In order to do so, the following questions need to be asked:

- Which businesses are involved?
- Who are their customers?
- What are the relevant products/ services?
- What are the characteristics of the players? (large/ small, imports/ domestic/ exports/ existing/ new that may enter the market)
- Will the regulation impact additional/ related markets?

An illustrative example: regulation of the supplier market may impact the distributor market.

Regulation stipulating that a supplier can only operate within a 50 km radius of its logistics center limits every

supplier to a defined geographic area. As a result of the geographic restriction on suppliers, results in the distributors (who buy from suppliers and sell to consumers) in each area are reliant on a single supplier. The distributors are hurt by the market strength of a single supplier as well as by the absence of product variety.

Following are issues that increase concerns about harm to the competition:

- In a concentrated market with few competitors, careful consideration must be given to the possible of harming competition (a simple rule of thumb is that a concentrated market is usually one with up to five players).
- The following types of policy arise increased concerns about harming competition: price control, production quotas (limiting and allotting quantities), import quotas and defining industry layout (size of competitors, number of competitors).

Leading questions	Illustrative examples
a. Will the policy directly limit the number of business?	
Does it set a limit on the number of businesses or grants exclusive rights to provide products or services?	• Granting a right to, or creating, a single supplier (for example: creating a national recycling corporation). • Allocating cellular frequencies constitutes granting of exclusive rights.
Does it establish a licensing regime in respect of the quality of the products or services?	• Standards in respect of characteristics unrelated to safety or health (for example: the minimum quantity of tomatoes in ketchup).
Does it create or increase entry barriers to a sector or market?	• Creation of threshold conditions for market entry: minimum manufacturing capacity, high business startup costs, high equity requirements in order to compete in a tender.
Does it create or increase import barriers? (Imposing restrictions on imports directly or restrictions that affect imported products only)	• A mandatory requirement to translate the user manual into the official language of the country of importation.. • A mandatory requirement that the manufacturing plant be approved by regulatory inspectors as a prerequisite for commencing marketing in Israel. Foreign manufacturing plants are dependent on an Israeli government inspector flying over to the foreign manufacturing plant to inspect it. • A prohibition on imports of non-Kosher beef may reduce competition in this industry.

Leading questions	Illustrative examples
b. Will the policy limit the number of businesses indirectly?	
Does it significantly increase the costs of the existing businesses to the extent that it causes them to exit the market?	- Significant regulatory or policy compliance costs for businesses to the extent that it causes them to exit the market. - Marking obligations, laboratory test obligations to prove compliance with standards and import duties.
Does it increase costs for new businesses compared with existing businesses or compared with another existing line of business?	Environmental legislation has provisions that impose significant costs on businesses. In order to reduce the impact, a decision was taken to grant relief to existing businesses, while new businesses will be required to comply with all the requirements.
Does it establish standards for product quality that gives certain businesses the edge over others?	- Special standards that give an artificial edge to Israeli businesses: a unique Israeli electrical socket that is not compatible with imported appliances. - Requirements to submit documents that only the official importer possesses, give it an advantage over comparable importers. - In the past, a specific Israel Standard required that tea bags be adhered without the use of staplers. Only one Israeli manufacturer complied with the standard, blocking out all its competitors.
c. Will the policy limit the ability of businesses to compete?	
Does it control or significantly impact the price that the business may charge or the characteristics of the products or the services?	- Price control through price setting. - A safety standard dictates a detailed engineering specification that determines the product structure.
Does it limit the sales channels that a business can use or the geographic area in which it can operate?	Grants exclusivity in a specific locality for defined clientele.
Does it limit the ability of businesses to advertise their products?	New regulation prohibits businesses from advertising their services. As a result, existing and well known businesses have an edge over new businesses that try to penetrate the market, but are unable to inform customers of the fact.

Leading questions	Illustrative examples
d. Will the policy limit the incentive of businesses to compete against each other?	
Does it help competitors to coordinate their behavior?	- The regulation sets out obligations or restrictions that help competitors to fix prices, quantities, service level, or split geographic areas between them, etc. - Joint ventures between competitors may enable them to coordinate their behavior.
Does it cause or make it easier for competitors to share information between them? (Publication of information on outputs of businesses or their prices, their sales, or their costs)	- The State certifies laboratories that are authorized to perform genetic testing. One of the terms of the certification requires laboratories publish the prices of all the tests they offer. - The State establishes a shared database for calculations of the insurance companies' premiums. The database enables all the insurance companies to view the data fed in by the other companies.
Does it create a regime of self-regulation or joint regulation? (enabling companies to participate in establishing the regulations)	Committees composed of business representatives convene in order to establish agreed product safety standards; the State will enforce the standards or grant benefits to manufacturers who comply with them.
e. Will the policy limit the customer options and information?	
Does it restrict the ability of customers to choose who they buy from?	- Regulation stipulates that deliveries of raw materials to restaurants shall not come from a food production plant more than to 30 km from the restaurant. As a result, restaurants are limited in the range of food production plants from which they can order raw materials. - Until the 1990s, a prohibition existed in Israel against opening a pharmacy within 500m of an existing pharmacy. This regulation reduced the number of pharmacies in a defined locality.
Does it limit the transfer barriers between businesses? (Diminishes the ability of customers to transfer to another business by increasing the overt or hidden costs of such a transfer)	- The regulation requires every Internet supplier to use a different model of modem; therefore, switching Internet supplier obliges the customer to purchase a new modem. - The regulation defines that the telephone number belongs to the telecom company and not to the subscriber; therefore, transferring to a different telecom company requires the customer relinquish its telephone number.

The rules in this Annex constitute a summary of select principles for identifying possible harm to competition. For more information, it is advisable to consult guides examining the competitive impact of regulation, and with an economist or antitrust expert. These rules are based on the guide to competitive influences issued by the British CMA and the OECD toolkit for assessing the impact of regulation on competition. (See references and other sources at the end of the guide).

ANNEX I: RULES FOR IDENTIFYING POSSIBLE HARM TO SMALL AND MEDIUM-SIZE BUSINESSES

Regulation and government policy affect businesses of different sizes in different ways. Small and medium-size businesses bear the burden of regulation more as they are smaller and more vulnerable.

"Small businesses": employ up to 20 workers and have an annual turnover of up to ILS 10 million.

"Medium-size businesses": employ up to 100 workers and have an annual turnover of up to ILS 100 million.

Small and medium-size businesses (SMBs) make a decisive contribution to the economy. In Israel, SMBs constitute 99.2% of all businesses, provide work for 70% of the employed, and contribute about half the economy's added value.

Regulation and public policy may impact SMBs directly and indirectly (for example: when the regulation provides supervision of a large business, but also has an impact on the suppliers or on its business customers who are SMBs).

The guiding principle is to think first

of the small businesses (Think Small First) – to formulate rules based on how small businesses will be impacted by them and examine costs from their perspective.

Examining risks that SMBs create

Strive to take the constraints and vulnerabilities of SMBs into account, subject to the risk assessment and to the influence of SMBs on the creation of the risks. To this end, clarify and examine:

- Does the activity of SMBs contribute to the creation of the risks that need to be addressed?
- What is the scale of the risk (frequency, severity of the economic damage) that might be created by SMBs? Quantifying the impact of the proposed method on SMBs
- If it is expected that SMBs will be impacted by the proposed method, it is important not to base the quantification on large businesses, but on SMBs.
- Focusing on SMBs during the gathering of information and consultation stages. For example, assistance can be obtained from the Small and Medium Business

Agency in the Ministry of Economy (including enlisting the support of MAOF, the Agency's field branch); organizations representing SMBs can be contacted; and a business sample created specifically focusing on SMBs.

Formulation of alternatives taking into account the characteristics of SMBs

The following are tools and considerations that can help decision-makers adapt the regulation or policy to the capabilities and limitations of SMBs:

- For SMBs, it is vital to match the level of risk they create with the costs of the measures for dealing with the risks. Excessive compliance costs may result in the closing down of SMBs. Special care must therefore be taken to avoid a policy of "exclusion" that includes provisions that prioritize large businesses or restrict the entry of small businesses into a specific market.
- After formulating the content of the regulation or the policy, the costs that will be imposed on small businesses can be compared with their total turnover. This way, it

is possible to determine whether significant expenses are added, or profit margins significantly hurt.

- Granting an exemption from the policy or from provisions of the regulation. A full exemption from the policy or regulation as a whole may be granted, or a partial exemption from some of the provisions.
- Relief can be granted from the information requirements. Information requirements include provisions about the gathering of information, documenting and reporting. Such relief may include reducing the frequency of information obligations. For example: switching from annual reporting to biennial reporting, reducing the scope of the information obligations (the scope of information that needs to be dealt with), and simplification of the obligations.
- Granting relief on a time scale. Such relief will grant an exemption period during which SMBs are not subject to the provisions. For example: establishing that the provisions will apply only to businesses that have been in existence for at least five years or granting a longer period

to prepare for implementation of the provisions (for example: large businesses are required to implement the provisions immediately; enforcement against SMBs begins after a year).

- Granting relief with respect to fees and other obligatory payments. The effect of every obligation and cost is amplified in SMBs.
- Ensuring that the policy and regulation are clear-cut and precise:

Clear-cut and simple provisions

When the provisions are clear-cut and simple, it is easy to prepare and implement them. Lack of clarity may result in flawed implementation of the provisions or necessitate approaches to the government for clarifications and explanations.

Certain, expected and consistent provisions

Uncertainty about the provisions makes it difficult to plan and prepare for the future. They may be perceived by supervised businesses as an unstable policy.

Advance analysis of the clarity and certainty

It is recommended to hold a public

discourse with SMBs during the planning and formulation of the policy or regulation.

Assimilation and accessibility of the provisions. After adoption of the policy or regulation, steps should be taken to ensure that the SMBs are acquainted with, understand, and are capable of implementing the provisions. This can be achieved by publishing guides and providing training and lectures.

Custom tailoring supervision and enforcement activities. Supervision and enforcement activities also impose costs on businesses. Therefore, ways should be examined on how to design the supervision and enforcement system so that SMBs are taken into consideration. For example: to establish that activities to supervise them will be focused and less burdensome.

Relief in the size of fines. SMBs also need to comply with the provisions and to be punished in the event of a breach. However, harsh financial penalties may cause the collapse of a

small business. Therefore, the financial strength of SMBs should be taken into account when defining the potency of the sanctions and applying them. In this regard, the relative financial weakness of SMBs means that lighter sanctions are sufficient to deter them.

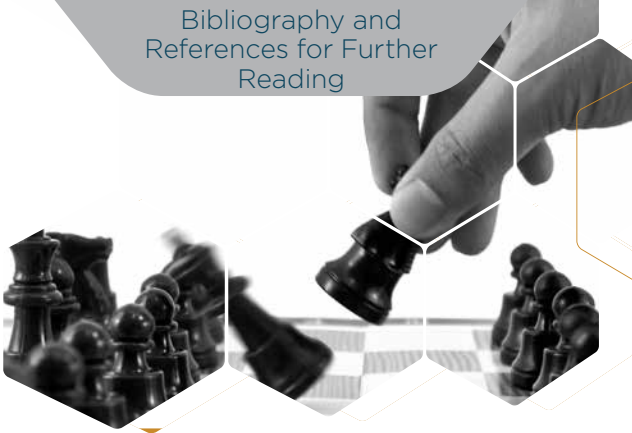
Examples for the application of some of these principles can be found in Government Resolution 890 of December 27, 2015 - "Encouragement of New Businesses".

In Israel, it is advisable to consult with and enlist the help of the Small and Medium Business Agency in the Ministry of Economy and Industry regarding risk management and the policy chosen for SMBs: asakim@economy.gov.il

ANNEX J: SUPERVISION TASK LIST

Risk Management Process in Supervision and Enforcement in the Field of [complete]: Supervision Task List

Serial Number	Requirement name	Supervised entities group	Requirement importance ranking	Description of element to be supervised



Size of the supervised entities group	Expected non-compliance rate	Main reason for non-compliance	Overall score	Category

ANNEX K: LINKS FOR DOWNLOADING TOOLS AND ACCESSORIES

The following files can be found on the website of the guide to risk management in regulation and public policy: risk registry, risk matrix, and a tool for planning the supervision and enforcement system.
<http://bit.ly/RegRiskManage>

A list of the regulatory methods and policy tools is provided in Addendum A to the Government Guide for Regulatory Impact Assessment - RIA Guide (2015):
<http://regulation.gov.il/RIAGUIDE>

For more on the subject of public participation and public discourse, see: the Prime Minister's Office "Guide to Public Participation and Public Discourse" (2017):
<http://www.pmo.gov.il/policyplanning/shituf/Documents/all%20web.pdf>



